

Analysis Fourier and Distributions

Exercise 1.1 — A meromorphic function and a Fourier series

Fix $\theta \in [0, 2\pi]$ and consider the meromorphic function

$$F(z) = \frac{1}{1+z^2} \frac{\cos((\pi - \theta)z)}{2\sin(\pi z)}.$$

Problem

- (a) Show that F has poles at $z = \pm i$ and at each integer $z = n$, $n \in \mathbb{Z}$. Compute the residues at these poles and verify that

$$\operatorname{Res}(F, \pm i) = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}, \quad \operatorname{Res}(F, n) = \frac{\cos(n\theta)}{2\pi(1+n^2)}.$$

- (b) For each $N \in \mathbb{N}$ define four curves $\Gamma_j^{(N)}$ on $[-1, 1]$ by

$$\Gamma_1^{(N)}(t) = \left(N + \frac{1}{2}\right)(1 + it), \quad \Gamma_2^{(N)}(t) = \left(N + \frac{1}{2}\right)(-t + i),$$

$$\Gamma_3^{(N)}(t) = \left(N + \frac{1}{2}\right)(-1 - it), \quad \Gamma_4^{(N)}(t) = \left(N + \frac{1}{2}\right)(t - i),$$

and let $\Gamma^{(N)}$ be the closed contour obtained by following $\Gamma_1^{(N)}, \Gamma_2^{(N)}, \Gamma_3^{(N)}, \Gamma_4^{(N)}$ in turn. Sketch $\Gamma^{(N)}$ in the complex plane and indicate the locations of the poles of F relative to this rectangle.

- (c) Show that there exists a constant $C > 0$, independent of N , such that for all integers $N \geq 1$ and all $\theta \in [0, 2\pi]$,

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \frac{C}{N}.$$

- (d) Using Cauchy's residue theorem applied to $\Gamma^{(N)}$, show that for $0 \leq \theta \leq 2\pi$,

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{e^{in\theta}}{1+n^2} \right| \leq \frac{C}{N},$$

and deduce that

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2},$$

with the series converging uniformly in $\theta \in [0, 2\pi]$.

Theory and background

The function F is meromorphic on $\mathbb{C}$, being the product of the entire function $\cos((\pi - \theta)z)$ with the factors $(1 + z^2)^{-1}$ and $(\sin \pi z)^{-1}$. The only singularities of F are poles arising from zeros of $1 + z^2$ and $\sin(\pi z)$.

The equation $1 + z^2 = 0$ has solutions $z = \pm i$, which give simple poles of F . The function $\sin(\pi z)$ vanishes simply at each integer $z = n \in \mathbb{Z}$, so $(\sin \pi z)^{-1}$ has simple poles there. Since $1 + z^2$ does not vanish at any integer, F also has simple poles at all $z = n \in \mathbb{Z}$.

For a simple pole at $a \in \mathbb{C}$, the residue is given by

$$\text{Res}(F, a) = \lim_{z \rightarrow a} (z - a) F(z).$$

At an integer n one uses the local behaviour $\sin(\pi z) = \pi(-1)^n(z - n) + O((z - n)^2)$ and evaluates the remaining factors at $z = n$. At $z = \pm i$ one factors $1 + z^2 = (z - i)(z + i)$ and again evaluates the remaining analytic part at the pole.

The contour $\Gamma^{(N)}$ is a rectangle with vertices

$$\pm \left(N + \frac{1}{2}\right) \pm i,$$

with horizontal sides at $\Im z = \pm 1$ and vertical sides at $\Re z = \pm(N + \frac{1}{2})$. For each fixed N , the contour encloses the poles at $z = \pm i$ and at the integers n with $|n| \leq N$.

Cauchy's residue theorem states that for such a closed contour,

$$\int_{\Gamma^{(N)}} F(z) dz = 2\pi i \sum_{a \in \mathcal{P}_N} \text{Res}(F, a),$$

where $\mathcal{P}_N$ is the finite set of poles of F inside $\Gamma^{(N)}$. Here $\mathcal{P}_N = \{-i, +i\} \cup \{n \in \mathbb{Z} : |n| \leq N\}$.

To estimate the integral along $\Gamma^{(N)}$ one uses the estimation lemma

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \sup_{z \in \Gamma^{(N)}} |F(z)| \cdot \text{length}(\Gamma^{(N)}).$$

On the vertical segments, $z = \pm(N + \frac{1}{2}) + it$ with $-1 \leq t \leq 1$, the factor $1/(1 + z^2)$ is of order $1/N^2$, whereas $\cos((\pi - \theta)z)$ and $1/\sin(\pi z)$ are uniformly bounded in N , since

$\Im z$ stays in a compact set. On the horizontal segments, $\Re z$ is bounded and $\Im z = \pm 1$, so F remains uniformly bounded. Combining these facts gives a bound of the form

$$\left| \int_{\Gamma(N)} F(z) dz \right| \leq \frac{C}{N}$$

for some constant C independent of N and θ .

Substituting the residue computations into the residue theorem gives an identity of the form

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{e^{in\theta}}{1 + n^2} = \frac{1}{2\pi i} \int_{\Gamma(N)} F(z) dz,$$

so that the absolute value of the difference is bounded by C/N . Letting $N \rightarrow \infty$ shows that the Fourier series

$$\sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2}$$

converges uniformly in $\theta \in [0, 2\pi]$ and represents $\cosh(\pi - \theta)/(2 \sinh \pi)$.

Exercise 1.1 — Solution

We consider, for fixed $\theta \in [0, 2\pi]$, the meromorphic function

$$F(z) = \frac{1}{1 + z^2} \frac{\cos((\pi - \theta)z)}{2 \sin(\pi z)}.$$

(a) Poles and residues

The factor $\cos((\pi - \theta)z)$ is entire, so the singularities of F arise from the denominators $1 + z^2$ and $\sin(\pi z)$.

- The zeros of $1 + z^2$ are $z = \pm i$, and these are simple.
- The zeros of $\sin(\pi z)$ are the integers $z = n \in \mathbb{Z}$, and each zero is simple, so $(\sin \pi z)^{-1}$ has simple poles at all integers.

Thus F has simple poles at $z = \pm i$ and at each $z = n \in \mathbb{Z}$.

For a simple pole at a , we have

$$\text{Res}(F, a) = \lim_{z \rightarrow a} (z - a)F(z).$$

Residues at $\pm i$. We factor

$$1 + z^2 = (z - i)(z + i).$$

Hence the residue of $(1 + z^2)^{-1}$ at $z = i$ is

$$\operatorname{Res}\left(\frac{1}{1 + z^2}, i\right) = \frac{1}{2i}, \quad \operatorname{Res}\left(\frac{1}{1 + z^2}, -i\right) = -\frac{1}{2i}.$$

Using $\cos(ix) = \cosh x$ and $\sin(ix) = i \sinh x$ we have

$$\cos((\pi - \theta)i) = \cosh(\pi - \theta), \quad \sin(\pi i) = i \sinh \pi, \quad \sin(-\pi i) = -i \sinh \pi.$$

Therefore

$$\operatorname{Res}(F, i) = \frac{1}{2i} \cdot \frac{\cosh(\pi - \theta)}{2 \sin(\pi i)} = \frac{1}{2i} \cdot \frac{\cosh(\pi - \theta)}{2i \sinh \pi} = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi},$$

and

$$\operatorname{Res}(F, -i) = -\frac{1}{2i} \cdot \frac{\cosh(\pi - \theta)}{2 \sin(-\pi i)} = -\frac{1}{2i} \cdot \frac{\cosh(\pi - \theta)}{-2i \sinh \pi} = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}.$$

Thus

$$\operatorname{Res}(F, \pm i) = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}.$$

Residues at integers. Let $n \in \mathbb{Z}$. Near $z = n$ we have

$$\sin(\pi z) = \sin(\pi n + \pi(z - n)) = \sin(\pi n) \cos(\pi(z - n)) + \cos(\pi n) \sin(\pi(z - n)).$$

Since $\sin(\pi n) = 0$ and $\cos(\pi n) = (-1)^n$, it follows that

$$\sin(\pi z) = (-1)^n \pi(z - n) + O((z - n)^2),$$

and therefore

$$\frac{1}{\sin(\pi z)} = \frac{(-1)^n}{\pi} \frac{1}{z - n} + O(1).$$

The other factors are analytic at $z = n$, so

$$\operatorname{Res}(F, n) = \frac{1}{1 + n^2} \cdot \frac{\cos((\pi - \theta)n)}{2} \cdot \frac{(-1)^n}{\pi}.$$

Using

$$\cos((\pi - \theta)n) = \cos(\pi n - n\theta) = \cos(\pi n) \cos(n\theta) + \sin(\pi n) \sin(n\theta) = (-1)^n \cos(n\theta),$$

we obtain

$$\operatorname{Res}(F, n) = \frac{(-1)^n}{2\pi(1 + n^2)} \cdot (-1)^n \cos(n\theta) = \frac{\cos(n\theta)}{2\pi(1 + n^2)}.$$

(b) The contour $\Gamma^{(N)}$

For $N \in \mathbb{N}$, the four curves

$$\Gamma_1^{(N)}(t) = \left(N + \frac{1}{2}\right)(1 + it), \quad \Gamma_2^{(N)}(t) = \left(N + \frac{1}{2}\right)(-t + i),$$

$$\Gamma_3^{(N)}(t) = \left(N + \frac{1}{2}\right)(-1 - it), \quad \Gamma_4^{(N)}(t) = \left(N + \frac{1}{2}\right)(t - i), \quad -1 \leq t \leq 1,$$

form the sides of the rectangle whose vertices are

$$\pm \left(N + \frac{1}{2}\right) \pm i.$$

We denote by $\Gamma^{(N)}$ the closed contour obtained by successively traversing $\Gamma_1^{(N)}, \dots, \Gamma_4^{(N)}$.

This rectangle encloses the poles at $z = \pm i$ and at $z = n$ for all integers n with $|n| \leq N$.

A schematic picture is shown in Figure 1.

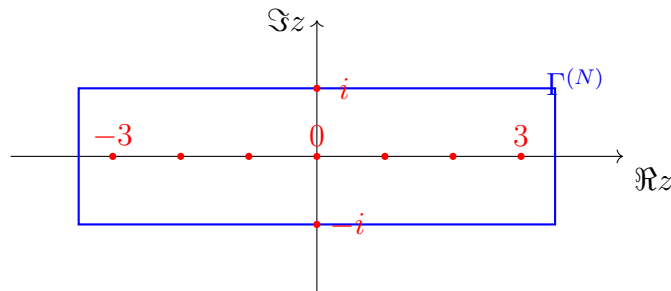


Figure 1: The contour $\Gamma^{(N)}$ and the poles of F .

(c) Estimate of the contour integral

We claim that there exists a constant $C > 0$, independent of N and θ , such that

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \frac{C}{N} \quad \text{for all } N \geq 1.$$

A standard way to see this is to decompose $\Gamma^{(N)}$ into its four sides and apply the estimation lemma.

On the vertical sides we have $z = \pm \left(N + \frac{1}{2}\right) + it$ with $|t| \leq 1$. Then $|z|^2 \asymp N^2$ and hence

$$\left| \frac{1}{1 + z^2} \right| \leq \frac{C_1}{N^2}.$$

Moreover, for such z the imaginary part is bounded, so both $\cos((\pi - \theta)z)$ and $1/\sin(\pi z)$ are uniformly bounded in N . The length of each vertical side is 2, so the total contribution of the vertical sides is $O(N^{-2})$.

On the horizontal sides we have $z = x \pm i$ with $-(N + \frac{1}{2}) \leq x \leq N + \frac{1}{2}$. Using explicit expressions for $\cos((\pi - \theta)(x \pm i))$ and $\sin(\pi(x \pm i))$, one checks that

$$|F(x + i) - F(x - i)| \leq \frac{C_2}{1 + |x|^3} \quad \text{for all } x \in \mathbb{R},$$

and that $|F(x \pm i)| \leq C_3/(1 + x^2)$ for large $|x|$. Writing the integral over the two horizontal sides as

$$\int_{-(N+1/2)}^{N+1/2} (F(x + i) - F(x - i)) dx,$$

one sees that the tail for $|x| \geq N$ is $O(N^{-2})$, while the integral over $|x| \leq N$ remains bounded independently of N . Altogether this yields a bound of the form

$$\left| \int_{\Gamma(N)} F(z) dz \right| \leq \frac{C}{N}$$

for some constant C independent of N .

(d) Residue theorem and the Fourier series

By Cauchy's residue theorem,

$$\int_{\Gamma(N)} F(z) dz = 2\pi i \sum_{a \in \mathcal{P}_N} \text{Res}(F, a),$$

where $\mathcal{P}_N = \{-i, +i\} \cup \{n \in \mathbb{Z} : |n| \leq N\}$ is the set of poles inside the rectangle. Using the residues computed in part (a),

$$\sum_{a \in \mathcal{P}_N} \text{Res}(F, a) = \text{Res}(F, i) + \text{Res}(F, -i) + \sum_{n=-N}^N \text{Res}(F, n) = -\frac{\cosh(\pi - \theta)}{2 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)}.$$

Therefore

$$\int_{\Gamma(N)} F(z) dz = 2\pi i \left(-\frac{\cosh(\pi - \theta)}{2 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \right),$$

or equivalently

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} = -\frac{1}{2\pi i} \int_{\Gamma(N)} F(z) dz.$$

Using the estimate from part (c) we obtain

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \right| \leq \frac{1}{2\pi} \left| \int_{\Gamma(N)} F(z) dz \right| \leq \frac{C}{N}.$$

Since

$$\frac{1}{2\pi} \sum_{n=-N}^N \frac{\cos(n\theta)}{1 + n^2} = \Re \left(\sum_{n=-N}^N \frac{e^{in\theta}}{1 + n^2} \right),$$

we can rewrite this in the complex exponential form

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{e^{in\theta}}{1 + n^2} \right| \leq \frac{C}{N}.$$

Letting $N \rightarrow \infty$ gives, uniformly in $\theta \in [0, 2\pi]$,

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2},$$

which is the desired Fourier series representation.

We first recall the residues already computed:

$$\begin{aligned} \operatorname{Res}(F, \pm i) &= -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}, \\ \operatorname{Res}(F, n) &= \frac{\cos(n\theta)}{2\pi(1 + n^2)}, \quad n \in \mathbb{Z}. \end{aligned}$$

The set of poles inside $\Gamma^{(N)}$ is

$$\mathcal{P}_N = \{-i, +i\} \cup \{n \in \mathbb{Z} : |n| \leq N\},$$

so

$$\sum_{a \in \mathcal{P}_N} \operatorname{Res}(F, a) = \operatorname{Res}(F, i) + \operatorname{Res}(F, -i) + \sum_{n=-N}^N \operatorname{Res}(F, n).$$

Now substitute the explicit formulas and simplify **step by step**:

$$\begin{aligned} \sum_{a \in \mathcal{P}_N} \operatorname{Res}(F, a) &= \left(-\frac{\cosh(\pi - \theta)}{4 \sinh \pi} \right) + \left(-\frac{\cosh(\pi - \theta)}{4 \sinh \pi} \right) + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \\ &= -\frac{\cosh(\pi - \theta)}{4 \sinh \pi} - \frac{\cosh(\pi - \theta)}{4 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \\ &= -\left(\frac{\cosh(\pi - \theta)}{4 \sinh \pi} + \frac{\cosh(\pi - \theta)}{4 \sinh \pi} \right) + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \\ &= -\frac{2 \cosh(\pi - \theta)}{4 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \\ &= -\frac{\cosh(\pi - \theta)}{2 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)}. \end{aligned}$$

Hence

$$\sum_{a \in \mathcal{P}_N} \operatorname{Res}(F, a) = -\frac{\cosh(\pi - \theta)}{2 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1+n^2)}.$$

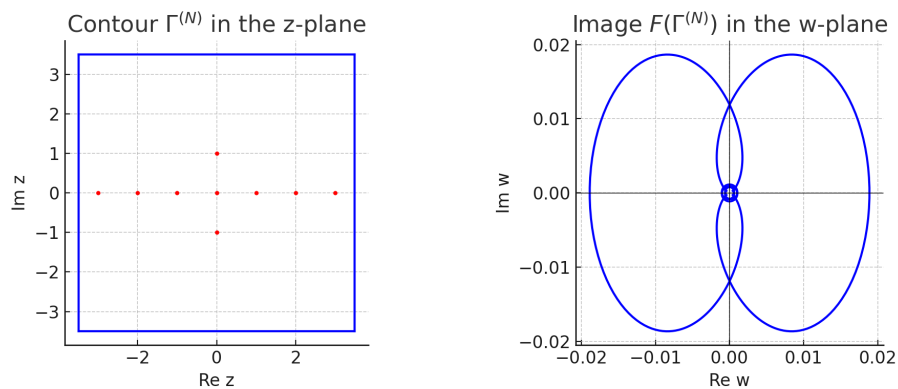


Figure 2: Left: the rectangular contour $\Gamma^{(N)}$ in the z -plane for $N = 3$, enclosing the poles of $F(z) = \frac{1}{1+z^2} \frac{\cos((\pi-\theta)z)}{2\sin(\pi z)}$ at $z = \pm i$ and at the integers with $|n| \leq N$. Right: the image $F(\Gamma^{(N)})$ in the w -plane for $\theta = \pi/3$. This visualises how the contour integral in Cauchy's residue theorem is transported under F .

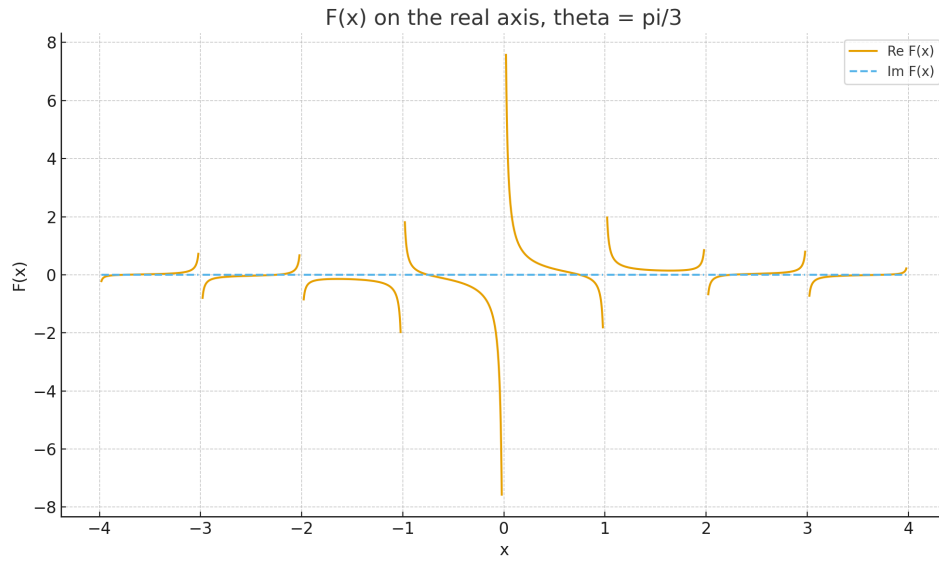


Figure 3: Real and imaginary parts of $F(x)$ for real x and $\theta = \pi/3$. The dashed curve shows $\Im F(x)$ and the solid curve shows $\Re F(x)$. The large spikes correspond to simple poles at the integers $x = n \in \mathbb{Z}$; small gaps have been inserted in the plot near the poles to avoid plotting the singular values.

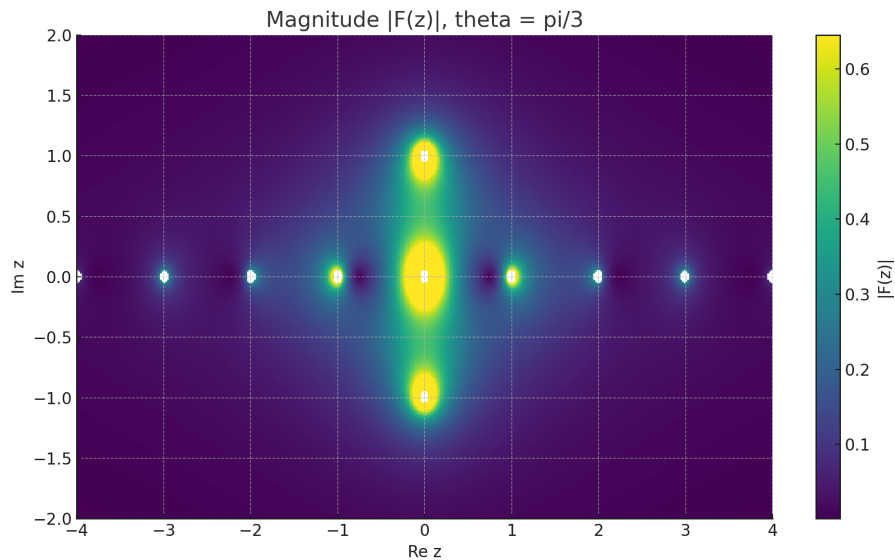


Figure 4: Density plot of the magnitude $|F(z)|$ for $-4 \leq \Re z \leq 4$ and $-2 \leq \Im z \leq 2$, with $\theta = \pi/3$. Brighter regions indicate larger values of $|F(z)|$, while dark spots mark the locations of the poles at integers $z = n$ and at $z = \pm i$. The plot illustrates the growth and singularity structure of F in the complex plane.

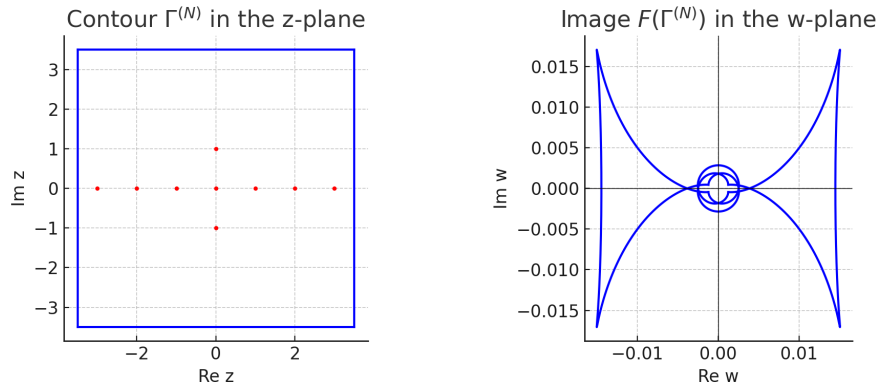


Figure 5: Left: the rectangular contour $\Gamma^{(N)}$ in the z -plane for $N = 3$, enclosing the poles of $F(z) = \frac{1}{1+z^2} \frac{\cos((\pi-\theta)z)}{2\sin(\pi z)}$ at $z = \pm i$ and the integers with $|n| \leq N$. Right: the image $F(\Gamma^{(N)})$ in the w -plane for $\theta = \pi/4$. Compared to $\theta = \pi/3$, the lobes in the w -plane rotate and change shape, illustrating the dependence of the contour image on the parameter θ .

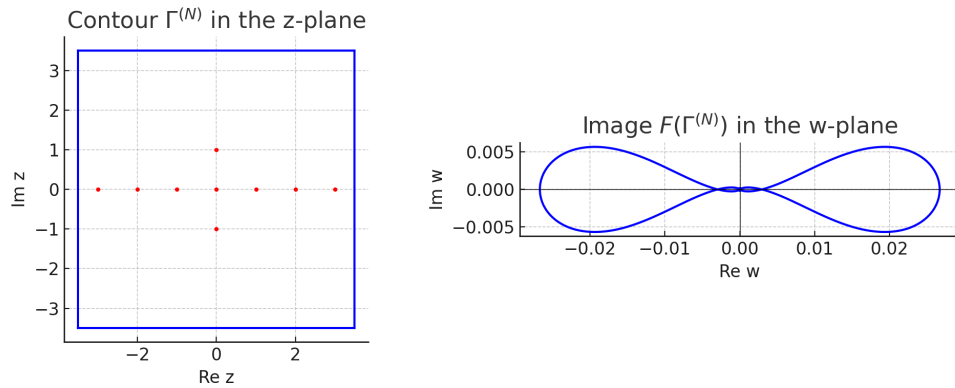


Figure 6: Left: the contour $\Gamma^{(N)}$ in the z -plane for $N = 3$. Right: the image $F(\Gamma^{(N)})$ in the w -plane for $\theta = \pi/2$. In this case $\pi - \theta = \pi/2$, so the cosine factor involves $\cos(\frac{\pi}{2}z)$ and the resulting image becomes more symmetric, with the loops in the w -plane reflecting this special choice of parameter.

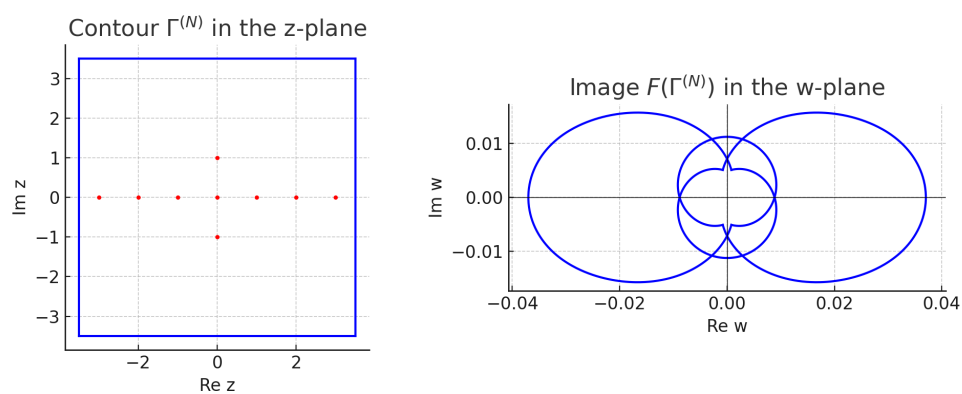


Figure 7: Left: the contour $\Gamma^{(N)}$ in the z -plane for $N = 3$. Right: the image $F(\Gamma^{(N)})$ in the w -plane for $\theta = \pi/8$. For a small value of θ the factor $\cos((\pi - \theta)z)$ is close to $\cos(\pi z)$, and the loops in the w -plane are correspondingly close to those for $\theta = 0$, while still showing a visible deformation.

$$\frac{\cosh \pi}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{1}{1+n^2}.$$

$$1 + 2 \sum_{n=1}^{\infty} \frac{1}{1+n^2} = \frac{\cosh \pi}{2 \sinh \pi}, \quad \sum_{n=1}^{\infty} \frac{1}{1+n^2} = \frac{1}{2} \left(\frac{\cosh \pi}{2 \sinh \pi} - 1 \right).$$

Exercise 1.2 — Periodic Green's function on the circle

Suppose that $f \in C^0(\mathbb{R})$ is a continuous function with period 2π , i.e. $f(\theta) = f(\theta + 2\pi)$. For $\theta \in [0, 2\pi]$ define

$$\psi(\theta) := \int_0^\theta f(\alpha) \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha + \int_\theta^{2\pi} f(\alpha) \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha,$$

and extend ψ to a function on $\mathbb{R}$ by periodicity:

$$\psi(\theta + 2\pi) = \psi(\theta), \quad \theta \in \mathbb{R}.$$

(a) Show that $\psi \in C^0(\mathbb{R})$.

(b) By differentiating the defining formula directly, show that

$$\psi'(\theta) = - \int_0^\theta f(\alpha) \frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha - \int_\theta^{2\pi} f(\alpha) \frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha,$$

and deduce that $\psi \in C^1(\mathbb{R})$.

(c) Differentiating once more, show that

$$\psi''(\theta) = -f(\theta) + \psi(\theta), \quad \theta \in \mathbb{R},$$

and conclude that $\psi \in C^2(\mathbb{R})$ is a 2π -periodic solution of the inhomogeneous ODE

$$\psi''(\theta) + f(\theta) = \psi(\theta), \quad \theta \in \mathbb{R}.$$

Suppose that $f \in C^0(\mathbb{R})$ is a continuous function with period 2π , i.e. $f(\theta) = f(\theta + 2\pi)$. For $\theta \in [0, 2\pi]$ define

$$\psi(\theta) := \int_0^\theta f(\alpha) \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha + \int_\theta^{2\pi} f(\alpha) \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha,$$

and extend ψ to a function on $\mathbb{R}$ by periodicity: $\psi(\theta) = \psi(\theta + 2\pi)$.

(a) Continuity of ψ

For $(\theta, \alpha) \in [0, 2\pi] \times [0, 2\pi]$ set

$$K_1(\theta, \alpha) = \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi}, \quad K_2(\theta, \alpha) = \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi}.$$

These kernels are continuous (indeed real-analytic) in both variables, and f is continuous. Thus the integrands

$$(\theta, \alpha) \mapsto f(\alpha)K_1(\theta, \alpha), \quad (\theta, \alpha) \mapsto f(\alpha)K_2(\theta, \alpha)$$

are continuous on the compact set $[0, 2\pi] \times [0, 2\pi]$. It follows from the standard theorem on parameter-dependent integrals that

$$\theta \mapsto \int_0^\theta f(\alpha)K_1(\theta, \alpha) d\alpha, \quad \theta \mapsto \int_\theta^{2\pi} f(\alpha)K_2(\theta, \alpha) d\alpha$$

are continuous on $(0, 2\pi)$; hence ψ is continuous on $(0, 2\pi)$.

At the endpoints one checks that the one-sided limits agree with the periodic extension. Using 2π -periodicity of f and the evenness of $\cosh$, we obtain

$$\lim_{\theta \rightarrow 0^+} \psi(\theta) = \int_0^{2\pi} f(\alpha) \frac{\cosh(-\pi - \alpha)}{2 \sinh \pi} d\alpha = \int_0^{2\pi} f(\alpha) \frac{\cosh(-\pi + \alpha)}{2 \sinh \pi} d\alpha = \psi(2\pi).$$

Together with the periodic extension $\psi(\theta + 2\pi) = \psi(\theta)$ this shows that $\psi \in C^0(\mathbb{R})$.

(b) First derivative and C^1 -regularity

Write

$$\psi(\theta) = I_1(\theta) + I_2(\theta),$$

where

$$I_1(\theta) = \int_0^\theta f(\alpha)K_1(\theta, \alpha) d\alpha, \quad I_2(\theta) = \int_\theta^{2\pi} f(\alpha)K_2(\theta, \alpha) d\alpha.$$

Since K_1, K_2 are smooth and f is continuous, we may differentiate under the integral sign using Leibniz' rule. For I_1 we obtain

$$I_1'(\theta) = f(\theta)K_1(\theta, \theta) + \int_0^\theta f(\alpha) \partial_\theta K_1(\theta, \alpha) d\alpha.$$

A direct computation yields

$$K_1(\theta, \theta) = \frac{\cosh \pi}{2 \sinh \pi}, \quad \partial_\theta K_1(\theta, \alpha) = -\frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi},$$

so that

$$I_1'(\theta) = f(\theta) \frac{\cosh \pi}{2 \sinh \pi} - \int_0^\theta f(\alpha) \frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha.$$

Similarly,

$$I_2'(\theta) = -f(\theta)K_2(\theta, \theta) + \int_\theta^{2\pi} f(\alpha) \partial_\theta K_2(\theta, \alpha) d\alpha,$$

with

$$K_2(\theta, \theta) = \frac{\cosh(-\pi)}{2 \sinh \pi} = \frac{\cosh \pi}{2 \sinh \pi}, \quad \partial_\theta K_2(\theta, \alpha) = -\frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi}.$$

Thus

$$I_2'(\theta) = -f(\theta) \frac{\cosh \pi}{2 \sinh \pi} - \int_{\theta}^{2\pi} f(\alpha) \frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha.$$

Adding $I_1'(\theta)$ and $I_2'(\theta)$, the boundary terms cancel and we obtain

$$\psi'(\theta) = - \int_0^{\theta} f(\alpha) \frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha - \int_{\theta}^{2\pi} f(\alpha) \frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha.$$

The integrands are continuous in (θ, α) , so the same argument as in part (a) shows that ψ' is continuous on $\mathbb{R}$. Hence $\psi \in C^1(\mathbb{R})$.

(c) Second derivative and the equation $\psi'' = -f + \psi$

Set

$$\widetilde{K}_1(\theta, \alpha) = -\frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi}, \quad \widetilde{K}_2(\theta, \alpha) = -\frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi},$$

so that the formula above can be written as

$$\psi'(\theta) = \int_0^{\theta} f(\alpha) \widetilde{K}_1(\theta, \alpha) d\alpha + \int_{\theta}^{2\pi} f(\alpha) \widetilde{K}_2(\theta, \alpha) d\alpha.$$

Differentiating again and using Leibniz' rule once more, we obtain

$$\psi''(\theta) = f(\theta) \widetilde{K}_1(\theta, \theta) - f(\theta) \widetilde{K}_2(\theta, \theta) + \int_0^{\theta} f(\alpha) \partial_{\theta} \widetilde{K}_1(\theta, \alpha) d\alpha + \int_{\theta}^{2\pi} f(\alpha) \partial_{\theta} \widetilde{K}_2(\theta, \alpha) d\alpha.$$

A direct computation shows that

$$\widetilde{K}_1(\theta, \theta) = -\frac{1}{2}, \quad \widetilde{K}_2(\theta, \theta) = +\frac{1}{2},$$

and

$$\partial_{\theta} \widetilde{K}_1(\theta, \alpha) = \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi}, \quad \partial_{\theta} \widetilde{K}_2(\theta, \alpha) = \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi}.$$

Hence

$$f(\theta) \widetilde{K}_1(\theta, \theta) - f(\theta) \widetilde{K}_2(\theta, \theta) = -\frac{1}{2}f(\theta) - \frac{1}{2}f(\theta) = -f(\theta),$$

while

$$\int_0^{\theta} f(\alpha) \partial_{\theta} \widetilde{K}_1(\theta, \alpha) d\alpha + \int_{\theta}^{2\pi} f(\alpha) \partial_{\theta} \widetilde{K}_2(\theta, \alpha) d\alpha = \psi(\theta).$$

Therefore

$$\psi''(\theta) = -f(\theta) + \psi(\theta) \quad (\theta \in (0, 2\pi)),$$

and by periodicity this holds for all $\theta \in \mathbb{R}$. The right-hand side is continuous, so $\psi'' \in C^0(\mathbb{R})$ and $\psi \in C^2(\mathbb{R})$.

Example: constant right-hand side

If $f(\theta) \equiv 1$, the equation becomes

$$\psi''(\theta) = -1 + \psi(\theta),$$

that is $(\psi'' - \psi)(\theta) = -1$ with ψ 2π -periodic. Solving the ordinary differential equation gives

$$\psi(\theta) = 1 + Ae^\theta + Be^{-\theta}.$$

The 2π -periodicity forces $A = B = 0$, so the unique periodic solution is $\psi(\theta) \equiv 1$, which agrees with the integral representation of ψ for the constant function $f \equiv 1$.

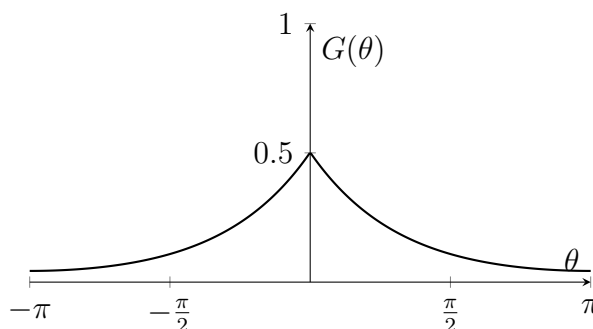


Figure 8: The periodic Green's kernel $G(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}$ on $[-\pi, \pi]$, extended periodically to $\mathbb{R}$. Convolution with G produces the unique 2π -periodic solution of $(-\psi'' + \psi) = f$.

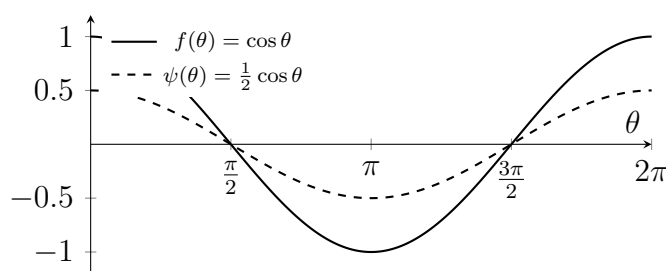


Figure 9: Example with $f(\theta) = \cos \theta$. The unique 2π -periodic solution of $\psi''(\theta) = -f(\theta) + \psi(\theta)$ is $\psi(\theta) = \frac{1}{2} \cos \theta$, shown as the dashed curve.

Periodic Green's kernel on the circle

Consider the differential operator

$$L := -\frac{d^2}{d\theta^2} + 1$$

acting on 2π -periodic functions. On the Fourier basis $e^{in\theta}$, $n \in \mathbb{Z}$, we have

$$L(e^{in\theta}) = (1 + n^2)e^{in\theta},$$

so L is invertible on the space of 2π -periodic functions, with bounded inverse

$$L^{-1}\left(\sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\theta}\right) = \sum_{n \in \mathbb{Z}} \frac{\hat{f}_n}{1 + n^2} e^{in\theta}.$$

The *periodic Green's function* for L is the 2π -periodic distribution G defined by

$$LG = \delta_0 \quad \text{on the circle,}$$

where δ_0 is the periodic Dirac mass at $\theta = 0$. Writing G as a Fourier series

$$G(\theta) = \sum_{n \in \mathbb{Z}} g_n e^{in\theta},$$

the equation $LG = \delta_0$ becomes

$$(1 + n^2)g_n = \frac{1}{2\pi}, \quad n \in \mathbb{Z},$$

since δ_0 has Fourier coefficients $1/(2\pi)$. Thus

$$g_n = \frac{1}{2\pi(1 + n^2)}, \quad G(\theta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1 + n^2}.$$

In Exercise 1.1 we showed that for $0 \leq \theta \leq 2\pi$,

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1 + n^2},$$

with uniform convergence in θ . Combining this identity with the 2π -periodicity of the Fourier series, and using the evenness of $\cosh$, one obtains the explicit formula

$$G(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

extended 2π -periodically to all $\theta \in \mathbb{R}$. The kernel G is even, $G(-\theta) = G(\theta)$, and C^∞ away from $\theta = 0$, with a logarithmic-type singularity encoded in the jump of G' at $\theta = 0$ corresponding to the Dirac mass.

Given a continuous 2π -periodic function f , the (formal) convolution

$$(G * f)(\theta) := \int_0^{2\pi} G(\theta - \alpha) f(\alpha) d\alpha$$

defines a 2π -periodic function. Since $G \in L^1([0, 2\pi])$ and $f \in C^0(\mathbb{R})$, this integral is well-defined for all θ , and the mapping $\theta \mapsto (G * f)(\theta)$ is continuous. Differentiating under the integral sign (justified by the smoothness of G away from $\alpha = \theta$ and the Fourier-series representation), one finds

$$\frac{d^2}{d\theta^2}(G * f)(\theta) = \int_0^{2\pi} G''(\theta - \alpha)f(\alpha) d\alpha,$$

so that

$$-(G * f)''(\theta) + (G * f)(\theta) = \int_0^{2\pi} (-G''(\theta - \alpha) + G(\theta - \alpha))f(\alpha) d\alpha.$$

Because G satisfies $(-G'' + G) = \delta_0$ on the circle, the integrand reduces to $\delta(\theta - \alpha)$ in the periodic sense, and hence

$$-(G * f)''(\theta) + (G * f)(\theta) = f(\theta), \quad \theta \in \mathbb{R}.$$

Thus the function $\psi := G * f$ is the unique 2π -periodic solution of the inhomogeneous ODE

$$-\psi''(\theta) + \psi(\theta) = f(\theta),$$

and the explicit formula used in Exercise 1.2 is precisely the representation of this convolution when one chooses the branch $\theta \in [0, 2\pi)$ and splits the integral at $\alpha = \theta$.

The Green's function G is defined by

$$LG = \delta_0,$$

i.e. G is the response of the system L to a unit point source at $\theta = 0$. The operator

$$L = -\frac{d^2}{d\theta^2} + 1$$

is translation invariant on the circle: for every $\alpha \in \mathbb{R}$ and every sufficiently smooth u we have

$$L(u(\theta - \alpha)) = (Lu)(\theta - \alpha).$$

Moreover, the Dirac mass at α is simply the translate of the Dirac mass at 0:

$$\delta_\alpha(\theta) = \delta_0(\theta - \alpha).$$

Starting from $(LG)(\theta) = \delta_0(\theta)$ and replacing θ by $\theta - \alpha$, we obtain

$$(LG)(\theta - \alpha) = \delta_0(\theta - \alpha) = \delta_\alpha(\theta).$$

By translation invariance,

$$(LG)(\theta - \alpha) = L(G(\theta - \alpha)),$$

and hence

$$L(G(\cdot - \alpha)) = \delta_\alpha.$$

Thus the response to a point source located at α is exactly the translate $G(\cdot - \alpha)$ of the Green's function.

Why we introduce $L := -\frac{d^2}{d\theta^2} + 1$ and what the Green's function means

The differential equation from Exercise 1.2 can be rewritten as

$$-\psi''(\theta) + \psi(\theta) = f(\theta),$$

so it is natural to introduce the linear operator

$$L := -\frac{d^2}{d\theta^2} + 1.$$

With this notation the problem takes the form

$$L\psi = f \quad \text{on the circle.}$$

The addition of the “+1” term is important: the operator $-\frac{d^2}{d\theta^2}$ is not invertible on 2π -periodic functions, since its kernel contains constants and trigonometric functions. However,

$$L(e^{in\theta}) = (1 + n^2)e^{in\theta},$$

and $1 + n^2 > 0$ for every $n \in \mathbb{Z}$, so L is invertible on the space of 2π -periodic functions.

The *periodic Green's function* for L is the unique 2π -periodic distribution G satisfying

$$LG = \delta_0,$$

where δ_0 is the periodic Dirac mass at $\theta = 0$. Thus G is the response of the system L to a unit point source at $\theta = 0$. A key property is that L is translation invariant on the circle:

$$L(u(\theta - \alpha)) = (Lu)(\theta - \alpha), \quad \alpha \in \mathbb{R},$$

for all sufficiently smooth u . Since $\delta_\alpha(\theta) = \delta_0(\theta - \alpha)$, it follows that

$$L(G(\theta - \alpha)) = \delta_\alpha(\theta).$$

Thus the response to a point source located at α is simply the translate $G(\cdot - \alpha)$.

A general 2π -periodic function f may be viewed formally as a superposition of point sources:

$$f(\theta) = \int_0^{2\pi} f(\alpha) \delta_\alpha(\theta) d\alpha.$$

By linearity of L , the corresponding solution must be the superposition of the individual responses:

$$\psi(\theta) = \int_0^{2\pi} f(\alpha) G(\theta - \alpha) d\alpha,$$

which is the periodic convolution $G * f$. A direct computation using translation invariance gives

$$L\psi(\theta) = \int_0^{2\pi} f(\alpha) L(G(\theta - \alpha)) d\alpha = \int_0^{2\pi} f(\alpha) \delta_\alpha(\theta) d\alpha = f(\theta),$$

so $\psi = G * f$ is indeed the unique 2π -periodic solution of $L\psi = f$.

In Exercise 1.1 it was shown that

$$\sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1 + n^2} = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

with uniform convergence in θ . Since the Fourier coefficients of δ_0 are $1/(2\pi)$, this identifies the periodic Green's function explicitly as

$$G(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

extended 2π -periodically. This is exactly the kernel that appears in the integral defining $\psi(\theta)$ in Exercise 1.2.

Fourier-series viewpoint on L^{-1}

Let $L = -\frac{d^2}{d\theta^2} + 1$ act on 2π -periodic functions. The complex exponentials

$$e_n(\theta) := e^{in\theta}, \quad n \in \mathbb{Z},$$

form an orthonormal basis of $L^2_{\text{per}}([0, 2\pi])$, and they are eigenfunctions of L :

$$Le_n = -\frac{d^2}{d\theta^2} e^{in\theta} + e^{in\theta} = (n^2 + 1)e^{in\theta} = (1 + n^2)e_n.$$

Thus L is *diagonal* in the Fourier basis, with eigenvalues $\lambda_n = 1 + n^2 > 0$.

Let f be a 2π -periodic function with Fourier series

$$f(\theta) = \sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\theta}, \quad \hat{f}_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta.$$

We look for a solution ψ of $L\psi = f$ in the same form,

$$\psi(\theta) = \sum_{n \in \mathbb{Z}} \hat{\psi}_n e^{in\theta}.$$

Applying L termwise gives

$$L\psi = \sum_{n \in \mathbb{Z}} (1 + n^2) \hat{\psi}_n e^{in\theta}.$$

Hence the equation $L\psi = f$ is equivalent to

$$(1 + n^2) \hat{\psi}_n = \hat{f}_n, \quad n \in \mathbb{Z},$$

so

$$\hat{\psi}_n = \frac{\hat{f}_n}{1+n^2}.$$

Therefore

$$\psi(\theta) = L^{-1}f(\theta) = \sum_{n \in \mathbb{Z}} \frac{\hat{f}_n}{1+n^2} e^{in\theta}.$$

This shows that the inverse operator L^{-1} multiplies each Fourier mode $e^{in\theta}$ by the factor $\frac{1}{1+n^2}$. In particular, the kernel G of L^{-1} (the periodic Green's function) has Fourier coefficients

$$\hat{G}_n = \frac{1}{2\pi(1+n^2)}, \quad n \in \mathbb{Z},$$

so

$$G(\theta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1+n^2}.$$

Using the identity from Exercise 1.1,

$$\sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1+n^2} = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

we recover the explicit formula

$$G(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

extended 2π -periodically. Thus the Green's function is nothing more than the integral kernel of L^{-1} when viewed in the Fourier basis.

Eigenvalues of L and diagonalisation in the Fourier basis

Let

$$L := -\frac{d^2}{d\theta^2} + 1$$

act on 2π -periodic functions. Consider the Fourier basis

$$e_n(\theta) := e^{in\theta}, \quad n \in \mathbb{Z}.$$

We first compute the action of L on e_n . Differentiating,

$$\frac{d}{d\theta} e^{in\theta} = in e^{in\theta}, \quad \frac{d^2}{d\theta^2} e^{in\theta} = (in)^2 e^{in\theta} = -n^2 e^{in\theta}.$$

Hence

$$Le_n = -\frac{d^2}{d\theta^2} e^{in\theta} + e^{in\theta} = -(-n^2 e^{in\theta}) + e^{in\theta} = (1+n^2) e^{in\theta}.$$

Thus e_n is an eigenfunction of L with eigenvalue

$$\lambda_n = 1 + n^2.$$

Because each basis vector e_n is an eigenfunction of L , the operator acts diagonally in this basis. That is, if we represent L as an infinite matrix with respect to the basis $\{e_n\}_{n \in \mathbb{Z}}$, then

$$Le_n = \lambda_n e_n \quad \Longleftrightarrow \quad (L)_{mn} = \lambda_n \delta_{mn},$$

so only the diagonal entries are nonzero. The Fourier basis therefore diagonalises L , with diagonal entries $\{1 + n^2 : n \in \mathbb{Z}\}$.

Why L is diagonal in the Fourier basis

Think of writing a linear operator as an infinite matrix with respect to a basis. If $\{v_k\}$ is a basis and

$$Tv_k = \sum_j a_{jk} v_j,$$

then the matrix of T is the array (a_{jk}) , where *column* k records how T acts on the k -th basis vector.

If each basis vector v_k is an eigenvector,

$$Tv_k = \lambda_k v_k,$$

then

$$Tv_k = \lambda_k v_k = \lambda_k v_k + 0 \cdot v_j \quad (j \neq k).$$

Therefore the matrix entries satisfy

$$a_{kk} = \lambda_k, \quad a_{jk} = 0 \quad (j \neq k).$$

Such a matrix has nonzero entries only on the diagonal; it is a *diagonal matrix*. Thus an operator is diagonal in a given basis precisely when every basis vector is an eigenvector.

For our operator

$$L := -\frac{d^2}{d\theta^2} + 1,$$

the relevant basis is the Fourier basis $\{e_n\}_{n \in \mathbb{Z}}$ with $e_n(\theta) = e^{in\theta}$. We showed earlier that

$$Le_n = (1 + n^2) e_n,$$

so e_n is an eigenfunction of L with eigenvalue $1 + n^2$.

Hence the “matrix” of L in the Fourier basis is

$$\text{diag}(\dots, 1 + (-2)^2, 1 + (-1)^2, 1 + 0^2, 1 + 1^2, 1 + 2^2, \dots),$$

i.e. diagonal with entries $1 + n^2$.

This is the precise sense in which we say:

L is diagonal in the Fourier basis, with eigenvalues $\lambda_n = 1 + n^2$.

Given a continuous 2π -periodic function f , the (formal) convolution

$$(G * f)(\theta) := \int_0^{2\pi} G(\theta - \alpha) f(\alpha) d\alpha$$

appears naturally from linearity and translation invariance of L .

Indeed, G is defined by $LG = \delta_0$, so G is the response of the system L to a unit point source at 0. Since L is translation invariant on the circle, the response to a point source at α is the translate $G(\theta - \alpha)$, satisfying

$$L(G(\theta - \alpha)) = \delta_\alpha(\theta).$$

A general right-hand side f may be viewed as a superposition of point sources,

$$f(\theta) = \int_0^{2\pi} f(\alpha) \delta_\alpha(\theta) d\alpha.$$

By linearity of L , the corresponding solution must be the superposition of the individual responses:

$$\psi(\theta) = \int_0^{2\pi} f(\alpha) G(\theta - \alpha) d\alpha = (G * f)(\theta).$$

Thus convolution with the Green's function G is exactly the operation that sends f to the unique 2π -periodic solution ψ of $L\psi = f$.

Green functions for simple one-dimensional operators

In this section we collect a few basic examples of Green functions for second-order operators in one dimension. Throughout, a Green function G for an operator L is a distributional solution of

$$LG = \delta_0,$$

with boundary conditions specified by the domain of L (whole line, finite interval, or circle). For a given right-hand side f , the solution of $Lu = f$ is then given by the convolution $u = G * f$ (or its appropriate periodic / bounded-interval analogue).

1. The operator $L_0 u = -u''$ on $\mathbb{R}$

We look for G_0 such that

$$-G_0''(x) = \delta_0(x) \quad \text{on } \mathbb{R}.$$

Away from $x = 0$ we have $-G_0'' = 0$, hence G_0 is affine on $(-\infty, 0)$ and on $(0, \infty)$:

$$G_0(x) = \begin{cases} a_+x + b_+, & x > 0, \\ a_-x + b_-, & x < 0. \end{cases}$$

We impose

- symmetry $G_0(-x) = G_0(x)$ (by uniqueness and symmetry of the operator), which gives $a_+ = -a_-$ and $b_+ = b_- =: b$;
- continuity at 0: $G_0(0^+) = G_0(0^-)$, already encoded above;
- the distributional equation $-G_0'' = \delta_0$, which is equivalent to the jump condition $G_0'(0^+) - G_0'(0^-) = -1$.

Here $G_0'(0^+) = a_+$ and $G_0'(0^-) = a_-$, so

$$a_+ - a_- = -1.$$

Together with $a_+ = -a_-$ this yields $2a_+ = -1$, hence $a_+ = -\frac{1}{2}$ and $a_- = \frac{1}{2}$. Thus

$$G_0(x) = -\frac{1}{2}|x|.$$

Figure. The graph of G_0 is a “V” shaped function.

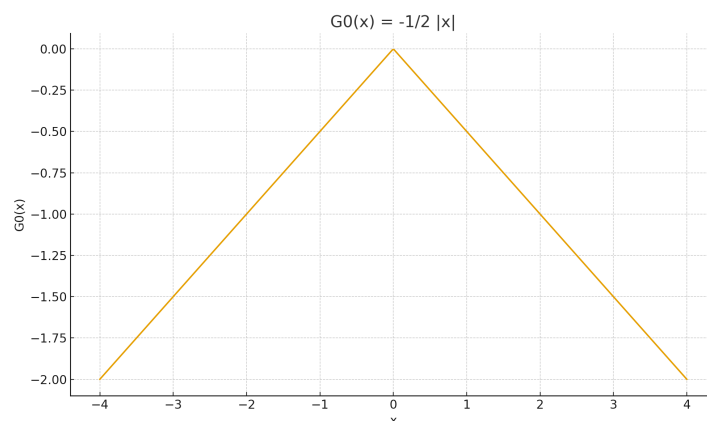


Figure 10: Green's function $G_0(x) = -\frac{1}{2}|x|$ for the operator $L_0 u = -u''$ on $\mathbb{R}$. The function is continuous and piecewise linear, and its second derivative is a Dirac mass at $x = 0$ in the distributional sense.

2. The operator $L_1 u = -u'' + u$ on $\mathbb{R}$

We now solve

$$-G_1''(x) + G_1(x) = \delta_0(x) \quad \text{on } \mathbb{R}.$$

For $x \neq 0$ this reduces to the homogeneous ODE

$$-G_1''(x) + G_1(x) = 0,$$

whose general solution is $G_1(x) = Ae^x + Be^{-x}$. To obtain a Green function that is bounded at $\pm\infty$, we choose

$$G_1(x) = \begin{cases} Ce^{-x}, & x > 0, \\ Ce^x, & x < 0, \end{cases}$$

so that G_1 decays at $\pm\infty$ and is even.

Continuity at 0 is automatic:

$$G_1(0^+) = C = G_1(0^-).$$

As before, the equation $-G_1'' + G_1 = \delta_0$ is equivalent to the jump condition

$$G_1'(0^+) - G_1'(0^-) = -1.$$

Differentiating,

$$G_1'(x) = \begin{cases} -Ce^{-x}, & x > 0, \\ Ce^x, & x < 0, \end{cases} \Rightarrow G_1'(0^+) = -C, \quad G_1'(0^-) = C.$$

Hence

$$G_1'(0^+) - G_1'(0^-) = -C - C = -2C = -1,$$

so $C = \frac{1}{2}$. Therefore

$$G_1(x) = \frac{1}{2}e^{-|x|}.$$

Figure. The Green function is a symmetric, exponentially decaying bump.

3. The operator $L_\lambda u = -u'' + \lambda u$ on $\mathbb{R}$

More generally, for $\lambda > 0$ we consider

$$-G_\lambda''(x) + \lambda G_\lambda(x) = \delta_0(x).$$

For $x \neq 0$ the homogeneous equation is

$$-G_\lambda''(x) + \lambda G_\lambda(x) = 0 \iff G_\lambda''(x) = \lambda G_\lambda(x),$$

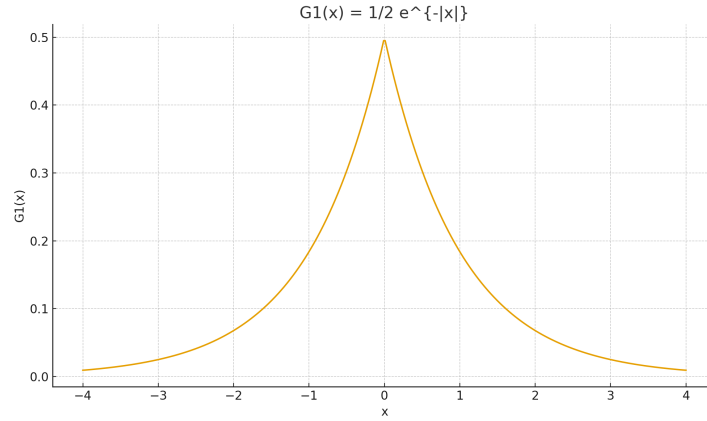


Figure 11: Green's function $G_1(x) = \frac{1}{2}e^{-|x|}$ for $L_1u = -u'' + u$ on $\mathbb{R}$. The exponential decay shows that high frequencies are strongly damped by L_1^{-1} .

with solutions $G_\lambda(x) = Ae^{\sqrt{\lambda}x} + Be^{-\sqrt{\lambda}x}$. As in the previous case, boundedness and evenness lead to

$$G_\lambda(x) = \begin{cases} Ce^{-\sqrt{\lambda}x}, & x > 0, \\ Ce^{\sqrt{\lambda}x}, & x < 0. \end{cases}$$

The jump condition for $-G_\lambda'' + \lambda G_\lambda = \delta_0$ again gives

$$G_\lambda'(0^+) - G_\lambda'(0^-) = -1.$$

Differentiating,

$$G_\lambda'(x) = \begin{cases} -C\sqrt{\lambda}e^{-\sqrt{\lambda}x}, & x > 0, \\ C\sqrt{\lambda}e^{\sqrt{\lambda}x}, & x < 0, \end{cases}$$

so $G_\lambda'(0^+) = -C\sqrt{\lambda}$ and $G_\lambda'(0^-) = C\sqrt{\lambda}$. Thus

$$G_\lambda'(0^+) - G_\lambda'(0^-) = -C\sqrt{\lambda} - C\sqrt{\lambda} = -2C\sqrt{\lambda} = -1 \implies C = \frac{1}{2\sqrt{\lambda}}.$$

Therefore

$$G_\lambda(x) = \frac{1}{2\sqrt{\lambda}} e^{-\sqrt{\lambda}|x|}.$$

4. Periodic operator $L_{\text{per}}u = -u'' + u$ on the circle

We finally consider the operator

$$L_{\text{per}}u := -\frac{d^2u}{d\theta^2} + u$$

on 2π -periodic functions. We seek a 2π -periodic Green function G_{per} such that

$$-G_{\text{per}}''(\theta) + G_{\text{per}}(\theta) = \delta_0(\theta)$$

on the circle. One way to construct G_{per} is to *periodise* the line kernel G_1 :

$$G_{\text{per}}(\theta) = \sum_{k \in \mathbb{Z}} G_1(\theta + 2\pi k) = \frac{1}{2} \sum_{k \in \mathbb{Z}} e^{-|\theta + 2\pi k|}.$$

This series converges absolutely and defines a 2π -periodic function which satisfies $(-G_{\text{per}}'' + G_{\text{per}}) = \delta_0$ in the periodic sense.

Using the Fourier-series computation of Exercise 1.1, one obtains the closed form

$$G_{\text{per}}(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

extended 2π -periodically.

Figure. The periodic Green function is a smooth bump on the circle, symmetric around 0.

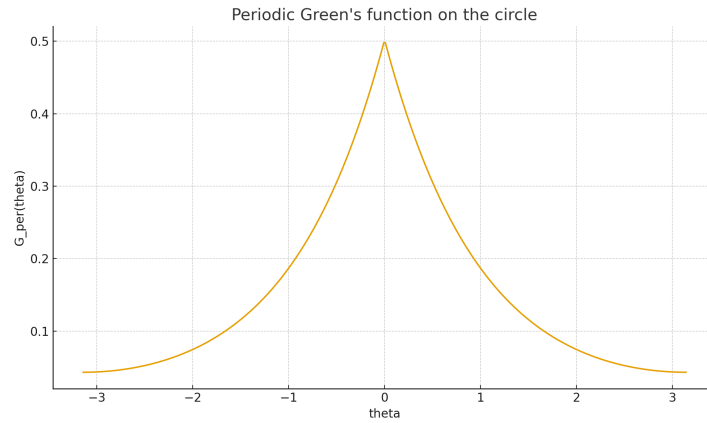


Figure 12: Periodic Green's function $G_{\text{per}}(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}$ for $L_{\text{per}}u = -u'' + u$ on the circle. It can be obtained either by periodising the line kernel $G_1(x) = \frac{1}{2}e^{-|x|}$ or by inverting L_{per} in the Fourier basis.

5. Summary table

Operator	Domain / BC	Green's function G with $LG = \delta_0$
$L_0 = -\frac{d^2}{dx^2}$	$\mathbb{R}$	$G_0(x) = -\frac{1}{2} x $
$L_1 = -\frac{d^2}{dx^2} + 1$	$\mathbb{R}$	$G_1(x) = \frac{1}{2}e^{- x }$
$L_\lambda = -\frac{d^2}{dx^2} + \lambda$	$\mathbb{R}, \lambda > 0$	$G_\lambda(x) = \frac{1}{2\sqrt{\lambda}}e^{-\sqrt{\lambda} x }$
$L_{\text{per}} = -\frac{d^2}{d\theta^2} + 1$	Circle, 2π -periodic	$G_{\text{per}}(\theta) = \frac{\cosh(\pi - \theta)}{2 \sinh \pi}$ on $[-\pi, \pi]$, periodic

6. Convolution representation

Given a continuous function f on the domain of L , the solution of

$$Lu = f$$

with the appropriate boundary conditions is obtained by convolving f with the corresponding Green function. For example, on $\mathbb{R}$,

$$u(x) = (G_\lambda * f)(x) := \int_{\mathbb{R}} G_\lambda(x - y)f(y) dy,$$

and on the circle

$$u(\theta) = \int_0^{2\pi} G_{\text{per}}(\theta - \alpha)f(\alpha) d\alpha.$$

In this way the Green function is the integral kernel of the inverse operator L^{-1} .

Dirichlet and Neumann Green functions on $(0, 1)$

We consider the second-order operator $L_0 u = -u''$ on the interval $(0, 1)$ with different boundary conditions.

1. Dirichlet Green function

We first impose Dirichlet boundary conditions

$$u(0) = u(1) = 0.$$

A Dirichlet Green function $G_D(x, \xi)$ is a function such that

$$-\partial_x^2 G_D(x, \xi) = \delta(x - \xi) \quad \text{for } 0 < x < 1,$$

with

$$G_D(0, \xi) = G_D(1, \xi) = 0.$$

For $x \neq \xi$ we have $-G_D'' = 0$, hence G_D is affine on each side:

$$G_D(x, \xi) = \begin{cases} a_-x + b_-, & 0 \leq x < \xi, \\ a_+x + b_+, & \xi < x \leq 1. \end{cases}$$

From $G_D(0, \xi) = 0$ we get $b_- = 0$, and from $G_D(1, \xi) = 0$ we obtain $a_+ + b_+ = 0$, i.e. $b_+ = -a_+$. Continuity at $x = \xi$ gives

$$a_- \xi = a_+ \xi + b_+ = a_+ (\xi - 1),$$

and the jump condition

$$G_{D,x}(\xi^+, \xi) - G_{D,x}(\xi^-, \xi) = -1$$

(from $-G_D'' = \delta$) reads $a_+ - a_- = -1$. Solving these equations yields

$$a_- = 1 - \xi, \quad a_+ = -\xi,$$

so that

$$G_D(x, \xi) = \begin{cases} x(1 - \xi), & 0 \leq x \leq \xi, \\ \xi(1 - x), & \xi \leq x \leq 1. \end{cases}$$

This kernel is a “tent” function with peak at $x = \xi$ and zeros at both endpoints.

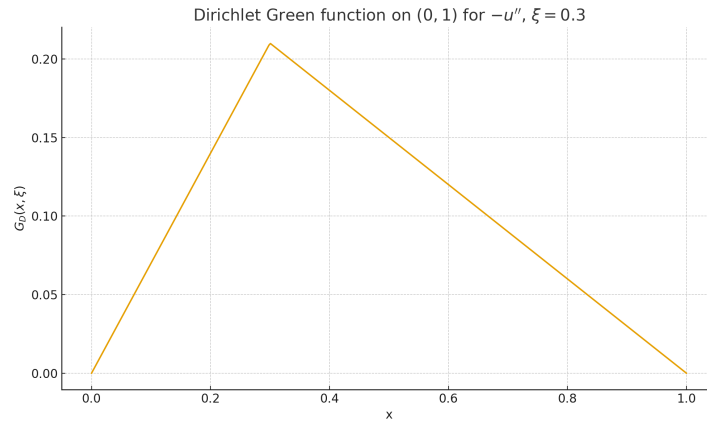


Figure 13: Dirichlet Green function $G_D(x, \xi)$ on $(0, 1)$ for $L_0 u = -u''$, plotted for $\xi = 0.3$. The function vanishes at $x = 0$ and $x = 1$ and is piecewise linear with a kink at $x = \xi$.

Given $f \in C([0, 1])$, the unique solution of $-u'' = f$ with $u(0) = u(1) = 0$ is

$$u(x) = \int_0^1 G_D(x, \xi) f(\xi) d\xi.$$

2. Neumann Green function

We now impose Neumann boundary conditions

$$u'(0) = u'(1) = 0.$$

In this case $L_0 u = -u''$ has constants in its kernel, so the equation $-u'' = f$ is solvable only if

$$\int_0^1 f(x) dx = 0,$$

and the solution is unique only up to an additive constant. To define a Green function we seek $G_N(x, \xi)$ satisfying

$$-\partial_x^2 G_N(x, \xi) = \delta(x - \xi) - 1$$

on $(0, 1)$, with Neumann boundary conditions

$$G_{N,x}(0, \xi) = G_{N,x}(1, \xi) = 0,$$

and the normalisation

$$\int_0^1 G_N(x, \xi) dx = 0 \quad \text{for each fixed } \xi.$$

For $x \neq \xi$ the equation reduces to $G_N''(x, \xi) = 1$, so G_N is quadratic on each side:

$$G_N(x, \xi) = \begin{cases} \frac{1}{2}x^2 + a_-x + b_-, & 0 \leq x < \xi, \\ \frac{1}{2}x^2 + a_+x + b_+, & \xi < x \leq 1. \end{cases}$$

The Neumann conditions give $G_{N,x}(0, \xi) = 0$ and $G_{N,x}(1, \xi) = 0$, that is

$$G_{N,x}(x, \xi) = \begin{cases} x + a_-, & x < \xi, \\ x + a_+, & x > \xi, \end{cases} \Rightarrow a_- = 0, \quad a_+ = -1.$$

Continuity at $x = \xi$ implies $b_- = b_+ - \xi$, and the jump condition coming from $-G_N'' = \delta - 1$,

$$G_{N,x}(\xi^+, \xi) - G_{N,x}(\xi^-, \xi) = -1,$$

is automatically satisfied with $a_- = 0$, $a_+ = -1$. The normalisation $\int_0^1 G_N(x, \xi) dx = 0$ then determines b_+ and b_- ; a short computation yields

$$b_+ = \frac{\xi^2}{2} + \frac{1}{3}, \quad b_- = \frac{\xi^2}{2} - \xi + \frac{1}{3}.$$

Thus

$$G_N(x, \xi) = \begin{cases} \frac{1}{2}x^2 + \frac{1}{2}\xi^2 - \xi + \frac{1}{3}, & 0 \leq x \leq \xi, \\ \frac{1}{2}x^2 - x + \frac{1}{2}\xi^2 + \frac{1}{3}, & \xi \leq x \leq 1. \end{cases}$$

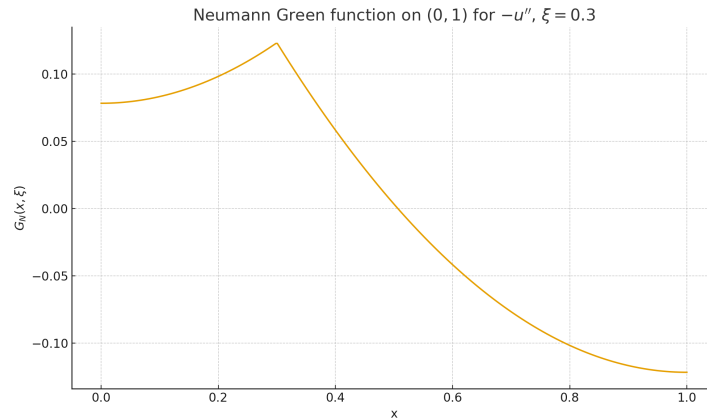


Figure 14: Neumann Green function $G_N(x, \xi)$ on $(0, 1)$ for $L_0 u = -u''$, plotted for $\xi = 0.3$. The derivative vanishes at $x = 0$ and $x = 1$, and G_N is piecewise quadratic with a slope jump of -1 at $x = \xi$. The normalisation $\int_0^1 G_N(x, \xi) dx = 0$ fixes the additive constant.

If f satisfies $\int_0^1 f(x) dx = 0$, then

$$u(x) = \int_0^1 G_N(x, \xi) f(\xi) d\xi$$

solves $-u'' = f$ with Neumann boundary conditions $u'(0) = u'(1) = 0$, and is determined up to an additive constant.

Helmholtz and damped one-dimensional operators

1. Helmholtz operator on $\mathbb{R}$

Consider the Helmholtz operator

$$L_H u := -u'' - k^2 u, \quad k > 0,$$

on the real line. A Green function G_H satisfies

$$-G_H''(x) - k^2 G_H(x) = \delta_0(x) \quad \text{on } \mathbb{R}.$$

For $x \neq 0$ the homogeneous equation $G_H'' + k^2 G_H = 0$ has solutions $A \cos(kx) + B \sin(kx)$. We look for an even Green function and try

$$G_H(x) = -\frac{1}{2k} \sin(k|x|).$$

For $x > 0$ we have $G_H(x) = -\frac{1}{2k} \sin(kx)$ and for $x < 0$, $G_H(x) = \frac{1}{2k} \sin(kx)$. In both cases $G_H'' + k^2 G_H = 0$, so the equation holds away from $x = 0$.

The derivative is

$$G_H'(x) = \begin{cases} -\frac{1}{2} \cos(kx), & x > 0, \\ \frac{1}{2} \cos(kx), & x < 0, \end{cases}$$

hence

$$G_H'(0^+) = -\frac{1}{2}, \quad G_H'(0^-) = \frac{1}{2},$$

and

$$G_H'(0^+) - G_H'(0^-) = -1.$$

This jump condition is equivalent to $-G_H'' = \delta_0$ in the distribution sense, while the $-k^2 G_H$ term integrates to zero across $x = 0$. Thus

$$G_H(x) = -\frac{1}{2k} \sin(k|x|)$$

is a Green function for L_H on $\mathbb{R}$. It is oscillatory and does not decay at infinity.

2. Damped operator on $\mathbb{R}$

Now consider the non-self-adjoint operator

$$L_D u := -u'' + cu', \quad c > 0.$$

We look for a one-sided (retarded) Green function G_D solving

$$-G_D''(x) + cG_D'(x) = \delta_0(x)$$

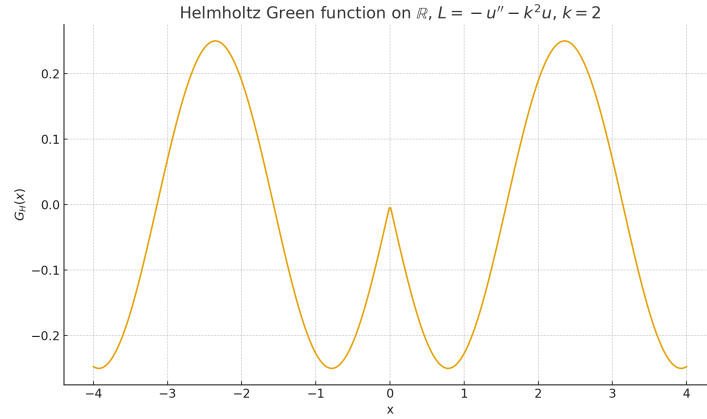


Figure 15: Helmholtz Green function $G_H(x) = -\frac{1}{2k} \sin(k|x|)$ for $L_H u = -u'' - k^2 u$, plotted for $k = 2$. The kernel is oscillatory and bounded.

and vanishing for $x > 0$.

For $x \neq 0$ we solve $-G_D'' + cG_D' = 0$, i.e. $G_D'' - cG_D' = 0$. Writing $v = G_D'$ gives $v' - cv = 0$ so $v(x) = Ae^{cx}$ and

$$G_D(x) = \frac{A}{c} e^{cx} + B.$$

We prescribe

$$G_D(x) = \begin{cases} \alpha + \beta e^{cx}, & x < 0, \\ 0, & x > 0, \end{cases}$$

and impose continuity at 0:

$$\alpha + \beta = 0 \quad \Rightarrow \quad \alpha = -\beta.$$

The jump condition from $-G_D'' + cG_D' = \delta_0$ reads

$$G_D'(0^+) - G_D'(0^-) = -1.$$

Here $G_D'(x) = 0$ for $x > 0$ and $G_D'(x) = c\beta e^{cx}$ for $x < 0$, so $G_D'(0^-) = c\beta$ and hence $-c\beta = -1$, i.e. $\beta = 1/c$ and $\alpha = -1/c$. Thus

$$G_D(x) = \begin{cases} -\frac{1}{c} + \frac{1}{c} e^{cx}, & x < 0, \\ 0, & x > 0. \end{cases}$$

This kernel is continuous at 0, has a slope jump of -1 , and vanishes for $x > 0$, so it is a retarded Green function for L_D .

Green function for $-\Delta$ in $\mathbb{R}^2$

In two dimensions the Laplacian is

$$\Delta u = u_{xx} + u_{yy}.$$

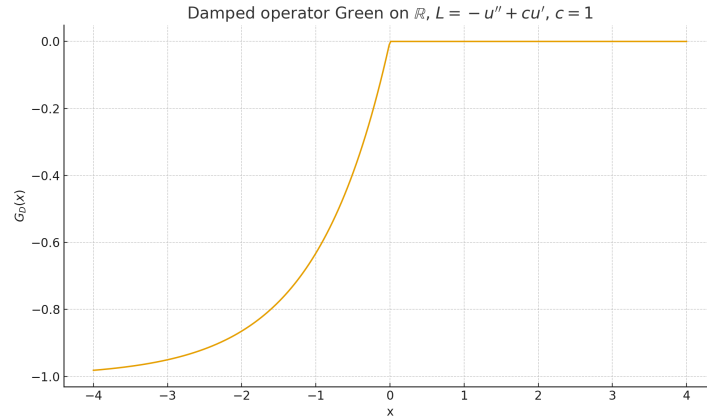


Figure 16: One-sided Green function G_D for the damped operator $L_D u = -u'' + cu'$ on $\mathbb{R}$, plotted for $c = 1$. The kernel vanishes for $x > 0$ and decays exponentially as $x \rightarrow -\infty$.

We seek a Green function $G : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$ such that

$$-\Delta G(x) = \delta_0(x)$$

in the sense of distributions. By rotational symmetry we look for a radial function $G(x) = g(r)$ with $r = |x|$. For a radial function $u(r)$ in $\mathbb{R}^2$ one has

$$\Delta u = u''(r) + \frac{1}{r}u'(r).$$

Thus away from the origin $-\Delta G = 0$ becomes

$$g''(r) + \frac{1}{r}g'(r) = 0.$$

Writing $v(r) = g'(r)$, this is $v'(r) + \frac{1}{r}v(r) = 0$, or equivalently $(rv(r))' = 0$, so $v(r) = C/r$ and hence

$$g(r) = C \log r + D.$$

To fix the constant C we integrate $-\Delta G$ over a disc B_R and apply the divergence theorem:

$$\int_{B_R} -\Delta G \, dx = \int_{\partial B_R} -\frac{\partial G}{\partial n} \, dS.$$

If $G(x) = A \log |x|$, then $\partial G / \partial r = A/r$ and on ∂B_R we obtain

$$-\frac{\partial G}{\partial n} = -\frac{A}{R}.$$

Since $dS = R \, d\theta$ on the circle of radius R ,

$$\int_{\partial B_R} -\frac{\partial G}{\partial n} \, dS = \int_0^{2\pi} \left(-\frac{A}{R}\right) R \, d\theta = -2\pi A.$$

We require this flux to equal 1 (because $-\Delta G = \delta_0$), hence $-2\pi A = 1$ and $A = -1/(2\pi)$. Therefore

$$G(x) = -\frac{1}{2\pi} \log |x|$$

is a Green function for the Laplacian in $\mathbb{R}^2$ (unique up to an additive constant).

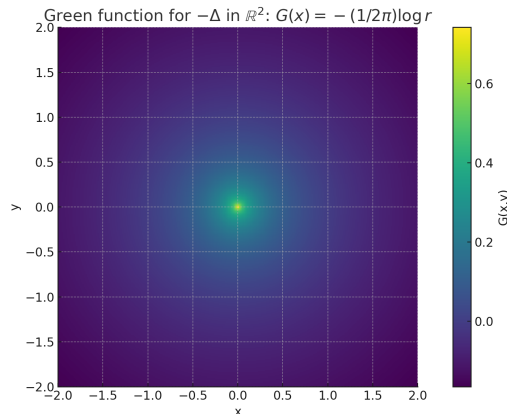


Figure 17: Green function $G(x) = -\frac{1}{2\pi} \log r$ for $-\Delta$ in $\mathbb{R}^2$ (plotted on $[-2, 2]^2$ with a small regularisation near $r = 0$). The level sets are concentric circles and G has a logarithmic singularity at the origin.

Green functions on the torus and sphere

1. Flat torus $\mathbb{T}^2$

Identify the flat torus $\mathbb{T}^2$ with $\mathbb{R}^2/(2\pi\mathbb{Z})^2$. The eigenfunctions of the Laplacian are

$$e_{m,n}(x, y) = e^{i(mx+ny)}, \quad (m, n) \in \mathbb{Z}^2,$$

with eigenvalues

$$-\Delta e_{m,n} = (m^2 + n^2)e_{m,n}.$$

The constant mode $(m, n) = (0, 0)$ has eigenvalue 0, so $-\Delta$ is not invertible on all of $L^2(\mathbb{T}^2)$. To define a Green function $G_{\mathbb{T}}$ one imposes

$$-\Delta G_{\mathbb{T}}(x, y) = \delta_0(x, y) - \frac{1}{(2\pi)^2},$$

together with the normalisation $\int_{\mathbb{T}^2} G_{\mathbb{T}} = 0$. Expanding in the Fourier basis and inverting mode by mode yields

$$G_{\mathbb{T}}(x, y) = \frac{1}{(2\pi)^2} \sum_{(m,n) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{e^{i(mx+ny)}}{m^2 + n^2}.$$

This series represents the periodisation of the planar Green function $-(1/2\pi) \log |x|$ and has a logarithmic singularity at the origin.

2. Round sphere $\mathbb{S}^2$

On the unit sphere $\mathbb{S}^2 \subset \mathbb{R}^3$ the eigenfunctions of the Laplace–Beltrami operator are the spherical harmonics Y_ℓ^m , $\ell = 0, 1, 2, \dots$ and $|m| \leq \ell$, with eigenvalues

$$-\Delta_{\mathbb{S}^2} Y_\ell^m = \ell(\ell + 1) Y_\ell^m.$$

Again the constant mode ($\ell = 0$) lies in the kernel. A Green function $G_{\mathbb{S}^2}(p, q)$ is defined by

$$-\Delta_{\mathbb{S}^2} G_{\mathbb{S}^2}(p, q) = \delta_q(p) - \frac{1}{4\pi},$$

together with $\int_{\mathbb{S}^2} G_{\mathbb{S}^2}(p, q) dp = 0$ for each fixed q . Using the completeness of the spherical harmonics one obtains the eigenfunction expansion

$$G_{\mathbb{S}^2}(p, q) = \sum_{\ell=1}^{\infty} \sum_{m=-\ell}^{\ell} \frac{1}{\ell(\ell+1)} Y_{\ell}^m(p) \overline{Y_{\ell}^m(q)}.$$

By rotational invariance, $G_{\mathbb{S}^2}(p, q)$ depends only on the geodesic distance γ between p and q (equivalently on $\cos \gamma = p \cdot q$), and one can write it as an explicit function of γ involving a logarithmic singularity as $\gamma \rightarrow 0$. Locally, the behaviour of $G_{\mathbb{S}^2}$ near the diagonal $p = q$ is therefore similar to the planar Green function $-(1/2\pi) \log |x|$.

Exercise 1.3 — Maximum principle for $-\psi'' + \psi$ on the circle

Let $\phi \in C^2(\mathbb{R})$ be 2π -periodic, i.e. $\phi(\theta + 2\pi) = \phi(\theta)$ for all $\theta \in \mathbb{R}$, and suppose that

$$-\phi''(\theta) + \phi(\theta) = 0, \quad \theta \in \mathbb{R}.$$

- (a) Show that if ϕ attains a maximum at some $\theta_0 \in \mathbb{R}$, then $\phi(\theta_0) \leq 0$.
- (b) Show that if ϕ attains a minimum at some $\theta_0 \in \mathbb{R}$, then $\phi(\theta_0) \geq 0$.
- (c) Deduce that $\phi \equiv 0$.
- (d) Let $f \in C^0(\mathbb{R})$ be a given 2π -periodic function. Conclude that there is at most one $\psi \in C^2(\mathbb{R})$ such that $\psi(\theta + 2\pi) = \psi(\theta)$ for all θ and

$$-\psi''(\theta) + \psi(\theta) = f(\theta), \quad \theta \in \mathbb{R}.$$

Let $\phi \in C^2(\mathbb{R})$ be 2π -periodic, i.e. $\phi(\theta + 2\pi) = \phi(\theta)$ for all $\theta \in \mathbb{R}$, and suppose that

$$-\phi''(\theta) + \phi(\theta) = 0, \quad \theta \in \mathbb{R}.$$

Equivalently, $\phi''(\theta) = \phi(\theta)$.

(a) Behaviour at a maximum

Suppose that ϕ attains a maximum at some $\theta_0 \in \mathbb{R}$. By elementary calculus,

$$\phi'(\theta_0) = 0, \quad \phi''(\theta_0) \leq 0.$$

Evaluating the differential equation at θ_0 gives

$$0 = -\phi''(\theta_0) + \phi(\theta_0) \implies \phi(\theta_0) = \phi''(\theta_0).$$

Since $\phi''(\theta_0) \leq 0$, it follows that

$$\phi(\theta_0) \leq 0.$$

Thus any maximum value of ϕ must be non-positive.

(b) Behaviour at a minimum

Suppose that ϕ attains a minimum at some $\theta_0 \in \mathbb{R}$. Then

$$\phi'(\theta_0) = 0, \quad \phi''(\theta_0) \geq 0.$$

Once more, evaluating the equation at θ_0 yields

$$0 = -\phi''(\theta_0) + \phi(\theta_0) \implies \phi(\theta_0) = \phi''(\theta_0) \geq 0.$$

Thus any minimum value of ϕ must be non-negative.

(c) Conclusion: $\phi \equiv 0$

Because ϕ is continuous and 2π -periodic, it attains both a maximum and a minimum on $\mathbb{R}$. Let

$$M := \max_{\theta \in \mathbb{R}} \phi(\theta), \quad m := \min_{\theta \in \mathbb{R}} \phi(\theta).$$

From part (a), $M \leq 0$, and from part (b), $m \geq 0$. Since always $m \leq M$, we obtain

$$m \leq M \leq 0 \quad \text{and} \quad m \geq 0.$$

Hence $m = 0$ and $M = 0$. A continuous function whose maximum and minimum are both equal to 0 must be identically zero, so

$$\phi(\theta) \equiv 0 \quad \text{for all } \theta \in \mathbb{R}.$$

(d) Uniqueness for $-\psi'' + \psi = f$

Let $f \in C^0(\mathbb{R})$ be a given 2π -periodic function and suppose that $\psi_1, \psi_2 \in C^2(\mathbb{R})$ are 2π -periodic solutions of

$$-\psi''(\theta) + \psi(\theta) = f(\theta), \quad \theta \in \mathbb{R}.$$

Consider their difference

$$\phi(\theta) := \psi_1(\theta) - \psi_2(\theta).$$

Then ϕ is 2π -periodic and satisfies

$$-\phi''(\theta) + \phi(\theta) = (-\psi_1''(\theta) + \psi_1(\theta)) - (-\psi_2''(\theta) + \psi_2(\theta)) = f(\theta) - f(\theta) = 0.$$

By part (c) we must have $\phi \equiv 0$, hence $\psi_1 = \psi_2$. Therefore there is at most one $\psi \in C^2(\mathbb{R})$ which is 2π -periodic and solves $-\psi'' + \psi = f$.

Remark (Why “on the circle”?). A 2π -periodic function $\phi \in C^2(\mathbb{R})$ naturally defines a function on the unit circle S^1 via

$$\Phi(e^{i\theta}) := \phi(\theta).$$

This is well defined because $\phi(\theta + 2\pi) = \phi(\theta)$, so the value depends only on the class of θ in $\mathbb{R}/(2\pi\mathbb{Z}) \simeq S^1$. Conversely, any function on S^1 can be regarded as a 2π -periodic function of θ . In this identification the operator

$$L = -\frac{d^2}{d\theta^2} + 1$$

is precisely $-\Delta_{S^1} + 1$, the Laplacian plus a mass term on the circle. Thus Exercise 1.3 can be viewed as a maximum principle for the operator $-\Delta_{S^1} + 1$ on the compact manifold S^1 .

Remark: periodic solutions of $-\phi'' = 0$ vs. $-\phi'' + \phi = 0$. It is important to distinguish between the operators $-\phi''$ and $-\phi'' + \phi$ when working on the circle $S^1 \cong \mathbb{R}/(2\pi\mathbb{Z})$.

1. The equation $-\phi'' = 0$. Solving $-\phi''(\theta) = 0$ gives

$$\phi(\theta) = a\theta + b.$$

If ϕ is 2π -periodic, i.e. $\phi(\theta + 2\pi) = \phi(\theta)$, then

$$a(\theta + 2\pi) + b = a\theta + b \implies a = 0.$$

Hence the only 2π -periodic solutions are *constants*. Therefore the kernel of $-\phi''$ on S^1 consists precisely of the constant functions:

$$\ker(-\phi'') = \{\text{constants}\}.$$

2. The equation $-\phi'' + \phi = 0$. Solving $\phi''(\theta) = \phi(\theta)$ gives

$$\phi(\theta) = Ae^\theta + Be^{-\theta}.$$

There is no nonzero 2π -periodic solution of this form. In Exercise 1.3(a)–(c) we proved a maximum principle showing that any 2π -periodic C^2 solution of

$$-\phi''(\theta) + \phi(\theta) = 0$$

must satisfy $\phi \equiv 0$. Thus the kernel of $-\phi'' + \phi$ on S^1 is *trivial*:

$$\ker(-\phi'' + \phi) = \{0\}.$$

3. Consequence for uniqueness in (d). Suppose ψ_1, ψ_2 are 2π -periodic C^2 solutions of

$$-\psi''(\theta) + \psi(\theta) = f(\theta).$$

Then their difference $\phi = \psi_1 - \psi_2$ satisfies

$$-\phi''(\theta) + \phi(\theta) = 0$$

and is 2π -periodic. By the result above, $\phi \equiv 0$, hence $\psi_1 = \psi_2$. Therefore the inhomogeneous equation $-\psi'' + \psi = f$ admits *at most one* 2π -periodic solution.

Summary.

Operator	General solution	2π -periodic solution
$-\phi''$	$a\theta + b$	constants ($a = 0$)
$-\phi'' + \phi$	$Ae^\theta + Be^{-\theta}$	only zero

This distinction explains why part (d) has uniqueness but not constant solutions.

Exercise 1.4

Exercise 1.4

For $t \in \mathbb{R}$ let

$$\chi(t) := \begin{cases} 0, & t \leq 0, \\ e^{-1/t}, & t > 0. \end{cases}$$

- (a) Show that $\chi \in C^\infty(\mathbb{R})$.
- (b) Show that there exists a function $\psi \in C_0^\infty(\mathbb{R}^n)$ such that:
 - (i) $0 \leq \psi(x) \leq 1$ for all $x \in \mathbb{R}^n$;
 - (ii) $\text{supp } \psi \subset B_2(0)$;
 - (iii) $\psi(x) = 1$ for all $|x| \leq 1$.

Hint: First construct a smooth function $\tilde{\chi} \in C^\infty(\mathbb{R})$ such that

$$\tilde{\chi}(x) = 0 \quad \text{for } x < -1, \quad \tilde{\chi}(x) = 1 \quad \text{for } x > 1.$$

For $t \in \mathbb{R}$ let

$$\chi(t) := \begin{cases} 0, & t \leq 0, \\ e^{-1/t}, & t > 0. \end{cases}$$

(a) Smoothness of χ

For $t < 0$ the function χ is identically zero, hence smooth. For $t > 0$ we have $\chi(t) = e^{-1/t}$, which is smooth as a composition of smooth functions. The only nontrivial point is $t = 0$. A straightforward induction shows that for each $k \in \mathbb{N}$ there is a polynomial P_k such that, for $t > 0$,

$$\chi^{(k)}(t) = P_k(1/t) e^{-1/t}.$$

Moreover, for every $m \geq 0$ one has

$$\lim_{t \rightarrow 0^+} t^m e^{-1/t} = 0,$$

so $P_k(1/t)e^{-1/t} \rightarrow 0$ as $t \rightarrow 0^+$. On the other hand, for $t < 0$ all derivatives of χ vanish identically. Therefore for each k we can define $\chi^{(k)}(0) := 0$ and obtain a continuous extension of $\chi^{(k)}$ to all of $\mathbb{R}$. It follows that $\chi \in C^\infty(\mathbb{R})$.

(b) Construction of a bump function ψ on $\mathbb{R}^n$

Step 1: a one-dimensional smooth step. Define

$$\eta(x) := \frac{\chi(x)}{\chi(x) + \chi(1-x)}, \quad x \in \mathbb{R}.$$

The denominator is strictly positive for all x (at least one of $\chi(x)$ or $\chi(1-x)$ is nonzero), and both numerator and denominator are smooth by part (a), so $\eta \in C^\infty(\mathbb{R})$. By inspection,

$$\eta(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x \geq 1. \end{cases}$$

Now set

$$\tilde{\chi}(x) := \eta\left(\frac{x+1}{2}\right).$$

Then $\tilde{\chi} \in C^\infty(\mathbb{R})$ and

$$\tilde{\chi}(x) = 0 \quad \text{for } x \leq -1, \quad \tilde{\chi}(x) = 1 \quad \text{for } x \geq 1,$$

as required in the hint.

Step 2: a radial cutoff on $\mathbb{R}^n$. Let $r = |x|$ and define

$$h(r) := \tilde{\chi}(2-r), \quad r \geq 0.$$

Then h is smooth on $[0, \infty)$ and satisfies

$$h(r) = \begin{cases} 1, & 0 \leq r \leq 1, \\ 0, & r \geq 2. \end{cases}$$

We now define a radial function ψ on $\mathbb{R}^n$ by

$$\psi(x) := h(|x|) = \tilde{\chi}(2 - |x|).$$

Since h is smooth and depends only on $|x|$, standard arguments show that $\psi \in C^\infty(\mathbb{R}^n)$. Moreover, $0 \leq \tilde{\chi} \leq 1$ implies $0 \leq \psi \leq 1$. If $|x| \leq 1$ then $2 - |x| \geq 1$ so $\tilde{\chi}(2 - |x|) = 1$ and hence $\psi(x) = 1$. If $|x| \geq 2$ then $2 - |x| \leq 0$ so $\tilde{\chi}(2 - |x|) = 0$ and hence $\psi(x) = 0$. Thus $\text{supp } \psi \subset B_2(0)$ and $\psi(x) = 1$ for $|x| \leq 1$.

We conclude that $\psi \in C_0^\infty(\mathbb{R}^n)$ and

$$0 \leq \psi \leq 1, \quad \text{supp } \psi \subset B_2(0), \quad \psi(x) = 1 \text{ for } |x| \leq 1,$$

as required.

Theory box: mollifiers and partitions of unity. A *mollifier* in $\mathbb{R}^n$ is a function $\rho \in C_0^\infty(\mathbb{R}^n)$ such that

$$\rho(x) \geq 0 \quad \text{and} \quad \int_{\mathbb{R}^n} \rho(x) dx = 1.$$

For $\varepsilon > 0$ we define the rescaled mollifier

$$\rho_\varepsilon(x) := \frac{1}{\varepsilon^n} \rho\left(\frac{x}{\varepsilon}\right).$$

If $f \in L_{\text{loc}}^1(\mathbb{R}^n)$, the mollified function $f_\varepsilon := \rho_\varepsilon * f$ is smooth, and as $\varepsilon \rightarrow 0$ one has $f_\varepsilon \rightarrow f$ in various senses (e.g. in L_{loc}^p , or uniformly when f is continuous). Normalised bump functions of the type constructed in Exercise 1.4 are the basic building blocks for such mollifiers.

A *partition of unity* on an open set $U \subset \mathbb{R}^n$ is a family $\{\varphi_j\}_{j \in J} \subset C^\infty(U)$ such that

1. $0 \leq \varphi_j \leq 1$ for all j ;
2. the family is locally finite (for each $x \in U$ there is a neighbourhood where only finitely many φ_j are nonzero);
3. $\sum_{j \in J} \varphi_j(x) = 1$ for all $x \in U$.

Given an open cover $U = \bigcup_j U_j$, one can construct a partition of unity $\{\varphi_j\}$ subordinate to this cover (i.e. $\text{supp } \varphi_j \subset U_j$ for each j). The bump function ψ from Exercise 1.4, with $\psi = 1$ on $B_1(0)$ and $\text{supp } \psi \subset B_2(0)$, is a model for such building blocks: by translating and rescaling ψ one obtains the pieces of a partition of unity. This construction is fundamental in analysis on manifolds, where partitions of unity allow one to glue local objects into global ones.

We use the standard Euclidean norm

$$|x| = \sqrt{x_1^2 + \cdots + x_n^2}, \quad x \in \mathbb{R}^n,$$

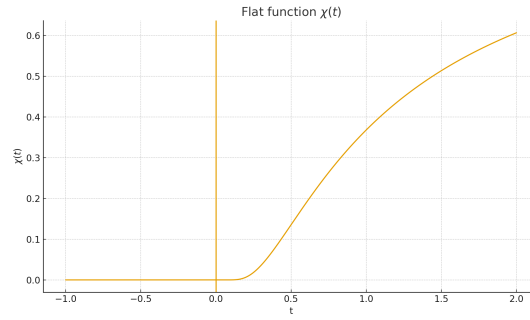


Figure 18: The flat function $\chi(t)$: 0 for $t \leq 0$, rapidly increasing for $t > 0$, with all derivatives vanishing at $t = 0$.

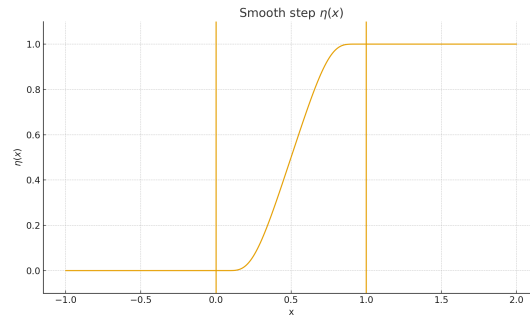


Figure 19: The smooth step $\eta(x) = \chi(x)/(\chi(x) + \chi(1-x))$: $\eta(x) = 0$ for $x \leq 0$ and $\eta(x) = 1$ for $x \geq 1$, with a smooth transition in between.

and denote by

$$B_r(0) := \{x \in \mathbb{R}^n : |x| < r\}$$

the open ball of radius r centred at the origin. Thus $B_1(0)$ is the unit ball and $B_2(0)$ is the ball of radius 2. The conditions

$$\psi(x) = 1 \text{ for } |x| \leq 1, \quad \text{supp } \psi \subset B_2(0),$$

mean that ψ is identically 1 on the closed unit ball $\overline{B_1(0)}$ and vanishes outside the ball $B_2(0)$.

Exercise 1.5

- (a) Suppose $\phi \in \mathcal{E}(\mathbb{R}^n)$. Let $\{x_l\}_{l=1}^\infty \subset \mathbb{R}^n$ be a sequence with $x_l \rightarrow 0$. Show that

$$\tau_{x_l} \phi \rightarrow \phi, \quad \text{as } l \rightarrow \infty,$$

in $\mathcal{E}(\mathbb{R}^n)$, where τ_x is the translation operator defined in equation (1.2).

- (b) Suppose $\phi \in \mathcal{E}(\mathbb{R}^n)$. Show that

$$\Delta_i^h \phi \rightarrow D_i \phi, \quad \text{as } h \rightarrow 0,$$

in $\mathcal{E}(\mathbb{R}^n)$, where Δ_i^h is the difference quotient defined in Example 2.

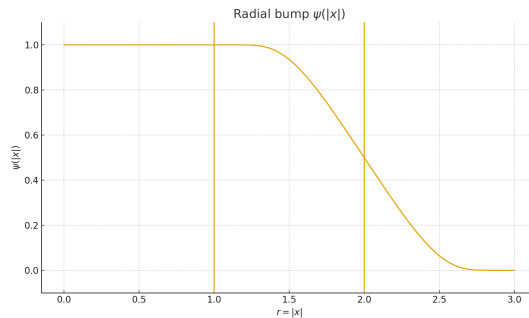


Figure 20: The radial bump $\psi(|x|)$ obtained from $\tilde{\chi}$: $\psi(r) = 1$ for $0 \leq r \leq 1$ and $\psi(r) = 0$ for $r \geq 2$, giving a smooth cutoff supported in $B_2(0)$.

Translation operator. For $x \in \mathbb{R}^n$ and $\phi : \mathbb{R}^n \rightarrow \mathbb{C}$ we define the translation operator τ_x by

$$(\tau_x \phi)(y) := \phi(y - x), \quad y \in \mathbb{R}^n.$$

Difference quotient in the i -th direction. Let e_i denote the i -th coordinate vector in $\mathbb{R}^n$. For $h \neq 0$ and $\phi : \mathbb{R}^n \rightarrow \mathbb{C}$ we define

$$(\Delta_i^h \phi)(x) := \frac{\phi(x + he_i) - \phi(x)}{h}, \quad x \in \mathbb{R}^n.$$

Formally, $\Delta_i^h \phi$ approximates the partial derivative $D_i \phi = \partial_{x_i} \phi$ as $h \rightarrow 0$.

Solution to Exercise 1.5

We use the standard Fréchet topology on $\mathcal{E}(\mathbb{R}^n)$ given by the seminorms

$$p_{K,m}(\phi) := \sup_{\substack{x \in K \\ |\alpha| \leq m}} |D^\alpha \phi(x)|,$$

where $K \subset \mathbb{R}^n$ is compact and $m \in \mathbb{N}$.

(a) Continuity of translations

Let $\phi \in \mathcal{E}(\mathbb{R}^n)$ and $x_l \rightarrow 0$ in $\mathbb{R}^n$. Recall that $(\tau_{x_l} \phi)(y) = \phi(y - x_l)$. We must show that $\tau_{x_l} \phi \rightarrow \phi$ in $\mathcal{E}(\mathbb{R}^n)$, i.e. that for every compact K and every $m \in \mathbb{N}$,

$$p_{K,m}(\tau_{x_l} \phi - \phi) \rightarrow 0 \quad \text{as } l \rightarrow \infty.$$

Fix such K and m , and let $|\alpha| \leq m$. Then

$$D^\alpha(\tau_{x_l} \phi)(y) = D^\alpha \phi(y - x_l),$$

so

$$\sup_{y \in K} |D^\alpha(\tau_{x_l} \phi)(y) - D^\alpha \phi(y)| = \sup_{y \in K} |D^\alpha \phi(y - x_l) - D^\alpha \phi(y)|.$$

Choose a compact set K' such that $K \cup (K - x_l) \subset K'$ for all l sufficiently large (for instance $K' = K + \overline{B_1(0)}$ if $|x_l| \leq 1$ eventually). The function $D^\alpha \phi$ is continuous on the compact set K' and hence uniformly continuous there. Thus for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|z - z'| < \delta$ implies $|D^\alpha \phi(z) - D^\alpha \phi(z')| < \varepsilon$ for all $z, z' \in K'$. Since $x_l \rightarrow 0$, for l large enough we have $|x_l| < \delta$, and then for all $y \in K$

$$|D^\alpha \phi(y - x_l) - D^\alpha \phi(y)| < \varepsilon.$$

Taking the supremum over $y \in K$ and $|\alpha| \leq m$ shows that $p_{K,m}(\tau_{x_l} \phi - \phi) \leq \varepsilon$ for all large l . Hence $p_{K,m}(\tau_{x_l} \phi - \phi) \rightarrow 0$ and $\tau_{x_l} \phi \rightarrow \phi$ in $\mathcal{E}(\mathbb{R}^n)$.

(b) Difference quotients converge to the derivative

Let $\phi \in \mathcal{E}(\mathbb{R}^n)$ and define, for $h \neq 0$,

$$\Delta_i^h \phi(x) := \frac{\phi(x + he_i) - \phi(x)}{h},$$

where e_i is the i -th coordinate vector. We must show that $\Delta_i^h \phi \rightarrow D_i \phi$ in $\mathcal{E}(\mathbb{R}^n)$ as $h \rightarrow 0$.

Fix a compact $K \subset \mathbb{R}^n$ and $m \in \mathbb{N}$. For a multi-index α with $|\alpha| \leq m$ we compute

$$D^\alpha(\Delta_i^h \phi)(x) = \frac{D^\alpha \phi(x + he_i) - D^\alpha \phi(x)}{h},$$

and hence

$$D^\alpha(\Delta_i^h \phi - D_i \phi)(x) = \frac{D^\alpha \phi(x + he_i) - D^\alpha \phi(x)}{h} - D_i D^\alpha \phi(x).$$

Set $g(x) := D^\alpha \phi(x)$, so $g \in C^\infty$ and the expression becomes

$$\frac{g(x + he_i) - g(x)}{h} - D_i g(x).$$

Let K' be a compact set containing K and $K + he_i$ for all $x \in K$ and $|h|$ small (e.g. $K' = K + \overline{B_1(0)}$). On K' the function $D_i g$ is continuous and therefore uniformly continuous. By the one-dimensional mean-value theorem, for each $x \in K$ there exists $\theta = \theta(x, h) \in (0, 1)$ such that

$$\frac{g(x + he_i) - g(x)}{h} = D_i g(x + \theta he_i),$$

so

$$\left| \frac{g(x + he_i) - g(x)}{h} - D_i g(x) \right| = |D_i g(x + \theta he_i) - D_i g(x)|.$$

Given $\varepsilon > 0$, uniform continuity of $D_i g$ on K' yields $\delta > 0$ such that $|z - z'| < \delta$ implies $|D_i g(z) - D_i g(z')| < \varepsilon$ for all $z, z' \in K'$. For $|h| < \delta$ we have $|x + \theta he_i - x| \leq |h| < \delta$, so the right hand side is $< \varepsilon$ for all $x \in K$. Thus

$$\sup_{x \in K} |D^\alpha(\Delta_i^h \phi - D_i \phi)(x)| < \varepsilon$$

for all $|\alpha| \leq m$ and $|h|$ sufficiently small. Hence $p_{K,m}(\Delta_i^h \phi - D_i \phi) \rightarrow 0$ as $h \rightarrow 0$, and therefore $\Delta_i^h \phi \rightarrow D_i \phi$ in $\mathcal{E}(\mathbb{R}^n)$.

Example (Exercise 1.5 for $\phi(x) = e^{-x^2}$ in $\mathbb{R}$). Let $\phi(x) = e^{-x^2}$. For a sequence $x_l \rightarrow 0$ the translation $\tau_{x_l}\phi$ is given by

$$(\tau_{x_l}\phi)(y) = \phi(y - x_l) = e^{-(y-x_l)^2}.$$

(a) Translation example. For each fixed $y \in \mathbb{R}$ we have

$$e^{-(y-x_l)^2} \longrightarrow e^{-y^2} = \phi(y) \quad \text{as } l \rightarrow \infty.$$

The convergence is uniform on every compact interval $K = [-R, R]$ since

$$(y - x_l)^2 \rightarrow y^2 \quad \text{uniformly on } [-R, R],$$

and the exponential function is uniformly continuous on bounded sets. Thus

$$\sup_{y \in [-R, R]} |(\tau_{x_l}\phi)(y) - \phi(y)| \rightarrow 0.$$

The same argument applies to all derivatives. For example,

$$\phi'(x) = -2xe^{-x^2}, \quad (\tau_{x_l}\phi)'(y) = -2(y - x_l)e^{-(y-x_l)^2},$$

and one checks that

$$(\tau_{x_l}\phi)'(y) \longrightarrow \phi'(y) \quad \text{uniformly on } [-R, R].$$

Hence $\tau_{x_l}\phi \rightarrow \phi$ in $\mathcal{E}(\mathbb{R})$.

(b) Difference quotient example. In one dimension the difference quotient is

$$(\Delta^h\phi)(x) = \frac{\phi(x+h) - \phi(x)}{h} = \frac{e^{-(x+h)^2} - e^{-x^2}}{h}.$$

We know

$$\phi'(x) = -2xe^{-x^2}.$$

By the mean value theorem, for each x and each small h , there exists $\theta = \theta(x, h) \in (0, 1)$ such that

$$\frac{e^{-(x+h)^2} - e^{-x^2}}{h} = \phi'(x + \theta h).$$

Thus

$$(\Delta^h\phi)(x) - \phi'(x) = \phi'(x + \theta h) - \phi'(x).$$

Fix a compact interval $K = [-R, R]$. For $|h|$ sufficiently small, all points $x + \theta h$ with $x \in K$ lie in a slightly larger compact interval, say $[-R-1, R+1]$. Since ϕ' is continuous

on this compact set, it is uniformly continuous there. Hence for any $\varepsilon > 0$ there exists $\delta > 0$ such that whenever $|h| < \delta$,

$$\sup_{x \in [-R, R]} |\phi'(x + \theta(x, h)h) - \phi'(x)| < \varepsilon.$$

Therefore

$$\sup_{x \in [-R, R]} |(\Delta^h \phi)(x) - \phi'(x)| \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

The same argument applies to all derivatives of $\Delta^h \phi$, since d^k/dx^k applied to the difference quotient produces the same formula with ϕ' replaced by a higher derivative. Consequently,

$$\Delta^h \phi \rightarrow \phi' \quad \text{in } \mathcal{E}(\mathbb{R}).$$

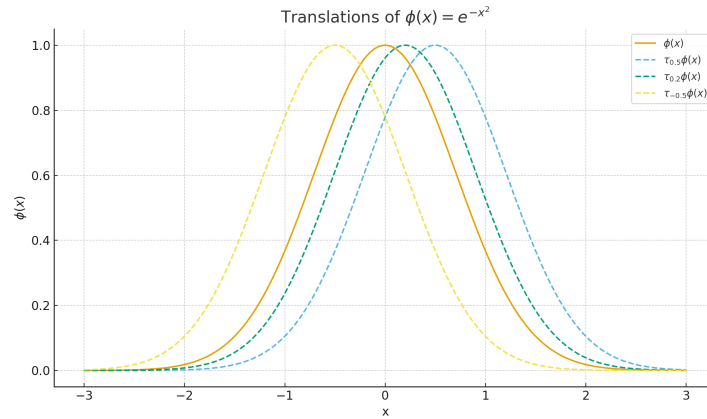


Figure 21: Translations $\tau_a \phi$ of $\phi(x) = e^{-x^2}$ for several values of a .

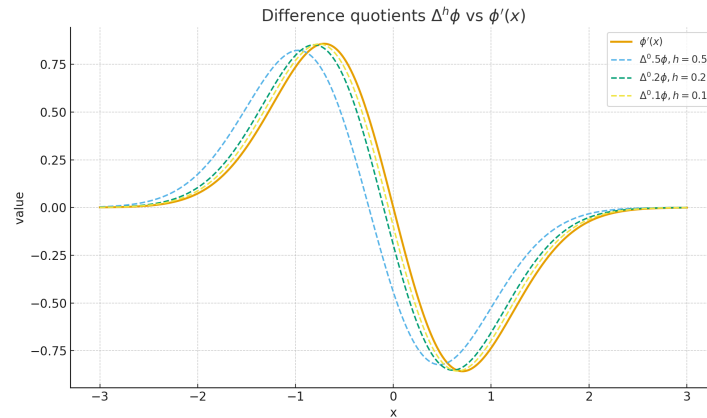


Figure 22: Difference quotients $\Delta^h \phi$ for $h = 0.5, 0.2, 0.1$ compared with the derivative $\phi'(x)$.

Theorem 0.1 (Continuity of translations and difference quotients in $\mathcal{E}(\mathbb{R}^n)$). *Let $\mathcal{E}(\mathbb{R}^n)$ denote the space of smooth functions on $\mathbb{R}^n$ endowed with the usual Fréchet topology given by the seminorms*

$$p_{K,m}(\phi) := \sup_{\substack{x \in K \\ |\alpha| \leq m}} |D^\alpha \phi(x)|,$$

where $K \subset \mathbb{R}^n$ is compact and $m \in \mathbb{N}$.

- (a) For each fixed $x \in \mathbb{R}^n$ the translation operator $\tau_x : \mathcal{E}(\mathbb{R}^n) \rightarrow \mathcal{E}(\mathbb{R}^n)$, defined by $(\tau_x \phi)(y) = \phi(y - x)$, is continuous. Moreover, if $x_l \rightarrow 0$ in $\mathbb{R}^n$ and $\phi \in \mathcal{E}$, then

$$\tau_{x_l} \phi \longrightarrow \phi \quad \text{in } \mathcal{E}(\mathbb{R}^n).$$

- (b) For each $i \in \{1, \dots, n\}$ and $h \neq 0$ the difference quotient operator

$$(\Delta_i^h \phi)(x) := \frac{\phi(x + he_i) - \phi(x)}{h}$$

is a continuous linear map on $\mathcal{E}(\mathbb{R}^n)$, and for every $\phi \in \mathcal{E}(\mathbb{R}^n)$ one has

$$\Delta_i^h \phi \longrightarrow D_i \phi \quad \text{in } \mathcal{E}(\mathbb{R}^n) \quad \text{as } h \rightarrow 0.$$

Proof. Fix a compact $K \subset \mathbb{R}^n$ and $m \in \mathbb{N}$.

- (a) *Translations.* For $|\alpha| \leq m$ we have

$$D^\alpha(\tau_x \phi)(y) = D^\alpha \phi(y - x).$$

If $x_l \rightarrow 0$, choose a compact set K' containing K and $K - x_l$ for all l large. Then $D^\alpha \phi$ is uniformly continuous on K' , so given $\varepsilon > 0$ there exists $\delta > 0$ such that $|z - z'| < \delta \Rightarrow |D^\alpha \phi(z) - D^\alpha \phi(z')| < \varepsilon$ for all $z, z' \in K'$. For l large enough we have $|x_l| < \delta$ and therefore

$$\sup_{y \in K} |D^\alpha \phi(y - x_l) - D^\alpha \phi(y)| \leq \varepsilon.$$

Taking the supremum over $|\alpha| \leq m$ yields $p_{K,m}(\tau_{x_l} \phi - \phi) \rightarrow 0$, i.e. $\tau_{x_l} \phi \rightarrow \phi$ in $\mathcal{E}(\mathbb{R}^n)$.

- (b) *Difference quotients.* Let $g(x) := D^\alpha \phi(x)$ for some $|\alpha| \leq m$. Then

$$D^\alpha(\Delta_i^h \phi)(x) = \frac{g(x + he_i) - g(x)}{h}.$$

By the mean value theorem,

$$\frac{g(x + he_i) - g(x)}{h} = D_i g(x + \theta he_i)$$

for some $\theta = \theta(x, h) \in (0, 1)$. Choosing a compact K' containing K and $K + he_i$ for all $|h|$ small, we see that $D_i g$ is uniformly continuous on K' . Hence for every $\varepsilon > 0$ there exists $\delta > 0$ such that for $|h| < \delta$,

$$\sup_{x \in K} |D^\alpha(\Delta_i^h \phi)(x) - D_i D^\alpha \phi(x)| = \sup_{x \in K} |D_i g(x + \theta he_i) - D_i g(x)| < \varepsilon.$$

Taking the supremum over $|\alpha| \leq m$ shows that $p_{K,m}(\Delta_i^h \phi - D_i \phi) \rightarrow 0$ as $h \rightarrow 0$, which is precisely $\Delta_i^h \phi \rightarrow D_i \phi$ in $\mathcal{E}(\mathbb{R}^n)$. $\square$

Exercise 1.6

- (a) Show that $\mathcal{S}$ is a vector subspace of $\mathcal{E}(\mathbb{R}^n)$. Show that if $\{\phi_j\}_{j=1}^\infty$ is a sequence of rapidly decreasing functions which tends to zero in $\mathcal{S}$, then $\phi_j \rightarrow 0$ in $\mathcal{E}(\mathbb{R}^n)$.
- (b) Show that $\mathcal{D}(\mathbb{R}^n)$ is a vector subspace of $\mathcal{S}$. Show that if $\{\phi_j\}_{j=1}^\infty$ is a sequence of compactly supported functions which tends to zero in $\mathcal{D}(\mathbb{R}^n)$, then $\phi_j \rightarrow 0$ in $\mathcal{S}$.
- (c) Give an example of a sequence $\{\phi_j\}_{j=1}^\infty \subset C_0^\infty(\mathbb{R}^n)$ such that
- (i) $\phi_j \rightarrow 0$ in $\mathcal{S}$, but $\{\phi_j\}$ has no limit in $\mathcal{D}(\mathbb{R}^n)$;
 - (ii) $\phi_j \rightarrow 0$ in $\mathcal{E}(\mathbb{R}^n)$, but $\{\phi_j\}$ has no limit in $\mathcal{S}$.

Exercise 1.7

- (a) Suppose $\phi \in \mathcal{S}$. Let $\{x_l\}_{l=1}^\infty \subset \mathbb{R}^n$ be a sequence with $x_l \rightarrow 0$. Show that

$$\tau_{x_l}\phi \rightarrow \phi, \quad \text{as } l \rightarrow \infty,$$

in $\mathcal{S}$, where τ_x is the translation operator defined in equation (1.2).

- (b) Suppose $\phi \in \mathcal{S}$, show that

$$\Delta_i^h \phi \rightarrow D_i \phi, \quad \text{as } h \rightarrow 0,$$

in $\mathcal{S}$, where Δ_i^h is the difference quotient defined in Example 2.

Exercise 1.6

- (a) Show that $\mathcal{S}(\mathbb{R}^n)$ is a vector subspace of $\mathcal{E}(\mathbb{R}^n)$. Show that if $\{\phi_j\}_{j=1}^\infty$ is a sequence of rapidly decreasing functions which tends to zero in $\mathcal{S}$, then $\phi_j \rightarrow 0$ in $\mathcal{E}(\mathbb{R}^n)$.

Solution. Every $\phi \in \mathcal{S}$ is smooth and satisfies

$$q_{\alpha,\beta}(\phi) = \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta \phi(x)| < \infty, \quad \forall \alpha, \beta,$$

so in particular each derivative is bounded on every compact set. Thus $\mathcal{S} \subset \mathcal{E}$ and is clearly a vector subspace.

If $\phi_j \rightarrow 0$ in $\mathcal{S}$, then

$$q_{\alpha,\beta}(\phi_j) \rightarrow 0 \quad \forall \alpha, \beta.$$

Fix a compact K and $m \in \mathbb{N}$. On K we have $|x^\alpha| \leq C_K$, hence

$$p_{K,m}(\phi_j) = \sup_{x \in K, |\beta| \leq m} |D^\beta \phi_j(x)| \leq \max_{|\beta| \leq m} q_{0,\beta}(\phi_j) \rightarrow 0.$$

Thus $\phi_j \rightarrow 0$ in $\mathcal{E}$.

- (b) Show that $\mathcal{D}(\mathbb{R}^n)$ is a vector subspace of $\mathcal{S}(\mathbb{R}^n)$. Show that if $\{\phi_j\}$ tends to 0 in $\mathcal{D}(\mathbb{R}^n)$, then $\phi_j \rightarrow 0$ in $\mathcal{S}$.

Solution. If $\phi \in C_0^\infty(\mathbb{R}^n)$ and $\text{supp } \phi \subset B_R(0)$, then

$$q_{\alpha,\beta}(\phi) = \sup_{x \in B_R(0)} |x^\alpha D^\beta \phi(x)| \leq R^{|\alpha|} \sup_{x \in B_R(0)} |D^\beta \phi(x)| < \infty,$$

so $\phi \in \mathcal{S}$. Linear combinations preserve smoothness and compact support, so $\mathcal{D} \subset \mathcal{S}$ as a subspace.

If $\phi_j \rightarrow 0$ in $\mathcal{D}$, then for some compact K we have $\text{supp } \phi_j \subset K$ and

$$\sup_{x \in K, |\beta| \leq m} |D^\beta \phi_j(x)| \rightarrow 0 \quad \forall m.$$

Hence

$$q_{\alpha,\beta}(\phi_j) = \sup_{x \in K} |x^\alpha D^\beta \phi_j(x)| \leq C_K \sup_{x \in K} |D^\beta \phi_j(x)| \rightarrow 0.$$

Thus $\phi_j \rightarrow 0$ in $\mathcal{S}$.

- (c) Give examples of sequences $\{\phi_j\} \subset C_0^\infty(\mathbb{R}^n)$ such that

- (i) $\phi_j \rightarrow 0$ in $\mathcal{S}$, but the sequence has no limit in $\mathcal{D}(\mathbb{R}^n)$;
- (ii) $\phi_j \rightarrow 0$ in $\mathcal{E}(\mathbb{R}^n)$, but the sequence has no limit in $\mathcal{S}$.

Solution.

Let $\psi \in C_0^\infty(\mathbb{R}^n)$, $\psi \not\equiv 0$, with $\text{supp } \psi \subset B_1(0)$.

(i) Converges in $\mathcal{S}$ but not in $\mathcal{D}$. Define

$$\phi_j(x) = e^{-j^2} \psi(x - je_1).$$

Then $\text{supp } \phi_j \subset B_1(je_1)$ escapes to infinity. For any α, β ,

$$|x^\alpha D^\beta \phi_j(x)| \leq C_{\alpha,\beta} (1+j)^{|\alpha|} e^{-j^2} \rightarrow 0.$$

Thus $\phi_j \rightarrow 0$ in $\mathcal{S}$. However, no convergent sequence in $\mathcal{D}$ may have supports escaping to infinity, so $\{\phi_j\}$ has no limit in $\mathcal{D}$.

(ii) Converges in $\mathcal{E}$ but not in $\mathcal{S}$. Define

$$\phi_j(x) = \psi(x - je_1).$$

For any compact K , $\phi_j \equiv 0$ on K for large j , hence $\phi_j \rightarrow 0$ in $\mathcal{E}$. However,

$$q_{\alpha,0}(\phi_j) = \sup_x |x^\alpha \phi_j(x)| \geq |j|^{|\alpha|} \max_{|y| \leq 1} |\psi(y)| \rightarrow \infty,$$

so the sequence does not converge in $\mathcal{S}$.

Exercise 1.7

(a) Suppose $\phi \in \mathcal{S}$. Let $\{x_l\} \subset \mathbb{R}^n$ with $x_l \rightarrow 0$. Show that

$$\tau_{x_l}\phi \rightarrow \phi \quad \text{in } \mathcal{S}.$$

Solution. Recall $\tau_x\phi(\xi) = \phi(\xi - x)$. Fix α, β . Using

$$\xi^\alpha = (\xi - x_l + x_l)^\alpha = \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} (\xi - x_l)^\gamma x_l^{\alpha-\gamma},$$

we obtain

$$q_{\alpha,\beta}(\tau_{x_l}\phi - \phi) \leq \sum_{\gamma < \alpha} C_{\alpha,\gamma} |x_l|^{\alpha-\gamma} q_{\gamma,\beta}(\phi) + \sup_{\xi} |\xi^\alpha (D^\beta \phi(\xi - x_l) - D^\beta \phi(\xi))|.$$

The first sum $\rightarrow 0$ because $x_l \rightarrow 0$. The second term $\rightarrow 0$ because $\xi^\alpha D^\beta \phi(\xi)$ is uniformly continuous (it is rapidly decaying). Thus $q_{\alpha,\beta}(\tau_{x_l}\phi - \phi) \rightarrow 0$, proving convergence in $\mathcal{S}$.

(b) Show that for $\phi \in \mathcal{S}$,

$$\Delta_i^h \phi \rightarrow D_i \phi \quad \text{in } \mathcal{S}, \quad h \rightarrow 0,$$

where

$$\Delta_i^h \phi(x) = \frac{\phi(x + he_i) - \phi(x)}{h}.$$

Solution. Fix α, β . Use the mean value theorem in the e_i direction:

$$\frac{D^\beta \phi(x + he_i) - D^\beta \phi(x)}{h} = D_i D^\beta \phi(x + \theta he_i)$$

for some $\theta \in (0, 1)$. Hence

$$x^\alpha D^\beta (\Delta_i^h \phi - D_i \phi)(x) = x^\alpha (D_i D^\beta \phi(x + \theta he_i) - D_i D^\beta \phi(x)).$$

The function $x^\alpha D_i D^\beta \phi(x)$ is continuous and decays rapidly, hence uniformly continuous. Therefore the supremum over x tends to 0 as $h \rightarrow 0$ and

$$q_{\alpha,\beta}(\Delta_i^h \phi - D_i \phi) \rightarrow 0.$$

Thus $\Delta_i^h \phi \rightarrow D_i \phi$ in $\mathcal{S}$.

Spaces of test functions and their seminorms.

- *Smooth functions:*

$$\mathcal{E}(\mathbb{R}^n) := C^\infty(\mathbb{R}^n)$$

with the Fréchet topology generated by seminorms

$$p_{K,m}(\phi) := \sup_{\substack{x \in K \\ |\alpha| \leq m}} |D^\alpha \phi(x)|, \quad K \subset \mathbb{R}^n \text{ compact, } m \in \mathbb{N}.$$

- *Compactly supported smooth functions:*

$$\mathcal{D}(\mathbb{R}^n) := C_0^\infty(\mathbb{R}^n),$$

endowed with the LF topology: a sequence $\phi_j \rightarrow 0$ in $\mathcal{D}$ iff there exists a compact K such that $\text{supp } \phi_j \subset K$ for all large j and

$$\sup_{x \in K, |\alpha| \leq m} |D^\alpha \phi_j(x)| \rightarrow 0 \quad \forall m \in \mathbb{N}.$$

- *Schwartz space:*

$$\mathcal{S}(\mathbb{R}^n) := \left\{ \phi \in C^\infty(\mathbb{R}^n) : q_{\alpha,\beta}(\phi) < \infty \quad \forall \alpha, \beta \right\},$$

where the seminorms are

$$q_{\alpha,\beta}(\phi) := \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta \phi(x)|, \quad \alpha, \beta \in \mathbb{N}^n.$$

They measure simultaneous control of all derivatives *and* all polynomial weights.

Continuous embeddings.

$$\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset \mathcal{E}(\mathbb{R}^n),$$

and both inclusions are continuous:

- Every compactly supported smooth function is rapidly decreasing, so $\mathcal{D} \subset \mathcal{S}$.
- Every Schwartz function is smooth, so $\mathcal{S} \subset \mathcal{E}$.
- Convergence in $\mathcal{D} \Rightarrow$ convergence in $\mathcal{S}$ (Exercise 1.6(b)).
- Convergence in $\mathcal{S} \Rightarrow$ convergence in $\mathcal{E}$ (Exercise 1.6(a)).

Examples of different convergences. Fix $\psi \in C_0^\infty(\mathbb{R})$, $\psi \neq 0$, $\text{supp } \psi \subset [-1, 1]$.

- (1) $\phi_j(x) = \frac{1}{j} \psi(x)$: converges to 0 in $\mathcal{D}, \mathcal{S}, \mathcal{E}$.

Space	Elements	Seminorms / topology	Convergence of a sequence $\phi_j \rightarrow 0$
$\mathcal{E}$	C^∞ functions on $\mathbb{R}^n$	Fréchet topology: $p_{K,m}(\phi) = \sup_{x \in K, \alpha \leq m} D^\alpha \phi(x) $	For every compact K and every m : $p_{K,m}(\phi_j) \rightarrow 0$, i.e. all derivatives converge to 0 <i>uniformly on each compact</i> .
$\mathcal{D}$	Compactly supported C^∞ functions	LF topology: inductive limit of $\mathcal{D}_K$ with fixed compact support K	There exists a compact K s.t. $\text{supp } \phi_j \subset K$ for all large j , and $p_{K,m}(\phi_j) \rightarrow 0$ for all m . Supports cannot escape to infinity.
$\mathcal{S}$	Rapidly decreasing C^∞ functions	Schwartz seminorms: $q_{\alpha,\beta}(\phi) = \sup_x x^\alpha D^\beta \phi(x) $	$q_{\alpha,\beta}(\phi_j) \rightarrow 0$ for every α, β , i.e. all derivatives tend to 0 faster than any inverse polynomial, uniformly on $\mathbb{R}^n$.

Table 1: Three topologies on smooth functions: $\mathcal{D} \subset \mathcal{S} \subset \mathcal{E}$.

- (2) $\phi_j(x) = e^{-j^2} \psi(x - j)$: the supports escape to $+\infty$, but the factors e^{-j^2} force all Schwartz seminorms $q_{\alpha,\beta}(\phi_j) \rightarrow 0$. Hence $\phi_j \rightarrow 0$ in $\mathcal{S}$ but the sequence has no limit in $\mathcal{D}$.
- (3) $\phi_j(x) = \psi(x - j)$: for any compact K , one has $\phi_j \equiv 0$ on K for all large j , so $\phi_j \rightarrow 0$ in $\mathcal{E}$. However $q_{1,0}(\phi_j) \sim j \rightarrow \infty$, so there is no convergence in $\mathcal{S}$.
- (4) $\phi_j(x) = e^{-(x-a_j)^2}$ with $a_j \rightarrow 0$: translations of a Gaussian. Then $\phi_j \rightarrow \phi$ in $\mathcal{S}$ (and hence in $\mathcal{E}$).
- (5) $\phi_h(x) = \frac{\phi(x+h) - \phi(x)}{h}$ with $\phi(x) = e^{-x^2}$: difference quotients; one checks that $\phi_h \rightarrow \phi'$ in $\mathcal{S}$ as $h \rightarrow 0$.

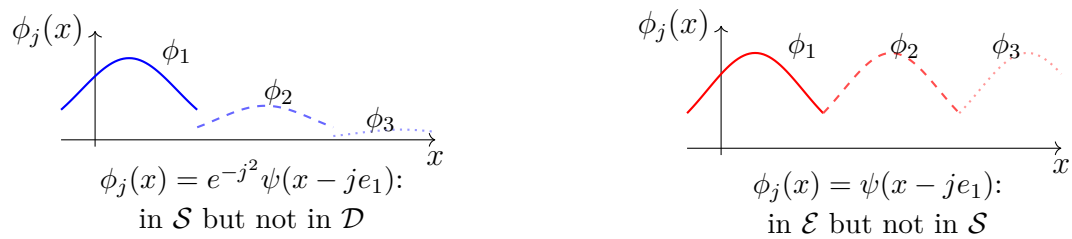


Figure 23: Schematic of the sequences in Exercise 1.6(c): left: scaled bumps converging in $\mathcal{S}$ but not in $\mathcal{D}$; right: bumps of fixed height converging in $\mathcal{E}$ but not in $\mathcal{S}$.

Exercise 2.1 (*): Schwartz functions and integrability

Assume $f, g, h \in \mathcal{S}(\mathbb{R}^n)$.

(a) Show that for any multi-index α and any $1 \leq p < \infty$, one has $D^\alpha f \in L^p(\mathbb{R}^n)$.

(b) Define

$$F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}, \quad F(x, y) = f(x) g(y - x).$$

Show that $F \in L^1(\mathbb{R}^n \times \mathbb{R}^n)$.

(c) For each $x \in \mathbb{R}^n$, set

$$G_x : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}, \quad G_x(y, z) = f(y) g(z) h(x - y - z).$$

Show that $G_x \in L^1(\mathbb{R}^n \times \mathbb{R}^n)$ for every x .

Solution

(a) $D^\alpha f \in L^p$. Since $f \in \mathcal{S}(\mathbb{R}^n)$, for each multi-index α and each integer $N > 0$ there exists $C_{\alpha, N} > 0$ such that

$$|D^\alpha f(x)| \leq \frac{C_{\alpha, N}}{(1 + |x|)^N} \quad \forall x \in \mathbb{R}^n.$$

Fix $1 \leq p < \infty$ and choose N with $Np > n$. Then

$$|D^\alpha f(x)|^p \leq \frac{C_{\alpha, N}^p}{(1 + |x|)^{Np}},$$

and the right-hand side is integrable on $\mathbb{R}^n$. Hence $\|D^\alpha f\|_{L^p} < \infty$, so $D^\alpha f \in L^p(\mathbb{R}^n)$.

(b) $F \in L^1(\mathbb{R}^{2n})$. We compute

$$\begin{aligned} \iint_{\mathbb{R}^n \times \mathbb{R}^n} |F(x, y)| \, dx \, dy &= \iint |f(x)g(y - x)| \, dx \, dy \\ &= \int_{\mathbb{R}^n} |f(x)| \left(\int_{\mathbb{R}^n} |g(y - x)| \, dy \right) dx \\ &\quad (\text{let } t = y - x \text{ so } dy = dt) \\ &= \int_{\mathbb{R}^n} |f(x)| \left(\int_{\mathbb{R}^n} |g(t)| \, dt \right) dx \\ &= \|g\|_{L^1(\mathbb{R}^n)} \|f\|_{L^1(\mathbb{R}^n)} < \infty, \end{aligned}$$

since $f, g \in \mathcal{S} \subset L^1$. Therefore $F \in L^1(\mathbb{R}^{2n})$.

(c) $G_x \in L^1(\mathbb{R}^{2n})$ for each fixed x . Fix $x \in \mathbb{R}^n$ and consider

$$I(x) := \iint_{\mathbb{R}^n \times \mathbb{R}^n} |G_x(y, z)| dy dz = \iint |f(y)g(z)h(x - y - z)| dy dz.$$

Make the change of variables $(y, z) \mapsto (y, w)$ with $w := y + z$. Then $z = w - y$, the Jacobian is 1, and

$$\begin{aligned} I(x) &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(y)| |g(w - y)| |h(x - w)| dy dw \\ &= \int_{\mathbb{R}^n} |h(x - w)| \left(\int_{\mathbb{R}^n} |f(y)| |g(w - y)| dy \right) dw \\ &= \int_{\mathbb{R}^n} |h(x - w)| (|f| * |g|)(w) dw \\ &= (|f| * |g| * |h|)(x). \end{aligned}$$

Since $f, g, h \in \mathcal{S} \subset L^1$, the convolution $|f| * |g| \in L^1$ and then $(|f| * |g|) * |h| \in L^1$. In particular, $(|f| * |g| * |h|)(x) < \infty$ for every x , so $I(x) < \infty$ and $G_x \in L^1(\mathbb{R}^{2n})$.

Schwartz functions: definition, examples, integrability

Definition 0.2 (Schwartz space). The Schwartz space $\mathcal{S}(\mathbb{R}^n)$ consists of all functions $f \in C^\infty(\mathbb{R}^n)$ such that

$$q_{\alpha, \beta}(f) := \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta f(x)| < \infty \quad \text{for all multi-indices } \alpha, \beta.$$

Equivalently, for every multi-index β and $N \in \mathbb{N}$ there exists $C_{N, \beta} > 0$ such that

$$|D^\beta f(x)| \leq \frac{C_{N, \beta}}{(1 + |x|)^N}, \quad x \in \mathbb{R}^n.$$

Examples.

- The Gaussian $f(x) = e^{-|x|^2}$ belongs to $\mathcal{S}(\mathbb{R}^n)$, since each derivative is a polynomial times $e^{-|x|^2}$.
- More generally, for any polynomial P we have $P(x)e^{-|x|^2} \in \mathcal{S}(\mathbb{R}^n)$.
- Every compactly supported smooth function is Schwartz, so $\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$.
- If $f \in \mathcal{S}$, then all derivatives $D^\alpha f$, translations $x \mapsto f(x - a)$, and polynomial multiples $x \mapsto P(x)f(x)$ also lie in $\mathcal{S}$.

Integrability properties. Let $f \in \mathcal{S}(\mathbb{R}^n)$.

- For every multi-index α and every $1 \leq p < \infty$,

$$D^\alpha f \in L^p(\mathbb{R}^n).$$

Indeed, for each N we have $|D^\alpha f(x)| \leq C_{\alpha,N}(1 + |x|)^{-N}$, and choosing N with $Np > n$ gives $|D^\alpha f|^p \in L^1$.

- For every multi-index γ and $1 \leq p < \infty$,

$$x^\gamma D^\alpha f(x) \in L^p(\mathbb{R}^n),$$

since we may take N large enough to dominate the polynomial growth of x^γ .

- In particular $f \in L^p(\mathbb{R}^n)$ for all $1 \leq p \leq \infty$, and the same holds for all derivatives $D^\alpha f$.
- If $f, g \in \mathcal{S}$ then $f, g \in L^1$ and the convolution $f * g$ is well-defined and belongs to $\mathcal{S}(\mathbb{R}^n)$.

Problem 3

Let $A + B := \{a + b : a \in A, b \in B\}$ and $B_r(x) = \{y \in \mathbb{R}^n : \|y - x\| < r\}$, $\overline{B_r(x)} = \{y : \|y - x\| \leq r\}$.

- (a) Prove the following identities for $r, s > 0$ and $x \in \mathbb{R}^n$:

- i) $B_r(x) + B_s(0) = B_{r+s}(x)$;
- ii) $\overline{B_r(x)} + B_s(0) = B_{r+s}(x)$;
- iii) $\overline{B_r(x)} + \overline{B_s(0)} = \overline{B_{r+s}(x)}$.

- (b) Suppose that $A, B \subset \mathbb{R}^n$. Show that:

- b) If one of A or B is open, then so is $A + B$.
- b) If A and B are both bounded, then so is $A + B$.
- b) If A is closed and B is compact, then $A + B$ is closed.
- b) If A and B are both compact, then so is $A + B$.

Proof. (a) (i) Let $z \in B_r(x) + B_s(0)$, so $z = y + u$ with $\|y - x\| < r$ and $\|u\| < s$. Then

$$\|z - x\| \leq \|y - x\| + \|u\| < r + s,$$

hence $z \in B_{r+s}(x)$.

Conversely, let $z \in B_{r+s}(x)$, so $\|z - x\| < r + s$. Set

$$y := x + \frac{r}{r+s}(z - x), \quad u := z - y = \frac{s}{r+s}(z - x).$$

Then

$$\|y - x\| = \frac{r}{r+s}\|z - x\| < r, \quad \|u\| = \frac{s}{r+s}\|z - x\| < s,$$

so $y \in B_r(x)$, $u \in B_s(0)$ and $z = y + u \in B_r(x) + B_s(0)$. Thus $B_r(x) + B_s(0) = B_{r+s}(x)$.

(ii) If $z = y + u$ with $\|y - x\| \leq r$ and $\|u\| < s$, then

$$\|z - x\| \leq \|y - x\| + \|u\| \leq r + \|u\| < r + s,$$

so $z \in B_{r+s}(x)$ and $\overline{B_r(x)} + B_s(0) \subset B_{r+s}(x)$.

Conversely, if $z \in B_{r+s}(x)$, part (i) gives $z = y + u$ with $y \in B_r(x) \subset \overline{B_r(x)}$ and $u \in B_s(0)$, hence $z \in \overline{B_r(x)} + B_s(0)$. Therefore $\overline{B_r(x)} + B_s(0) = B_{r+s}(x)$.

(iii) If $z = y + u$ with $\|y - x\| \leq r$, $\|u\| \leq s$, then $\|z - x\| \leq r + s$, so $\overline{B_r(x)} + \overline{B_s(0)} \subset \overline{B_{r+s}(x)}$.

For the reverse inclusion, let $\|z - x\| < r + s$. Then $z \in B_{r+s}(x)$ and by (i) we have $z = y + u$ with $y \in B_r(x) \subset \overline{B_r(x)}$ and $u \in B_s(0) \subset \overline{B_s(0)}$, so $z \in \overline{B_r(x)} + \overline{B_s(0)}$.

If $\|z - x\| = r + s$, set $z_t = x + t(z - x)$, $0 < t < 1$. Then $\|z_t - x\| = t(r + s) < r + s$, hence $z_t \in B_{r+s}(x)$ and thus $z_t \in \overline{B_r(x)} + \overline{B_s(0)}$. The set $\overline{B_r(x)} + \overline{B_s(0)}$ is compact (sum of two compact sets), hence closed, so $z = \lim_{t \rightarrow 1} z_t$ also belongs to it. Therefore $\overline{B_{r+s}(x)} \subset \overline{B_r(x)} + \overline{B_s(0)}$ and

$$\overline{B_r(x)} + \overline{B_s(0)} = \overline{B_{r+s}(x)}.$$

(b) Assume A is open (the case where B is open is analogous). Let $z \in A + B$, so $z = a + b$ with $a \in A$, $b \in B$. Since A is open, there exists $r > 0$ with $B_r(a) \subset A$. If $\|y - z\| < r$ and we set $a' := a + (y - z)$, then $\|a' - a\| = \|y - z\| < r$, so $a' \in B_r(a) \subset A$, and $y = a' + b \in A + B$. Thus $B_r(z) \subset A + B$ and $A + B$ is open.

(c) Suppose $A \subset B_M(0)$ and $B \subset B_N(0)$ for some $M, N > 0$. If $z = a + b \in A + B$, then

$$\|z\| \leq \|a\| + \|b\| \leq M + N,$$

so $A + B \subset B_{M+N}(0)$ and $A + B$ is bounded.

(d) Assume A is closed and B is compact. Let (z_k) be a sequence in $A + B$ with $z_k \rightarrow z$. Write $z_k = a_k + b_k$ with $a_k \in A$ and $b_k \in B$.

By compactness of B , there is a subsequence (b_{k_j}) such that $b_{k_j} \rightarrow b \in B$. Then

$$a_{k_j} = z_{k_j} - b_{k_j} \longrightarrow z - b.$$

Since A is closed and $a_{k_j} \in A$, we have $z - b \in A$, hence $z = (z - b) + b \in A + B$. Therefore $A + B$ is closed.

(e) If A and B are compact, then $A \times B$ is compact and the map

$$f : A \times B \rightarrow \mathbb{R}^n, \quad f(a, b) = a + b$$

is continuous. Hence $A + B = f(A \times B)$ is compact as the continuous image of a compact set. $\square$

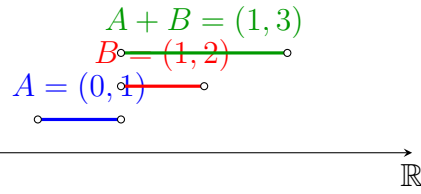


Figure 24: Minkowski sum of intervals on the real line.

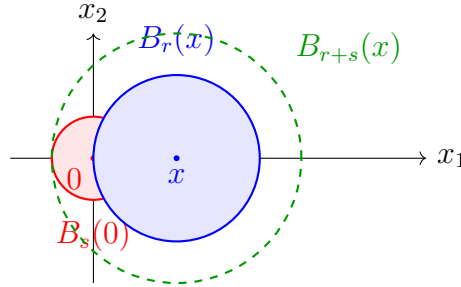
Figure 25: Minkowski sum of balls: $B_r(x) + B_s(0) = B_{r+s}(x)$.

Figure 26 gives a concrete picture for part (e): the Minkowski sum $A + B$ of two compact sets is again compact. Here A is a closed triangle and B is a small closed ball; the sum is just a uniformly thickened version of A .

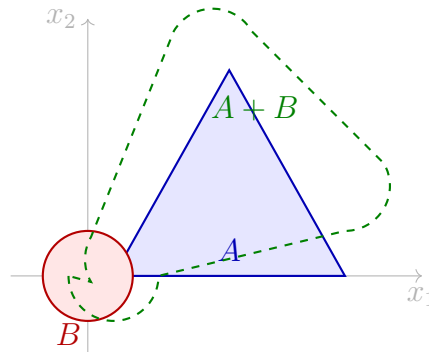


Figure 26: Minkowski sum of a compact set A (triangle) with a small ball B . Geometrically, $A + B$ is obtained by “thickening” A by the radius of B . Since both A and B are compact, their sum $A + B$ is compact as well.

Example (openness of $A + B$). Let $A = B_1(0) = \{x \in \mathbb{R}^2 : \|x\| < 1\}$ and let $b \in \mathbb{R}^2$. Define

$$B = \{0, b\}.$$

Then

$$A + B = \{a + p : a \in A, p \in B\} = A + \{0, b\} = A \cup (A + b).$$

In particular, for the two specific choices

$$b_1 = (2, 0), \quad b_2 = (0, 2),$$

we obtain

$$A + \{0, b_1\} = B_1(0) \cup B_1((2, 0)), \quad A + \{0, b_2\} = B_1(0) \cup B_1((0, 2)).$$

In each case A is open, B is finite, and $A + B$ is the union of two open discs, hence open but not connected.

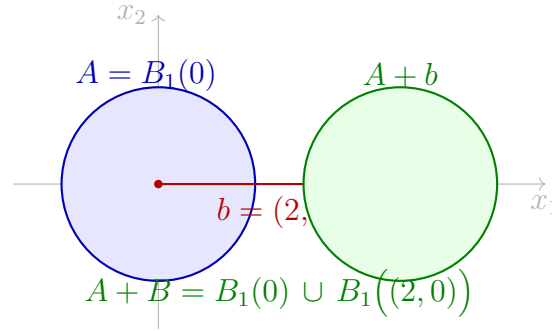


Figure 27: Example for part (b) with $b = (2, 0)$. The set A is the blue open disc, $B = \{0, b\}$ consists of the two red points, and $A + B$ is the union of the two open discs.

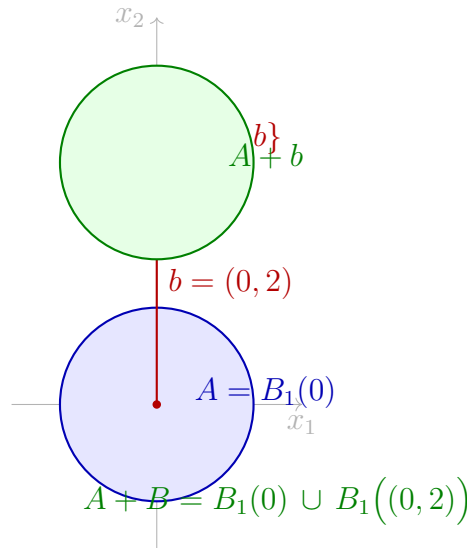


Figure 28: Example for part (b) with $b = (0, 2)$. Again A is the blue disc, $B = \{0, b\}$ is the two-point set, and $A + B$ is the union of two open discs, now vertically separated.

Example (two-point set B). Let

$$A = B_1(0) = \{x \in \mathbb{R}^2 : \|x\| < 1\}$$

and choose two vectors

$$b_1 = (2, 0), \quad b_2 = (0, 2).$$

Define

$$B = \{b_1, b_2\}.$$

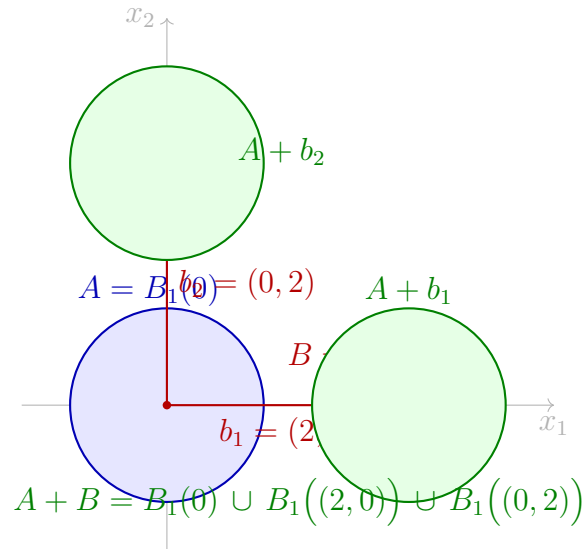


Figure 29: Example for part (b) with $A = B_1(0)$ and $B = \{0, b_1, b_2\}$. The Minkowski sum $A + B$ is the union of three open discs, hence open but not connected.

Then the Minkowski sum is

$$\begin{aligned}
 A + B &= \{a + p : a \in A, p \in B\} = A + \{b_1, b_2\} \\
 &= (A + b_1) \cup (A + b_2) \\
 &= B_1((2, 0)) \cup B_1((0, 2)).
 \end{aligned}$$

Thus $A + B$ is the union of two open discs, hence open but not connected. Note that, in contrast with the case $B = \{0, b_1, b_2\}$, there is now no disc centred at the origin, because $0 \notin B$.

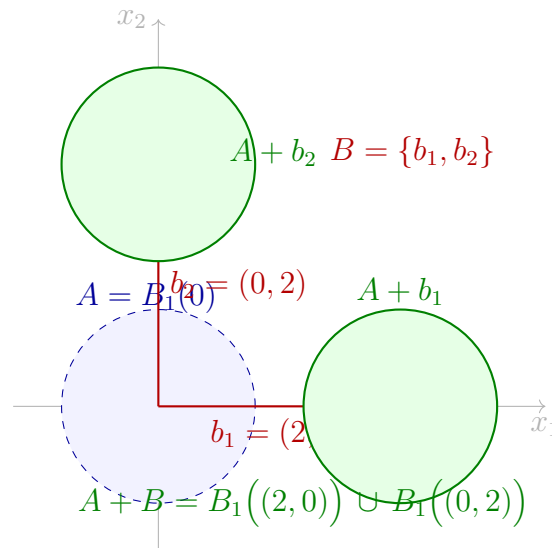


Figure 30: Example with $A = B_1(0)$ and $B = \{b_1, b_2\}$, where $b_1 = (2, 0)$ and $b_2 = (0, 2)$. The Minkowski sum $A + B$ is the union of the two translated discs $A + b_1$ and $A + b_2$, hence open but not connected.

Exercise 2.4 (Convolution preserves smoothness)

Show that if $f \in C_0^k(\mathbb{R}^n)$ and $g \in C_0^\ell(\mathbb{R}^n)$ then $f * g \in C_0^{k+\ell}(\mathbb{R}^n)$. Conclude that $\mathcal{D}(\mathbb{R}^n)$ is closed under convolution.

Solution. Let

$$(f * g)(x) := \int_{\mathbb{R}^n} f(x - y) g(y) dy.$$

Step 1: Compact support. Let $K_f = \text{supp } f$ and $K_g = \text{supp } g$. If $(f * g)(x) \neq 0$, then there exists y such that $f(x - y) \neq 0$ and $g(y) \neq 0$, so $x - y \in K_f$ and $y \in K_g$. Hence

$$x = (x - y) + y \in K_f + K_g := \{a + b : a \in K_f, b \in K_g\}.$$

Thus

$$\text{supp}(f * g) \subset K_f + K_g.$$

Since K_f and K_g are compact, their Minkowski sum $K_f + K_g$ is compact. Therefore $f * g$ has compact support.

Step 2: Differentiation under the integral. Let α be a multiindex. Since f has compact support, the integrand in

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy$$

vanishes outside a fixed compact set, so we may differentiate under the integral:

$$\partial^\alpha (f * g)(x) = \int_{\mathbb{R}^n} \partial^\alpha f(x - y) g(y) dy = (\partial^\alpha f) * g.$$

Similarly,

$$\partial^\alpha (f * g) = f * (\partial^\alpha g).$$

Both expressions are continuous, since convolution of continuous compactly supported functions is continuous.

Step 3: Smoothness of order $k + \ell$. Let α be a multiindex with $|\alpha| \leq k + \ell$. Choose multiindices β, γ with $\alpha = \beta + \gamma$, $|\beta| \leq k$, $|\gamma| \leq \ell$. Then:

$$\partial^\alpha (f * g) = \partial^\beta \partial^\gamma (f * g) = \partial^\beta (f * (\partial^\gamma g)) = (\partial^\beta f) * (\partial^\gamma g).$$

Here $\partial^\beta f$ and $\partial^\gamma g$ are continuous and compactly supported, so their convolution is continuous. Thus $f * g \in C_0^{k+\ell}(\mathbb{R}^n)$.

Step 4: Closure of $\mathcal{D}(\mathbb{R}^n)$ under convolution. Recall:

$$\mathcal{D}(\mathbb{R}^n) = C_0^\infty(\mathbb{R}^n) = \bigcap_{m=0}^{\infty} C_0^m(\mathbb{R}^n).$$

If $f, g \in \mathcal{D}(\mathbb{R}^n)$, then for each m we have $f, g \in C_0^m$. Applying the previous result with $k = \ell = m$ yields $f * g \in C_0^{2m}$ for every m , so $f * g \in C_0^r$ for all r . Hence

$$f * g \in C_0^\infty(\mathbb{R}^n) = \mathcal{D}(\mathbb{R}^n),$$

and $\mathcal{D}(\mathbb{R}^n)$ is closed under convolution.

Example: hat functions in $\mathbb{R}$

Take

$$f(x) = \max(1 - |x|, 0), \quad g(x) = \max(1 - |x|, 0).$$

- Both f and g are continuous, piecewise linear.
- They vanish outside the interval $[-1, 1]$.
- Hence $f, g \in C_0^0(\mathbb{R})$ (continuous with compact support, $k = 0$).

Their convolution is

$$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y) dy,$$

which is a continuous “rounded tent” supported in the interval $[-2, 2]$.

- If $|x| > 2$, then the supports of $y \mapsto f(x - y)$ and $y \mapsto g(y)$ do not overlap, so $(f * g)(x) = 0$.
- If $|x| \leq 2$, the integrand is nonzero on an interval in y , giving $(f * g)(x) > 0$.

Thus

$$\text{supp } f = [-1, 1], \quad \text{supp } g = [-1, 1], \quad \text{supp}(f * g) \subset [-1, 1] + [-1, 1] = [-2, 2],$$

and $f * g$ is “smoother” than f and g (it has no sharp corner at 0).

For higher regularity one can replace the hat by a standard smooth bump

$$\varphi(x) = \begin{cases} e^{-1/(1-x^2)}, & |x| < 1, \\ 0, & |x| \geq 1, \end{cases}$$

and take $f = \varphi, g = \varphi$. Then $f, g \in C_0^\infty(\mathbb{R})$, and the result of Exercise 2.4 says that

$$f * g \in C_0^\infty(\mathbb{R}) \quad \text{with} \quad \text{supp}(f * g) \subset [-2, 2].$$

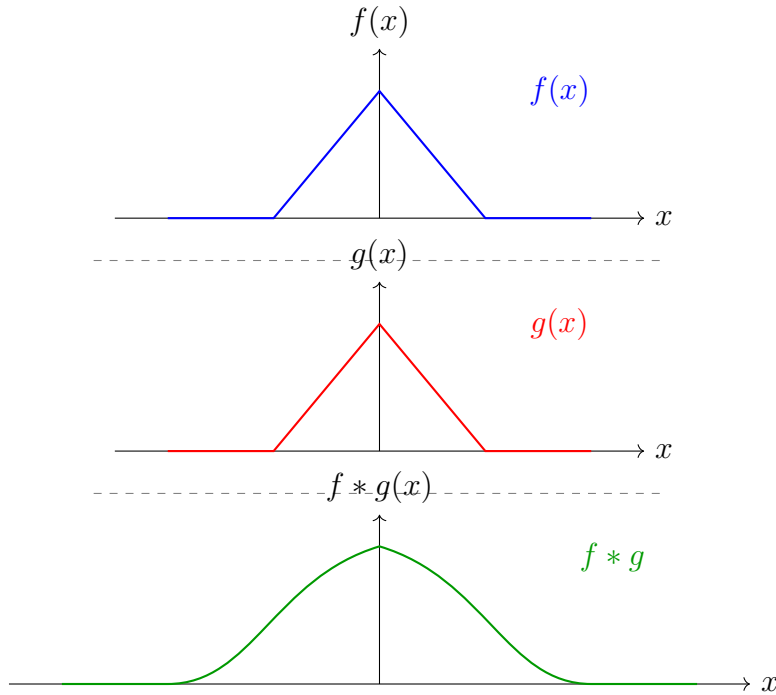


Figure 31: Example of convolution of compactly supported functions on $\mathbb{R}$. Two hat functions f, g with support in $[-1, 1]$ produce a wider, smoother convolution $f * g$ supported in $[-2, 2]$.

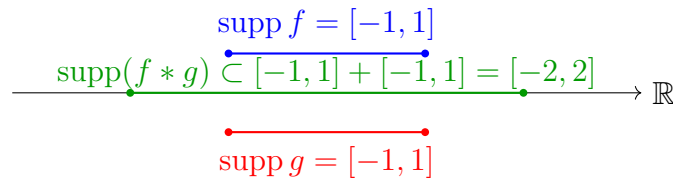


Figure 32: Support of the convolution $f * g$ is contained in the Minkowski sum of the supports of f and g .

(i) Supports of f and g

Consider the hat functions

$$f(x) = \max(1 - |x|, 0), \quad g(x) = \max(1 - |x|, 0).$$

Both vanish outside the interval $[-1, 1]$, hence

$$\text{supp } f = \text{supp } g = [-1, 1].$$

(ii) Support of the convolution $f * g$

The convolution is

$$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y) dy.$$

The integrand is nonzero only when

$$x - y \in [-1, 1] \quad \text{and} \quad y \in [-1, 1],$$

which implies

$$x \in [-1, 1] + [-1, 1] = [-2, 2].$$

Thus

$$\text{supp}(f * g) \subset [-2, 2].$$

(iii) Shape and smoothness of $f * g$

The hat functions f and g are continuous but not differentiable at $x = 0$. Their convolution $f * g$ is smoother: it is continuous, piecewise quadratic, and has no corner at $x = 0$. Convolution therefore has a regularizing effect even when the input functions are only continuous.

(iv) A C^∞ example

Let

$$\varphi(x) = \begin{cases} e^{-1/(1-x^2)}, & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

Then $\varphi \in C_0^\infty(\mathbb{R})$. Setting $f = \varphi$, $g = \varphi$ gives

$$f, g \in C_0^\infty(\mathbb{R}) \implies f * g \in C_0^\infty(\mathbb{R}),$$

and

$$\text{supp}(f * g) \subset [-2, 2].$$

(v) Interpretation via Minkowski sums

For compactly supported functions f and g one always has

$$\text{supp}(f * g) \subset \text{supp } f + \text{supp } g.$$

In the present case,

$$[-1, 1] + [-1, 1] = [-2, 2],$$

which geometrically explains the growth of the support under convolution.

The Meaning and Applications of Convolution

Given two functions f and g on $\mathbb{R}^n$, their convolution is

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy.$$

Conceptually, we *flip* one function, *slide* it across the other, and measure their weighted overlap. Convolution describes how two shapes interact under translation. Below we present the main interpretations.

1. Convolution as overlapping shapes

Convolution measures how much of the shape of f overlaps with the shape of g when g is shifted by x . If the overlap is strong, $(f * g)(x)$ is large; if the functions barely overlap, $(f * g)(x)$ is small; if there is no overlap, the convolution is zero. Thus $(f * g)(x)$ describes the overlap as one function slides across the other.

2. Convolution as smoothing or averaging

If g is smooth, then $f * g$ is smoother than f . Indeed, for each x ,

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy = \text{a weighted average of } f \text{ near } x.$$

Thus convolution removes noise and irregularities in f . This is the basis of smoothing kernels, mollifiers, and the regularizing effects of the heat equation.

3. Convolution as propagation of influence (PDE)

If g is a Green's function, then convolution represents the solution of a partial differential equation. For instance,

$$u = f * G$$

solves the equation $Lu = f$ for a differential operator L . Examples: heat kernels solve the heat equation; Poisson kernels give harmonic extensions; the fundamental solution of Laplace gives Newtonian potentials. In this sense convolution spreads or propagates the influence of f through the medium described by g .

4. Convolution in probability

If X and Y are independent random variables with densities f and g , then the density of $X + Y$ is the convolution $f * g$. Thus convolution describes the distribution of sums of random variables. In particular, adding many independent random variables produces repeated convolutions, leading toward Gaussian limits (the Central Limit Theorem).

5. Convolution in signal processing and Fourier analysis

Convolution corresponds to filtering. If g is a “filter” (for instance a low-pass, high-pass, or band-pass filter), then $f * g$ is the input signal f passed through that filter. In the Fourier domain,

$$\mathcal{F}(f * g) = \widehat{f} \widehat{g},$$

so convolution in physical space becomes multiplication in frequency space.

Meaningful examples of convolution

Example 1: Averaging window. Let

$$g(t) = \frac{1}{2h} 1_{[-h, h]}(t).$$

Then

$$(f * g)(x) = \frac{1}{2h} \int_{x-h}^{x+h} f(t) dt,$$

which is just the *average* of f around x over an interval of length $2h$.

$\Rightarrow$ *Smoothing*: local averaging of the signal.

Example 2: Gaussian smoothing. Let

$$g_\varepsilon(y) = \frac{1}{\sqrt{\pi} \varepsilon} e^{-y^2/\varepsilon^2}.$$

Then $f * g_\varepsilon$ is the *Gaussian-blurred* version of f .

$\Rightarrow$ *Effect*: removes noise, increases differentiability.

Example 3: Derivative filters. If $g = f'$, then formally

$$f * f' = \frac{d}{dx} (f * f).$$

More generally, for a multiindex α ,

$$\partial^\alpha (f * g) = (\partial^\alpha f) * g = f * (\partial^\alpha g).$$

$\Rightarrow$ *Effect*: convolution propagates derivatives and can act as a differentiating filter.

Example 4: Probability (sum of random variables). Let f be the density of a random variable X and g the density of an independent random variable Y . Then the density of $X + Y$ is $f * g$:

$$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y) dy.$$

$\Rightarrow$ *Effect*: convolution describes the law of sums of independent random variables.

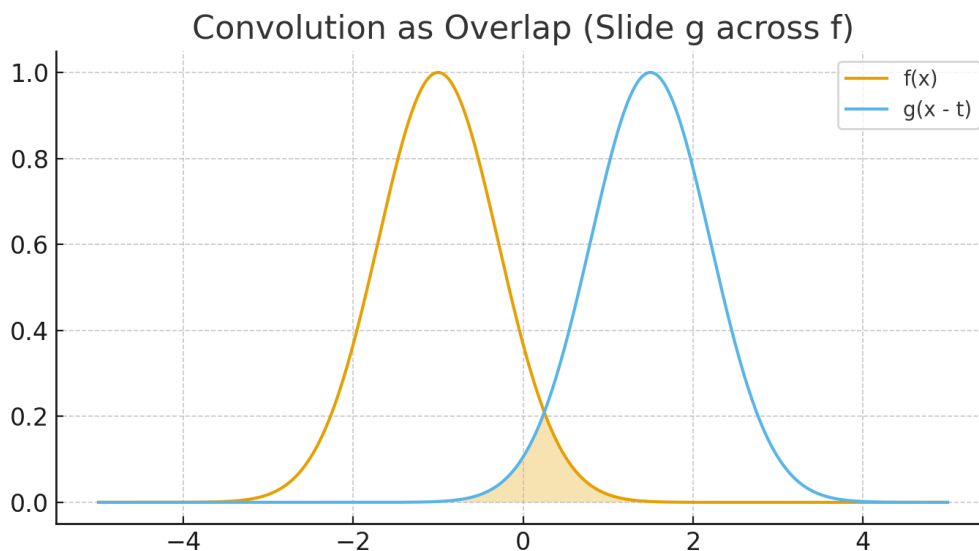


Figure 33: Convolution as overlap: the function g is shifted horizontally and its overlap with f determines the value of $(f * g)(t)$.

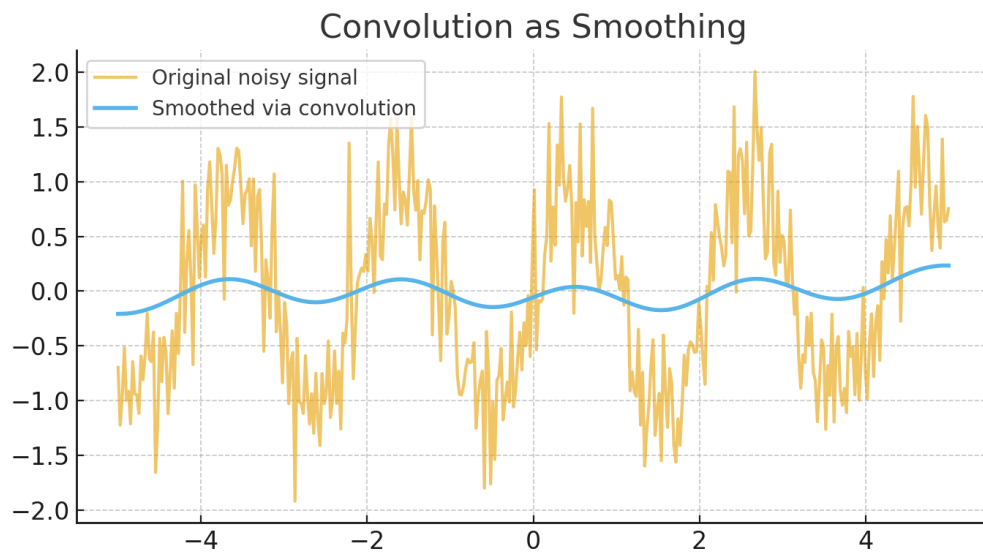


Figure 34: Convolution as smoothing: a noisy signal (blue) is convolved with a Gaussian kernel, producing a smoother function (orange).

Exercise 2.5 (Young, Hölder and Minkowski inequalities)

Suppose that $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is a positive simple function.

- (a) Show that Minkowski's integral inequality holds in the case $p = 1$:

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx \leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx.$$

- (b) Prove Young's inequality: if $a, b \geq 0$ and $p, q > 1$ satisfy $p^{-1} + q^{-1} = 1$, then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Hint: Set $t = p^{-1}$, consider the function $\log[ta^p + (1-t)b^q]$ and use the concavity of $\log$.

- (c) With $p, q > 1$ such that $p^{-1} + q^{-1} = 1$, show that if $\|f\|_{L^p} = 1$ and $\|g\|_{L^q} = 1$ then

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq 1.$$

Deduce Hölder's inequality

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \|f\|_{L^p} \|g\|_{L^q}, \quad f \in L^p(\mathbb{R}^n), \quad g \in L^q(\mathbb{R}^n).$$

- (d) Set

$$G(y) := \left(\int_{\mathbb{R}^n} F(x, y) dx \right)^{p-1}.$$

- i) Show that if $q = \frac{p}{p-1}$, then

$$\|G\|_{L^q(\mathbb{R}^n)} = \left\| \int_{\mathbb{R}^n} F(x, \cdot) dx \right\|_{L^p(\mathbb{R}^n)}^{p-1}.$$

- ii) Show that

$$\left\| \int_{\mathbb{R}^n} F(x, \cdot) dx \right\|_{L^p(\mathbb{R}^n)}^p = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} G(y) F(x, y) dy \right) dx.$$

- iii) Applying Hölder's inequality, deduce a Minkowski-type inequality for F .

- (e) Deduce Minkowski's integral inequality:

$$\left(\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right|^p dx \right)^{1/p} \leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x, y)|^p dx \right)^{1/p} dy,$$

which holds for any measurable function $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}$ and all $1 \leq p < \infty$.

Deep Theory / Roadmap

This exercise reconstructs the chain of ideas leading from convexity to Young's inequality, Hölder's inequality, and finally to Minkowski's integral inequality. We outline the conceptual structure.

Part (a): Minkowski's inequality for $p = 1$

Suppose $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is a positive simple function. We want to show

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx \leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx.$$

Solution. Since $F(x, y) \geq 0$ for all (x, y) , we have for each fixed x :

$$\int_{\mathbb{R}^n} F(x, y) dy \geq 0 \implies \left| \int_{\mathbb{R}^n} F(x, y) dy \right| = \int_{\mathbb{R}^n} F(x, y) dy.$$

Therefore

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x, y) dy dx.$$

On the other hand $F \geq 0$ implies $|F| = F$, hence

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x, y) dy dx.$$

Thus the two sides are equal, and in particular the desired inequality holds.

Example. Let

$$F(x, y) = 1_{[0,1]^n}(x) 1_{[0,1]^n}(y).$$

Then

$$\int_{\mathbb{R}^n} F(x, y) dy = 1_{[0,1]^n}(x),$$

so

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx = \int_{[0,1]^n} 1 dx = 1.$$

Moreover $|F| = F$, and

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx = \int_{[0,1]^n} \int_{[0,1]^n} 1 dy dx = 1.$$

This concrete case illustrates that for $p = 1$ and $F \geq 0$ the inequality is actually an equality.

Additional examples for part (a)

The inequality

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx \leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx$$

holds with equality whenever $F \geq 0$. Here are three regular examples.

Example 1: Gaussian. Let $F(x, y) = e^{-x^2-y^2}$ on $\mathbb{R}$. Then

$$\int_{\mathbb{R}} F(x, y) dy = \sqrt{\pi} e^{-x^2}.$$

Hence

$$\int_{\mathbb{R}} \left| \int_{\mathbb{R}} F(x, y) dy \right| dx = \sqrt{\pi} \int_{\mathbb{R}} e^{-x^2} dx = \pi.$$

Since $F \geq 0$,

$$\int_{\mathbb{R}} \int_{\mathbb{R}} |F(x, y)| dy dx = \pi.$$

Example 2: Polynomial times Gaussian. Let

$$F(x, y) = (1 + x^2)(1 + y^2)e^{-x^2-y^2}.$$

A direct computation gives

$$\int_{\mathbb{R}} (1 + t^2)e^{-t^2} dt = \frac{3\sqrt{\pi}}{2}.$$

Thus

$$\int_{\mathbb{R}} \left| \int_{\mathbb{R}} F(x, y) dy \right| dx = \left(\frac{3\sqrt{\pi}}{2} \right)^2 = \frac{9\pi}{4},$$

and the same value is obtained on the right-hand side.

Example 3: Trigonometric example. On $[0, \pi]$ let

$$F(x, y) = \sin^2 x \sin^2 y.$$

Since $\int_0^\pi \sin^2 t dt = \frac{\pi}{2}$,

$$\int_0^\pi \left| \int_0^\pi F(x, y) dy \right| dx = \frac{\pi^2}{4} = \int_0^\pi \int_0^\pi F(x, y) dy dx.$$

These examples illustrate how regular, smooth functions behave exactly like simple functions when $p = 1$ and $F \geq 0$: the inequality becomes an equality.

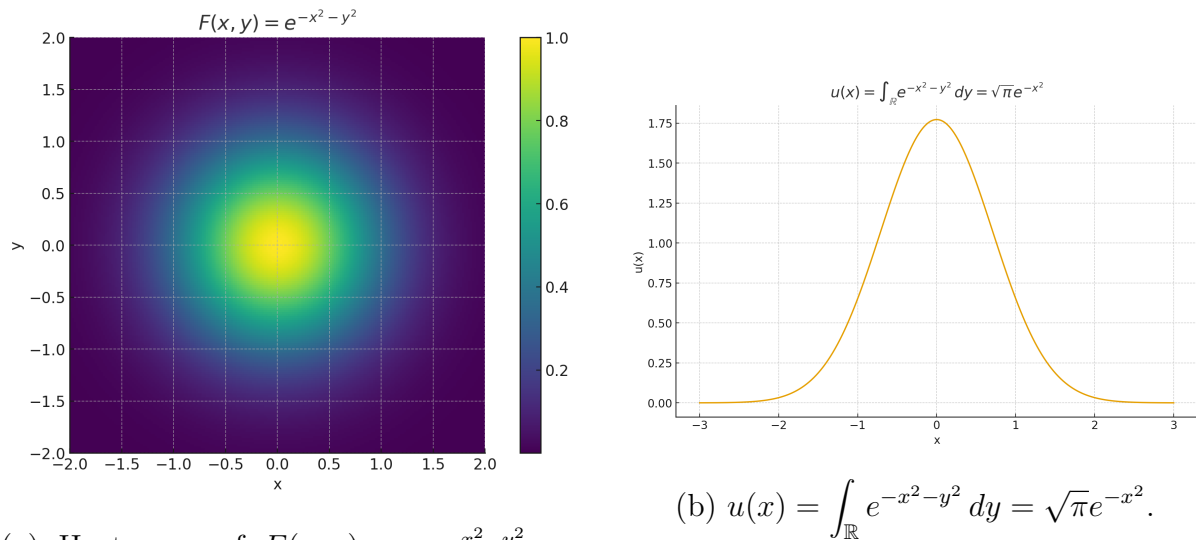


Figure 35: Gaussian example for Minkowski's integral inequality with $p = 1$: the area under $u(x)$ equals the double integral of $F(x, y)$.

How simple functions and regular functions are related

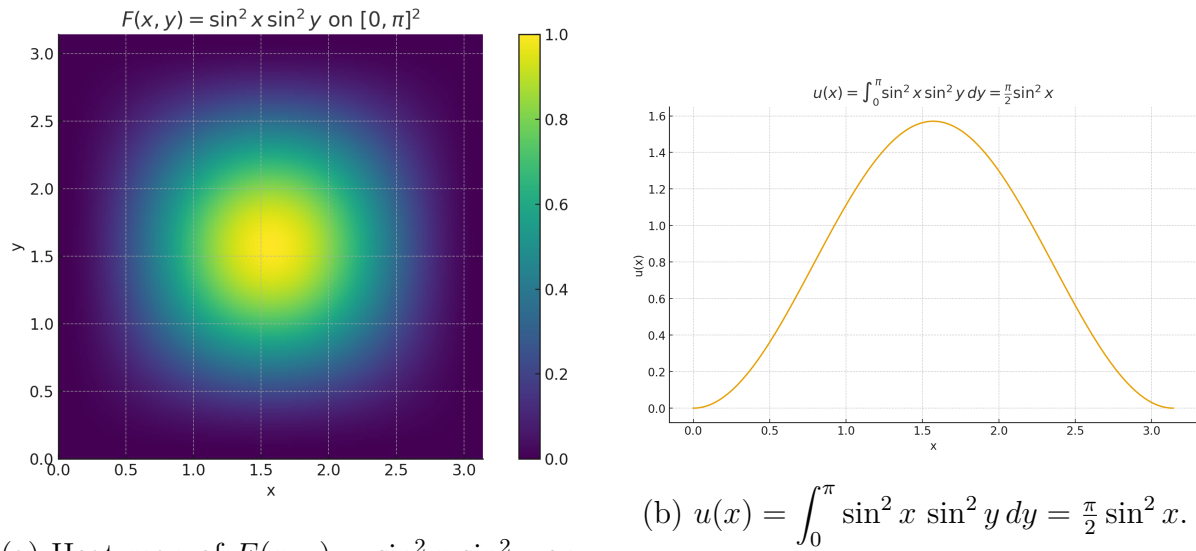
The Minkowski inequality (and many foundational results in integration theory) is first proved for *simple functions*, because these have a completely transparent structure:

- simple functions are finite linear combinations of indicator sets;
- all integrals reduce to finite sums;
- Tonelli–Fubini can be verified directly;
- no pathological behaviour can occur.

Once the inequality is established for simple functions, one extends it to general measurable functions (continuous, C^∞ , polynomials, Gaussians, trigonometric functions, etc.) by:

- approximation from below by increasing sequences of simple functions;
- the Monotone or Dominated Convergence Theorem;
- density of simple functions in L^p .

We now describe this relation step by step.



(a) Heat map of $F(x, y) = \sin^2 x \sin^2 y$ on $[0, \pi]^2$.

Figure 36: Trigonometric example for Minkowski's inequality with $p = 1$: the integral of $u(x)$ over $[0, \pi]$ equals the double integral of $F(x, y)$.

1. Simple functions as the foundational case

A simple function has the form

$$F(x, y) = \sum_{k=1}^m a_k 1_{A_k}(x) 1_{B_k}(y), \quad a_k \geq 0.$$

For such F , all integrals become finite sums:

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x, y) \, dy \, dx = \sum_{k=1}^m a_k |A_k| |B_k|.$$

Thus Minkowski's inequality for $p = 1$ is an equality for simple functions.

2. Approximation of general nonnegative measurable functions

Every nonnegative measurable function F on $\mathbb{R}^n \times \mathbb{R}^n$ admits an increasing sequence of simple functions F_n such that

$$F_n(x, y) \uparrow F(x, y) \quad \text{for all } (x, y).$$

This construction is canonical and works for every $F \geq 0$.

3. The Monotone Convergence Theorem connects the two

By the Monotone Convergence Theorem,

$$\int F_n \rightarrow \int F, \quad \int \left| \int F_n(\cdot, y) \, dy \right| \rightarrow \int \left| \int F(\cdot, y) \, dy \right|.$$

Since Minkowski holds exactly for each F_n , taking limits shows it holds for F as well.

4. Sign-changing regular functions

If F is a smooth or regular function that takes both signs, use the triangle inequality:

$$\left| \int_{\mathbb{R}^n} F(x, y) dy \right| \leq \int_{\mathbb{R}^n} |F(x, y)| dy.$$

Thus the sign-changing case reduces to the nonnegative case by replacing F with $|F|$.

5. Regular functions (Gaussians, trigonometric, polynomials)

Regular functions such as

$$e^{-x^2-y^2}, \quad (1+x^2)(1+y^2)e^{-x^2-y^2}, \quad \sin^2 x \sin^2 y,$$

lie in every L^p , and satisfy

$$F_n \uparrow F \quad \text{or} \quad F_n \rightarrow F \text{ in } L^p.$$

Therefore all results established for simple functions extend to these regular functions by approximation.

Summary. Simple functions form the atomic building blocks. Every regular function is a limit of simple ones. Minkowski's inequality is verified first on the atoms, and then passed to the limit using monotone or dominated convergence.

(i) L^p -spaces and Hölder conjugates.

- For $1 \leq p < \infty$, the space $L^p(\mathbb{R}^n)$ consists of all measurable functions f such that

$$\int_{\mathbb{R}^n} |f(x)|^p dx < \infty.$$

- The Hölder conjugate exponent q of p is defined by

$$\frac{1}{p} + \frac{1}{q} = 1, \quad 1 < p < \infty.$$

- The pair (p, q) is fundamental: these are exactly the exponents for which Young's and Hölder's inequalities hold and for which the duality $(L^p)^* \cong L^q$ is valid.

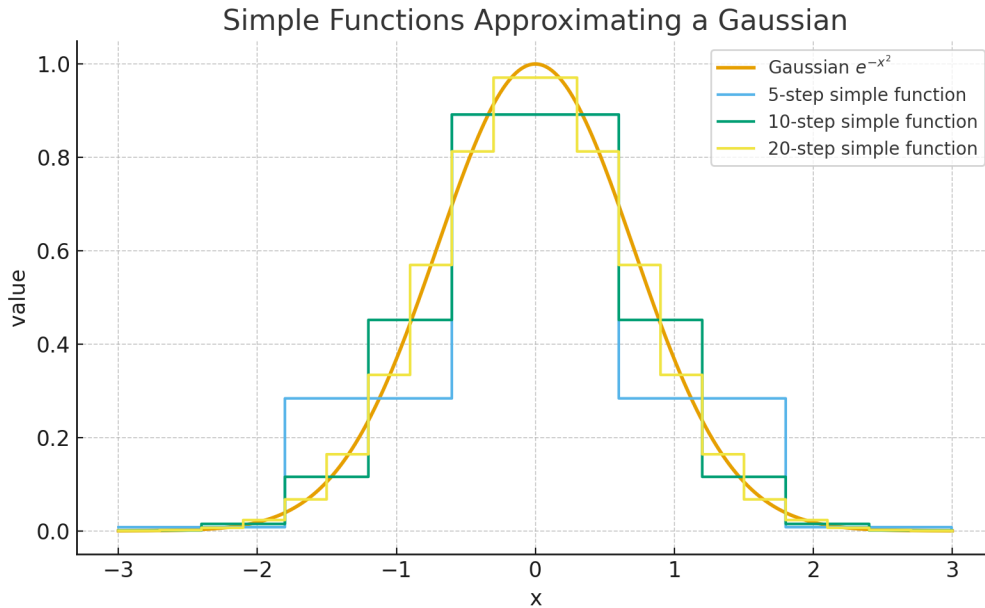


Figure 37: Approximation of the Gaussian e^{-x^2} by simple functions. The step functions with 5, 10, and 20 intervals converge monotonically to the Gaussian, illustrating how general integrable functions are obtained as limits of simple functions.

(ii) **Young's inequality as convex duality.**

- The function $\phi(t) = e^t$ is convex and the logarithm is concave. The hint in part (b) suggests proving Young's inequality using concavity of $\log$.
- The inequality

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

is equivalent to the statement

$$\log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right).$$

Because $\log$ is concave, this inequality follows from Jensen's inequality applied to $\log$.

- More conceptually: the functions $x \mapsto x^p/p$ and $y \mapsto y^q/q$ are Legendre transforms of each other. Young's inequality expresses this duality.

Young's inequality and concavity of $\log$.

- The inequality

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

is equivalent (since $\log$ is increasing) to the statement

$$\log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right).$$

- Set $t = p^{-1}$ and consider the function

$$\varphi(t) = \log[ta^p + (1-t)b^q].$$

Because $\log$ is concave, Jensen's inequality yields

$$\log[ta^p + (1-t)b^q] \geq t \log(a^p) + (1-t) \log(b^q) = p t \log a + q(1-t) \log b.$$

Choosing $t = 1/p$ (so $1-t = 1/q$) gives

$$\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \log a + \log b = \log(ab),$$

which is precisely Young's inequality.

Legendre-transform viewpoint. Let

$$\Phi(x) = \frac{x^p}{p}, \quad x \geq 0.$$

Then Φ is convex and its Legendre transform (convex conjugate) is

$$\Phi^*(y) = \sup_{x \geq 0} (xy - \Phi(x)) = \frac{y^q}{q}, \quad y \geq 0,$$

where q is the Hölder conjugate of p , i.e. $\frac{1}{p} + \frac{1}{q} = 1$. The Fenchel–Young inequality asserts that

$$xy \leq \Phi(x) + \Phi^*(y), \quad x, y \geq 0.$$

Substituting $\Phi(x) = x^p/p$ and $\Phi^*(y) = y^q/q$ yields

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q},$$

which is Young's inequality (with $x = a$, $y = b$). Thus Young's inequality expresses the Legendre duality between the functions $x \mapsto x^p/p$ and $y \mapsto y^q/q$.

(iii) Hölder's inequality.

- Using Young's inequality, apply it pointwise to $|f(x)|$ and $|g(x)|$:

$$|f(x)g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q}.$$

- Integrating over $\mathbb{R}^n$ gives

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \frac{1}{p} \int |f|^p + \frac{1}{q} \int |g|^q.$$

- If $\|f\|_{L^p} = \|g\|_{L^q} = 1$, the right side equals 1.
- Scaling f and g yields the general form of Hölder's inequality:

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \|f\|_{L^p} \|g\|_{L^q}.$$

- This identity shows that multiplication by $g \in L^q$ defines a bounded linear functional on L^p with operator norm $\|g\|_{L^q}$ — the starting point of L^p duality.

(iv) Minkowski's inequality and L^p -norms.

- The classical Minkowski inequality

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$$

is the triangle inequality for the L^p -norm. Hölder's inequality is the main tool in its proof.

- In this exercise we prove a *two-variable* version:

$$\left\| \int_{\mathbb{R}^n} F(\cdot, y) dy \right\|_{L^p(\mathbb{R}^n)} \leq \int_{\mathbb{R}^n} \|F(\cdot, y)\|_{L^p(\mathbb{R}^n)} dy.$$

- The key idea is to define

$$u(x) := \int_{\mathbb{R}^n} F(x, y) dy, \quad G(y) := |u(y)|^{p-1} \operatorname{sgn}(u(y)),$$

so that

$$\|u\|_{L^p}^p = \int_{\mathbb{R}^n} u(x)G(x) dx.$$

- Applying Hölder's inequality to the right-hand side gives

$$\|u\|_{L^p}^p \leq \|G\|_{L^q} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x, y)|^p dx \right)^{1/p} dy,$$

from which Minkowski's integral inequality follows.

(v) Simple functions and general measurable functions.

- Parts (a)–(d) assume F is a *positive simple function*: a finite linear combination of characteristic functions. On these, one may freely permute and decompose integrals.
- For general nonnegative measurable F , approximate F from below by simple functions and use the Monotone Convergence Theorem.
- For complex-valued F , apply the inequality to $|F|$.
- This yields Minkowski's integral inequality in full generality for all $1 \leq p < \infty$.

From Fenchel–Young to Hölder's inequality. The Fenchel–Young inequality for the convex function $\Phi(x) = x^p/p$ and its conjugate $\Phi^*(y) = y^q/q$ reads

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}, \quad x, y \geq 0.$$

Applying this pointwise with $x = |f(x)|$ and $y = |g(x)|$ gives

$$|f(x)g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q}, \quad x \in \mathbb{R}^n.$$

Integrating over $\mathbb{R}^n$ yields

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \frac{1}{p} \int_{\mathbb{R}^n} |f(x)|^p dx + \frac{1}{q} \int_{\mathbb{R}^n} |g(x)|^q dx = \frac{\|f\|_{L^p}^p}{p} + \frac{\|g\|_{L^q}^q}{q}.$$

If $\|f\|_{L^p}, \|g\|_{L^q} > 0$, apply this to $\tilde{f} = f/\|f\|_{L^p}$ and $\tilde{g} = g/\|g\|_{L^q}$ to obtain

$$\int_{\mathbb{R}^n} |\tilde{f}(x)\tilde{g}(x)| dx \leq \frac{1}{p} + \frac{1}{q} = 1,$$

and hence

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \|f\|_{L^p} \|g\|_{L^q}.$$

This recovers Hölder's inequality directly from the Fenchel–Young inequality for $\Phi(x) = x^p/p$.

Examples for Exercise 2.5

(i) Example for Young's inequality

Take $p = q = 2$ and $a = 3, b = 4$. Young's inequality states that

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Here

$$ab = 3 \cdot 4 = 12,$$

while

$$\frac{a^2}{2} + \frac{b^2}{2} = \frac{9}{2} + \frac{16}{2} = \frac{25}{2} = 12.5.$$

Thus

$$12 = ab \leq 12.5 = \frac{a^2}{2} + \frac{b^2}{2},$$

which illustrates the inequality for a specific choice of parameters.

(ii) Example for Hölder's inequality

Let $p = q = 2$ and define functions on $\mathbb{R}$ by

$$f(x) = 1_{[0,1]}(x), \quad g(x) = x 1_{[0,1]}(x).$$

Then

$$f(x)g(x) = x 1_{[0,1]}(x),$$

and

$$\int_{\mathbb{R}} |f(x)g(x)| dx = \int_0^1 x dx = \frac{1}{2}.$$

The norms are

$$\|f\|_{L^2}^2 = \int_0^1 1^2 dx = 1 \implies \|f\|_{L^2} = 1,$$

and

$$\|g\|_{L^2}^2 = \int_0^1 x^2 dx = \frac{1}{3} \implies \|g\|_{L^2} = \frac{1}{\sqrt{3}}.$$

Thus

$$\int_{\mathbb{R}} |f(x)g(x)| dx = \frac{1}{2} \leq \frac{1}{\sqrt{3}} = \|f\|_{L^2} \|g\|_{L^2},$$

which confirms Hölder's inequality in this concrete case (with strict inequality).

(iii) Example for Minkowski's integral inequality

Let $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$F(x, y) = 1_{[0,1]}(x) 1_{[0,1]}(y),$$

and take $p = 2$. Define

$$u(x) := \int_{\mathbb{R}} F(x, y) dy.$$

Then

$$u(x) = \int_{\mathbb{R}} 1_{[0,1]}(x) 1_{[0,1]}(y) dy = 1_{[0,1]}(x) \cdot 1,$$

so $u(x) = 1$ for $x \in [0, 1]$ and $u(x) = 0$ otherwise. Hence

$$\left(\int_{\mathbb{R}} |u(x)|^2 dx \right)^{1/2} = \left(\int_0^1 1 dx \right)^{1/2} = 1.$$

On the other hand,

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} |F(x, y)|^2 dx \right)^{1/2} dy$$

is computed as follows. Since $F^2 = F$,

$$\int_{\mathbb{R}} |F(x, y)|^2 dx = \int_{\mathbb{R}} 1_{[0,1]}(x) 1_{[0,1]}(y) dx = 1_{[0,1]}(y) \cdot 1.$$

Thus

$$\left(\int_{\mathbb{R}} |F(x, y)|^2 dx \right)^{1/2} = 1_{[0,1]}(y),$$

and therefore

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} |F(x, y)|^2 dx \right)^{1/2} dy = \int_0^1 1 dy = 1.$$

In this example both sides of Minkowski's integral inequality are equal to 1, so equality holds.

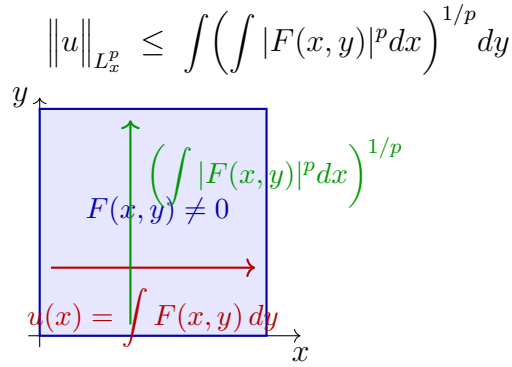


Figure 38: Geometric view of Minkowski's integral inequality. Horizontal integration in y gives $u(x) = \int F(x, y) dy$, whose L^p -norm appears on the left-hand side. For each fixed y , the vertical arrow represents the L^p -norm in x of the slice $F(\cdot, y)$; these norms are then integrated over y on the right-hand side.

Exercise 2.6. Distributions and their orders

- (a) Show that δ_x , as defined in Example 7, is continuous and linear, hence a distribution. Find its order.
- (b) Show that T_f , as defined in Example 7, is continuous and linear, hence a distribution. Find its order.
- (c) By constructing a suitable sequence of smooth functions show that the order of $\text{P.V.}\left(\frac{1}{x}\right)$ is one.

(a) **The Dirac mass δ_x .** For fixed $x \in \mathbb{R}^n$ the functional δ_x is defined by

$$\delta_x(\varphi) := \varphi(x), \quad \varphi \in \mathcal{D}(\mathbb{R}^n).$$

It is clearly linear:

$$\delta_x(\alpha\varphi + \beta\psi) = (\alpha\varphi + \beta\psi)(x) = \alpha\varphi(x) + \beta\psi(x) = \alpha\delta_x(\varphi) + \beta\delta_x(\psi).$$

To show continuity and determine its order, let $K \subset \mathbb{R}^n$ be compact and suppose $\text{supp } \varphi \subset K$. If $x \notin K$ then $\delta_x(\varphi) = 0$. If $x \in K$, then

$$|\delta_x(\varphi)| = |\varphi(x)| \leq \sup_{y \in K} |\varphi(y)|.$$

Thus for each compact K we can choose $C_K = 1$ and obtain

$$|\delta_x(\varphi)| \leq C_K \sup_K |\varphi|.$$

Hence δ_x is a distribution of order 0.

(b) The regular distribution T_f . Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and define

$$T_f(\varphi) := \int_{\mathbb{R}^n} f(x) \varphi(x) dx.$$

Linearity is immediate from linearity of the integral. For continuity, fix a compact set $K \subset \mathbb{R}^n$ and suppose $\text{supp } \varphi \subset K$. Then

$$|T_f(\varphi)| = \left| \int_K f(x) \varphi(x) dx \right| \leq \int_K |f(x)| |\varphi(x)| dx \leq \left(\int_K |f(x)| dx \right) \sup_K |\varphi(x)|.$$

Setting $C_K := \int_K |f(x)| dx < \infty$, we obtain

$$|T_f(\varphi)| \leq C_K \sup_K |\varphi|,$$

so T_f is a distribution of order 0.

(c) The order of P.V.(1/x). The principal value distribution P.V.(1/x) acts on $\varphi \in \mathcal{D}(\mathbb{R})$ by

$$\langle \text{P.V.}(1/x), \varphi \rangle := \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

Step 1: P.V.(1/x) has order at most 1. One shows (by integration by parts on the truncated integrals and passing to the limit) that

$$\langle \text{P.V.}(1/x), \varphi \rangle = - \int_{\mathbb{R}} \log |x| \varphi'(x) dx.$$

Let $K \subset \mathbb{R}$ be compact and $\text{supp } \varphi \subset K$. Then

$$\left| \langle \text{P.V.}(1/x), \varphi \rangle \right| \leq \int_K |\log |x|| |\varphi'(x)| dx \leq \left(\int_K |\log |x|| dx \right) \sup_K |\varphi'(x)|.$$

Thus there is a constant C_K such that

$$\left| \langle \text{P.V.}(1/x), \varphi \rangle \right| \leq C_K \sup_K |\varphi'|,$$

which means that P.V.(1/x) is a distribution of order ≤ 1 .

Step 2: P.V.(1/x) is not of order 0. Suppose, towards a contradiction, that it had order 0. Then for $K = [-1, 1]$ there would exist $C > 0$ such that for all $\varphi \in \mathcal{D}(\mathbb{R})$ with $\text{supp } \varphi \subset [-1, 1]$,

$$\left| \langle \text{P.V.}(1/x), \varphi \rangle \right| \leq C \sup_{[-1, 1]} |\varphi|. \quad (*)$$

Choose $\eta \in C_c^\infty(\mathbb{R})$ with $0 \leq \eta \leq 1$, $\eta(x) = 0$ for $x \leq 0$ and $\eta(x) = 1$ for $x \geq 1$, and let $\chi \in C_c^\infty((-1, 1))$ satisfy $\chi \equiv 1$ on $[-1/2, 1/2]$. Define

$$\varphi_n(x) := \eta(nx) \chi(x).$$

Then $\varphi_n \in C_c^\infty((-1, 1))$, $0 \leq \varphi_n \leq 1$ and $\sup |\varphi_n| = 1$ for all n . Moreover, for $x \in [1/n, 1/2]$ we have $\varphi_n(x) = 1$.

For these test functions the principal value reduces to an improper integral:

$$\langle \text{P.V.}(1/x), \varphi_n \rangle = \int_0^1 \frac{\varphi_n(x)}{x} dx.$$

Hence

$$\langle \text{P.V.}(1/x), \varphi_n \rangle \geq \int_{1/n}^{1/2} \frac{1}{x} dx - \int_0^{1/n} \frac{|\varphi_n(x)|}{x} dx = \log n + \log\left(\frac{1}{2}\right) - R_n,$$

where R_n is a bounded remainder. Consequently

$$\left| \langle \text{P.V.}(1/x), \varphi_n \rangle \right| \rightarrow \infty \quad \text{as } n \rightarrow \infty,$$

while $\sup_{[-1,1]} |\varphi_n| = 1$. This contradicts (*), so $\text{P.V.}(1/x)$ cannot be of order 0.

Combining the two steps, we conclude that the order of $\text{P.V.}(1/x)$ is exactly 1.

Regular distributions, order, and singularity hierarchy

Definition 0.3 (Regular distributions). Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. The functional

$$T_f(\varphi) := \int_{\mathbb{R}^n} f(x) \varphi(x) dx, \quad \varphi \in \mathcal{D}(\mathbb{R}^n),$$

is called the *regular distribution* associated with f . Every regular distribution has order 0.

Definition 0.4 (Order of a distribution). A distribution $T \in \mathcal{D}'(\mathbb{R}^n)$ has *order* $\leq m$ if for every compact $K \subset \mathbb{R}^n$ there exists $C_K > 0$ such that

$$|\langle T, \varphi \rangle| \leq C_K \sum_{|\alpha| \leq m} \sup_{x \in K} |\partial^\alpha \varphi(x)|, \quad \text{supp } \varphi \subset K.$$

The smallest such m is called the *order* of T .

Examples and their orders

1. Dirac mass δ_x (order 0).

$$\delta_x(\varphi) := \varphi(x).$$

For compact $K \ni x$,

$$|\delta_x(\varphi)| = |\varphi(x)| \leq \sup_K |\varphi|.$$

Thus δ_x is linear, continuous, and of **order 0**.

2. Regular distribution T_f (order 0). If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$,

$$T_f(\varphi) = \int f(x)\varphi(x) dx.$$

If $\text{supp } \varphi \subset K$,

$$|T_f(\varphi)| \leq \left(\int_K |f| \right) \sup_K |\varphi|.$$

Thus T_f is of **order 0**.

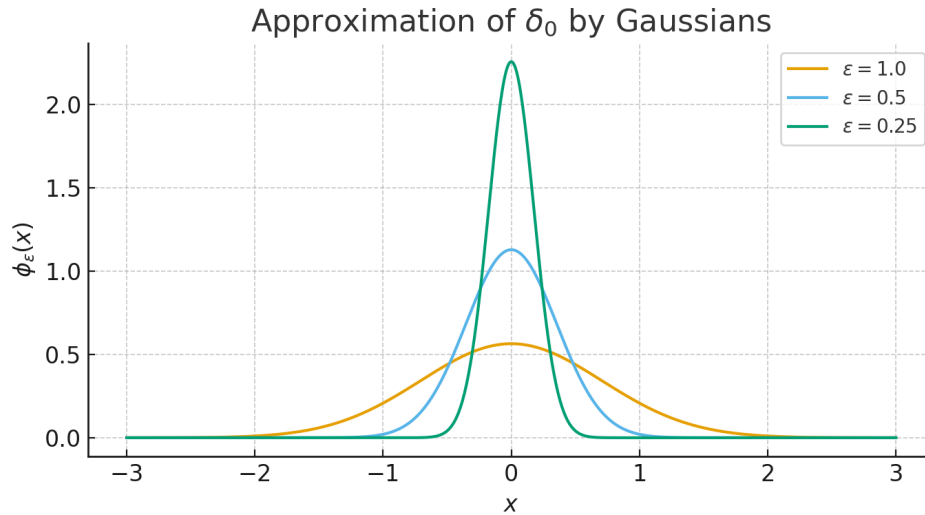


Figure 39: Approximation of the Dirac mass δ_0 by Gaussians $\phi_\varepsilon(x) = \frac{1}{\sqrt{\pi\varepsilon}} e^{-x^2/\varepsilon^2}$ as $\varepsilon \rightarrow 0$. Each ϕ_ε is a regular function, but the limit distribution has order 0.

Example (b): regular distribution T_f . Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and define

$$T_f(\varphi) = \int_{\mathbb{R}^n} f(x) \varphi(x) dx.$$

If $\text{supp } \varphi \subset K$, then

$$|T_f(\varphi)| \leq \int_K |f(x)| |\varphi(x)| dx \leq \left(\int_K |f(x)| dx \right) \sup_K |\varphi(x)|.$$

Hence T_f is linear, continuous, and of order 0.

3. Principal value distribution P.V.(1/x) (order 1).

$$\left\langle \text{P.V.}\left(\frac{1}{x}\right), \varphi \right\rangle := \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

One shows that

$$\langle \text{P.V.}(1/x), \varphi \rangle = - \int_{\mathbb{R}} \log |x| \varphi'(x) dx.$$

Thus

$$|\langle \text{P.V.}(1/x), \varphi \rangle| \leq C_K \sup_K |\varphi'|.$$

So P.V.(1/x) has **order** ≤ 1 . It is not of order 0, hence the order is **exactly 1**.

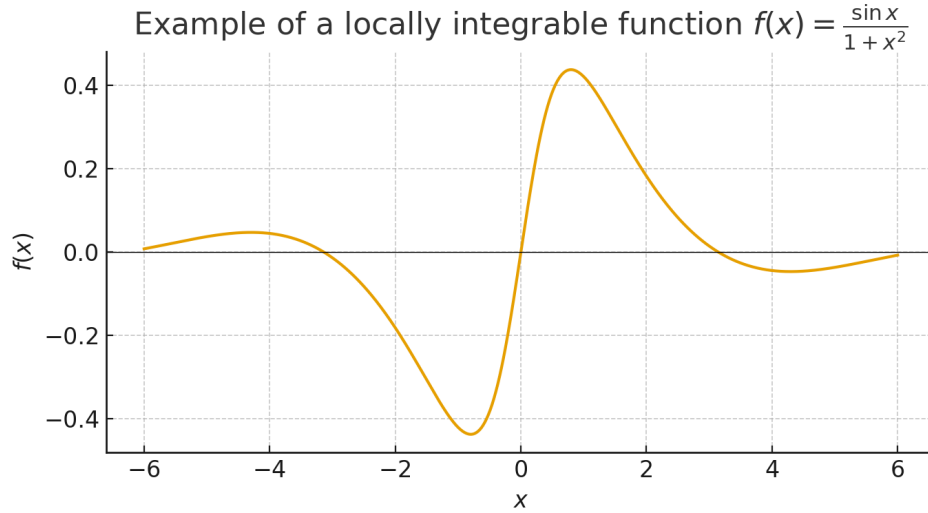


Figure 40: Example of a locally integrable function $f(x) = \frac{\sin x}{1+x^2}$. The associated regular distribution T_f has order 0: it only depends on the size of φ , not on its derivatives.

4. Derivative of the Dirac mass δ'_0 (order 1). By definition,

$$\langle \delta'_0, \varphi \rangle := -\varphi'(0).$$

Therefore, for compact $K \ni 0$,

$$|\langle \delta'_0, \varphi \rangle| = |\varphi'(0)| \leq \sup_K |\varphi'|.$$

Hence δ'_0 has **order 1**.

Singularity hierarchy: lower order means more regular

functions $\subset$ measures (order 0) $\subset$ order 1 distributions $\subset$ order 2 distributions $\subset \dots$

- Order 0: only $\sup |\varphi|$ matters (regular T_f , δ_x).
- Order 1: depends on first derivatives of φ (δ' , P.V.($1/x$)).
- Order m : depends on derivatives up to order m — increasingly singular.

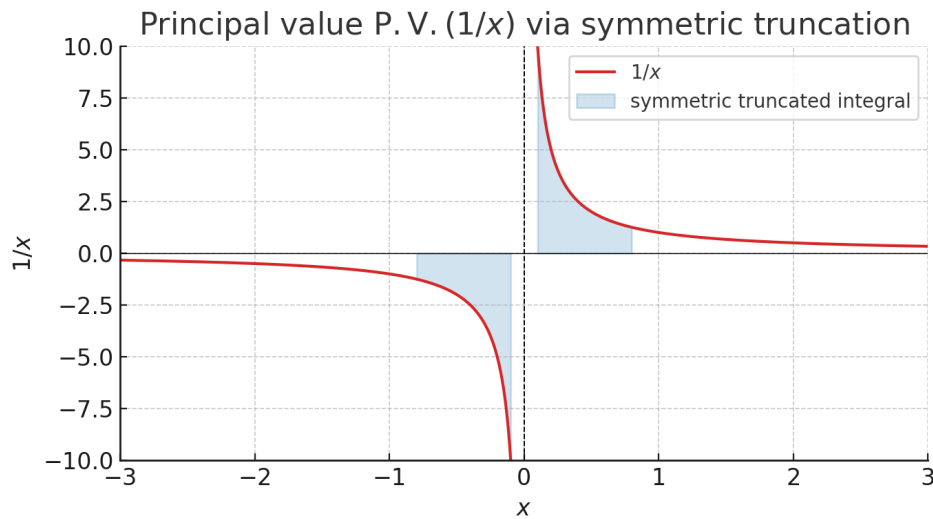


Figure 41: Graph of $1/x$ with symmetric truncation around the origin, illustrating the principal value definition of $\text{P.V.}(1/x)$. The distribution cannot be represented by a locally integrable function, and it has order 1, depending on first derivatives of test functions.

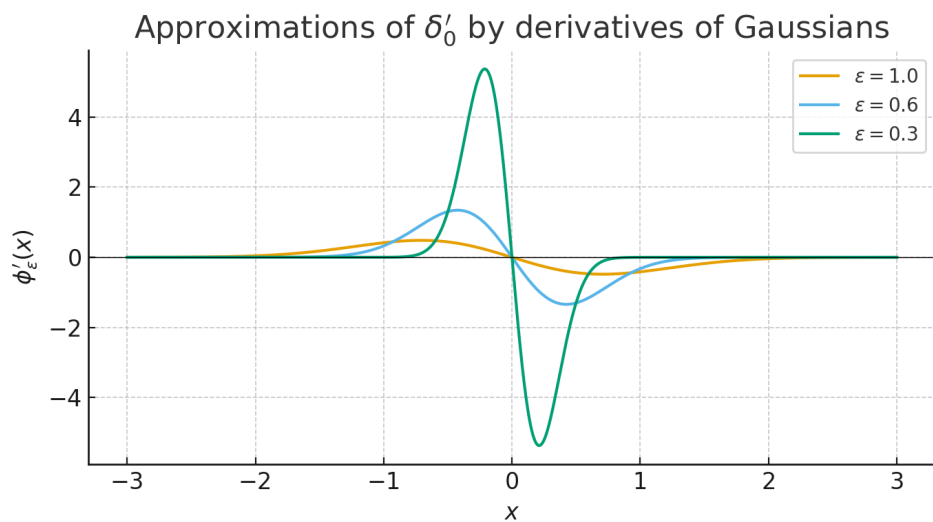


Figure 42: Approximations of the derivative of the Dirac mass δ'_0 by derivatives of Gaussians $\phi'_\varepsilon(x) = -(2x/\varepsilon^2)\phi_\varepsilon(x)$. The positive and negative lobes concentrate near 0, consistent with $\langle \delta'_0, \varphi \rangle = -\varphi'(0)$.

Exercise 2.7. Local mollification in L^1_{loc}

Suppose $f \in L^1_{\text{loc}}(\Omega)$. Let ϕ_ε be the standard mollifier from Theorem 1.13 and, for x with $d(x, \partial\Omega) \geq \varepsilon$, define

$$f_\varepsilon(x) := T_f[T_x\check{\phi}_\varepsilon],$$

where $\check{\phi}_\varepsilon(y) := \phi_\varepsilon(-y)$ and $T_x\psi(y) := \psi(y - x)$.

Show that for any compact $K \subset \Omega$,

$$\|f_\varepsilon - f\|_{L^1(K)} \longrightarrow 0 \quad \text{as } \varepsilon \rightarrow 0.$$

Hint: Follow the proof of Theorem 2.2, but use part (b) of Theorem 1.13.

Solution

First note that for x with $d(x, \partial\Omega) \geq \varepsilon$,

$$\begin{aligned} f_\varepsilon(x) &= T_f[T_x\check{\phi}_\varepsilon] = \int_{\Omega} f(y) (T_x\check{\phi}_\varepsilon)(y) dy \\ &= \int_{\Omega} f(y) \check{\phi}_\varepsilon(y - x) dy = \int_{\Omega} f(y) \phi_\varepsilon(x - y) dy. \end{aligned}$$

Thus $f_\varepsilon(x) = (f * \phi_\varepsilon)(x)$ wherever $B(x, \varepsilon) \subset \Omega$.

Let $K \subset \Omega$ be compact. There exists an open set U with

$$K \subset U, \quad \overline{U} \subset \Omega.$$

Choose $\chi \in C_c^\infty(\Omega)$ such that $\chi \equiv 1$ on U . Extend χf by 0 to a function in $L^1(\mathbb{R}^n)$.

For $\varepsilon > 0$ sufficiently small we have $\varepsilon < \text{dist}(K, \partial\Omega)$, so that for all $x \in K$,

$$f_\varepsilon(x) = \int_{\Omega} f(y) \phi_\varepsilon(x - y) dy = \int_{\Omega} \chi(y) f(y) \phi_\varepsilon(x - y) dy = ((\chi f) * \phi_\varepsilon)(x).$$

Hence, for such ε ,

$$\|f_\varepsilon - f\|_{L^1(K)} = \|(\chi f) * \phi_\varepsilon - \chi f\|_{L^1(K)} \leq \|(\chi f) * \phi_\varepsilon - \chi f\|_{L^1(\mathbb{R}^n)}.$$

By Theorem 1.13(b), since $\chi f \in L^1(\mathbb{R}^n)$, we have

$$\|(\chi f) * \phi_\varepsilon - \chi f\|_{L^1(\mathbb{R}^n)} \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0.$$

Therefore $\|f_\varepsilon - f\|_{L^1(K)} \rightarrow 0$ as $\varepsilon \rightarrow 0$, which shows that $f_\varepsilon \rightarrow f$ in $L^1_{\text{loc}}(\Omega)$.

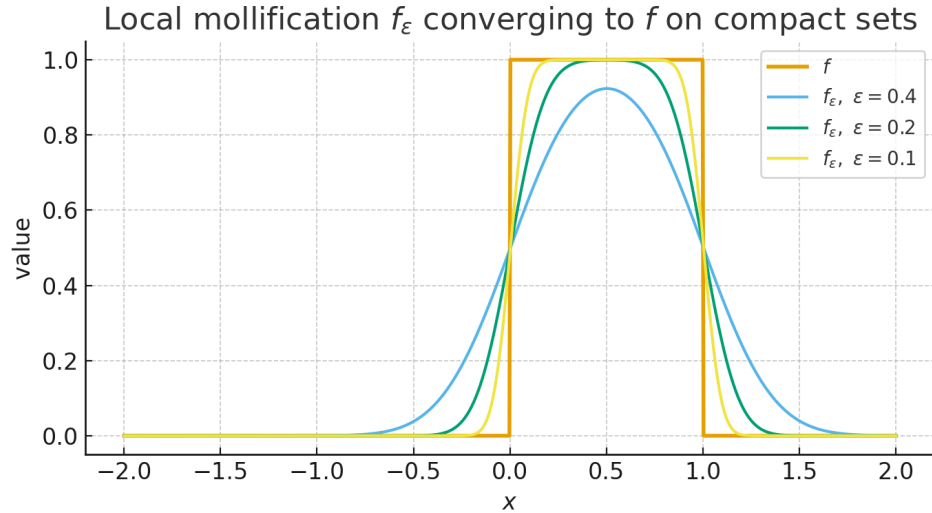


Figure 43: Example of local mollification. Here $f(x) = 1$ on $(0, 1)$ and 0 elsewhere, and $f_\varepsilon = f * \phi_\varepsilon$ is plotted for several values of ε . On compact sets away from the boundary, f_ε converges to f in L^1 .

Mollifiers, test functions and tempered test functions

Standard mollifiers as test functions. A *standard mollifier* is built from a function $\phi \in C_c^\infty(\mathbb{R}^n)$ such that

- $\phi \geq 0$,
- $\text{supp } \phi \subset B(0, 1)$,
- $\int_{\mathbb{R}^n} \phi(x) dx = 1$.

For $\varepsilon > 0$ we set

$$\phi_\varepsilon(x) := \frac{1}{\varepsilon^n} \phi\left(\frac{x}{\varepsilon}\right).$$

Each ϕ_ε is smooth and compactly supported, hence

$$\phi_\varepsilon \in C_c^\infty(\mathbb{R}^n) = \mathcal{D}(\mathbb{R}^n).$$

Thus, in the framework of $\mathcal{D}'(\mathbb{R}^n)$, *standard mollifiers are particular test functions* which additionally satisfy the approximate identity properties

$$\phi_\varepsilon \geq 0, \quad \int \phi_\varepsilon = 1, \quad \phi_\varepsilon \rightarrow \delta_0 \text{ in } \mathcal{D}'.$$

Gaussian mollifiers. One often also uses the Gaussian kernels

$$\eta_\varepsilon(x) := \frac{1}{(2\pi\varepsilon^2)^{n/2}} e^{-\frac{|x|^2}{2\varepsilon^2}}, \quad \varepsilon > 0.$$

These functions are smooth and rapidly decaying, but not compactly supported. Hence

$$\eta_\varepsilon \notin \mathcal{D}(\mathbb{R}^n),$$

where compact support is required, but

$$\eta_\varepsilon \in \mathcal{S}(\mathbb{R}^n),$$

the Schwartz space of rapidly decreasing smooth functions. They also form an approximate identity, now in the sense of tempered distributions.

Relation to tempered test functions and tempered distributions. Recall that

$$\mathcal{D}(\mathbb{R}^n) = C_c^\infty(\mathbb{R}^n)$$

is the test function space for *distributions* $\mathcal{D}'(\mathbb{R}^n)$, while

$$\mathcal{S}(\mathbb{R}^n)$$

is the test function space for *tempered distributions* $\mathcal{S}'(\mathbb{R}^n)$ (distributions of at most polynomial growth).

- Compactly supported mollifiers ϕ_ε live in $\mathcal{D}(\mathbb{R}^n)$ and are suitable for regularizing arbitrary distributions in $\mathcal{D}'(\mathbb{R}^n)$ (at least away from the boundary in a domain).
- Gaussian-type mollifiers η_ε live in $\mathcal{S}(\mathbb{R}^n)$ and are particularly well adapted to the theory of tempered distributions $\mathcal{S}'(\mathbb{R}^n)$, since convolution with η_ε preserves temperateness and interacts nicely with the Fourier transform.

In summary:

“standard mollifier” in Theorem 1.13 $\in \mathcal{D}$, Gaussian mollifier $\in \mathcal{S}$,

and both act as approximate identities in their respective distribution spaces $\mathcal{D}'$ and $\mathcal{S}'$.

Mollifiers acting on distributions and tempered distributions

Lemma 0.5 (Mollifiers in $\mathcal{D}'(\mathbb{R}^n)$). *Let $(\phi_\varepsilon)_{\varepsilon>0} \subset \mathcal{D}(\mathbb{R}^n)$ be a standard approximate identity as in Theorem 1.13, and let $T \in \mathcal{D}'(\mathbb{R}^n)$ be a distribution. Define the convolution $\phi_\varepsilon * T$ by*

$$\langle \phi_\varepsilon * T, \psi \rangle := \langle T, \check{\phi}_\varepsilon * \psi \rangle, \quad \psi \in \mathcal{D}(\mathbb{R}^n),$$

*where $\check{\phi}_\varepsilon(x) := \phi_\varepsilon(-x)$. Then $\phi_\varepsilon * T \in C^\infty(\mathbb{R}^n)$ for every $\varepsilon > 0$, and*

$$\phi_\varepsilon * T \longrightarrow T \quad \text{in } \mathcal{D}'(\mathbb{R}^n) \text{ as } \varepsilon \rightarrow 0.$$

Proof. First, for each fixed x , the function $y \mapsto \phi_\varepsilon(x - y)$ is a test function, so

$$(\phi_\varepsilon * T)(x) := \langle T, \phi_\varepsilon(x - \cdot) \rangle$$

defines a smooth function of x . The distributional formulation is

$$\langle \phi_\varepsilon * T, \psi \rangle = \langle T, \check{\phi}_\varepsilon * \psi \rangle, \quad \psi \in \mathcal{D}(\mathbb{R}^n).$$

By Theorem 1.13(b), $(\check{\phi}_\varepsilon)_{\varepsilon>0}$ is also an approximate identity, so for each $\psi \in \mathcal{D}$ we have

$$\check{\phi}_\varepsilon * \psi \longrightarrow \psi \quad \text{in } \mathcal{D}(\mathbb{R}^n).$$

Since T is continuous on $\mathcal{D}$, it follows that

$$\langle \phi_\varepsilon * T, \psi \rangle = \langle T, \check{\phi}_\varepsilon * \psi \rangle \longrightarrow \langle T, \psi \rangle \quad (\varepsilon \rightarrow 0).$$

This is precisely the convergence $\phi_\varepsilon * T \rightarrow T$ in $\mathcal{D}'(\mathbb{R}^n)$. □

Lemma 0.6 (Mollifiers in $\mathcal{S}'(\mathbb{R}^n)$). *Let $(\eta_\varepsilon)_{\varepsilon>0} \subset \mathcal{S}(\mathbb{R}^n)$ be a Schwartz approximate identity (e.g. Gaussian mollifiers), and let $U \in \mathcal{S}'(\mathbb{R}^n)$ be a tempered distribution. Define*

$$\langle \eta_\varepsilon * U, \psi \rangle := \langle U, \check{\eta}_\varepsilon * \psi \rangle, \quad \psi \in \mathcal{S}(\mathbb{R}^n),$$

*with $\check{\eta}_\varepsilon(x) := \eta_\varepsilon(-x)$. Then $\eta_\varepsilon * U \in \mathcal{S}(\mathbb{R}^n)$ for each $\varepsilon > 0$, and*

$$\eta_\varepsilon * U \longrightarrow U \quad \text{in } \mathcal{S}'(\mathbb{R}^n) \text{ as } \varepsilon \rightarrow 0.$$

Proof. For each $\varepsilon > 0$ and $\psi \in \mathcal{S}$, the convolution $\check{\eta}_\varepsilon * \psi$ belongs to $\mathcal{S}$ and, by the approximate identity property in the Schwartz topology,

$$\check{\eta}_\varepsilon * \psi \longrightarrow \psi \quad \text{in } \mathcal{S}(\mathbb{R}^n).$$

Since U is continuous on $\mathcal{S}$, we obtain

$$\langle \eta_\varepsilon * U, \psi \rangle = \langle U, \check{\eta}_\varepsilon * \psi \rangle \longrightarrow \langle U, \psi \rangle.$$

This is exactly the convergence $\eta_\varepsilon * U \rightarrow U$ in $\mathcal{S}'(\mathbb{R}^n)$. □

Fourier transform viewpoint for Gaussian mollifiers

Let $\eta_\varepsilon \in \mathcal{S}(\mathbb{R}^n)$ be the Gaussian mollifier

$$\eta_\varepsilon(x) = \frac{1}{(2\pi\varepsilon^2)^{n/2}} e^{-\frac{|x|^2}{2\varepsilon^2}}.$$

Up to normalisation conventions, its Fourier transform is again a Gaussian:

$$\widehat{\eta_\varepsilon}(\xi) = e^{-\frac{\varepsilon^2|\xi|^2}{2}}.$$

In particular, for any tempered distribution $U \in \mathcal{S}'(\mathbb{R}^n)$ we have

$$\mathcal{F}(\eta_\varepsilon * U) = \widehat{\eta_\varepsilon * U} = \widehat{\eta_\varepsilon} \widehat{U} = e^{-\frac{\varepsilon^2|\xi|^2}{2}} \widehat{U}(\xi).$$

Thus convolution with the Gaussian mollifier corresponds in the frequency domain to multiplication by the *low-pass filter*

$$m_\varepsilon(\xi) := e^{-\frac{\varepsilon^2|\xi|^2}{2}}.$$

For each fixed $\varepsilon > 0$, the factor $m_\varepsilon(\xi)$ damps high frequencies strongly, so that $\eta_\varepsilon * U$ is a very smooth (indeed Schwartz) function. As $\varepsilon \rightarrow 0$,

$$m_\varepsilon(\xi) \longrightarrow 1 \quad \text{for all } \xi \in \mathbb{R}^n,$$

so $\widehat{\eta_\varepsilon * U} \rightarrow \widehat{U}$ in $\mathcal{S}'$, which is equivalent to the convergence

$$\eta_\varepsilon * U \longrightarrow U \quad \text{in } \mathcal{S}'(\mathbb{R}^n).$$

In summary, Gaussian mollification can be viewed either as *smoothing in physical space* (convolution with η_ε) or as *frequency cut-off* (multiplication by $e^{-\varepsilon^2|\xi|^2/2}$) in the Fourier domain.

Example A: regular tempered distribution T_f . Let $f(x) = \frac{\sin x}{1+x^2}$. Then $f \in L^1_{\text{loc}}(\mathbb{R})$ and even $f \in \mathcal{S}(\mathbb{R})$, so the functional

$$\langle T_f, \varphi \rangle := \int_{\mathbb{R}} f(x) \varphi(x) dx, \quad \varphi \in \mathcal{D}(\mathbb{R}),$$

defines a regular distribution of order 0. Since f is rapidly decreasing, T_f is a tempered distribution as well ($T_f \in \mathcal{S}'(\mathbb{R})$).

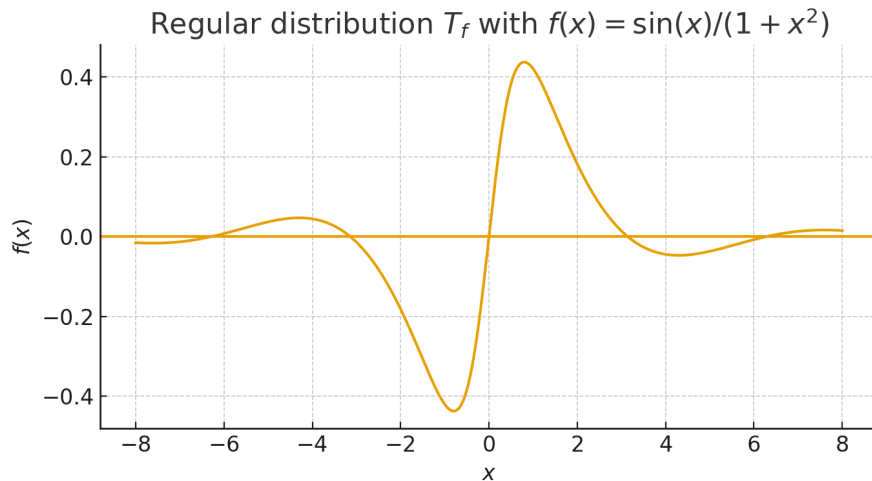


Figure 44: Graph of $f(x) = \sin(x)/(1+x^2)$, inducing the regular tempered distribution T_f .

Concrete action of a regular distribution. Let

$$f(x) = \frac{\sin x}{1 + x^2},$$

and let $\varphi \in C_c^\infty(\mathbb{R})$ be a smooth bump supported in $(-2, 2)$ (for instance $\varphi(x) = C \exp(-1/(1 - (x/2)^2))$ for $|x| < 2$ and 0 otherwise, with C chosen so that $\int \varphi = 1$). The regular distribution T_f acts by

$$\langle T_f, \varphi \rangle = \int_{\mathbb{R}} f(x) \varphi(x) dx.$$

The integrand is the pointwise product $f(x)\varphi(x)$.

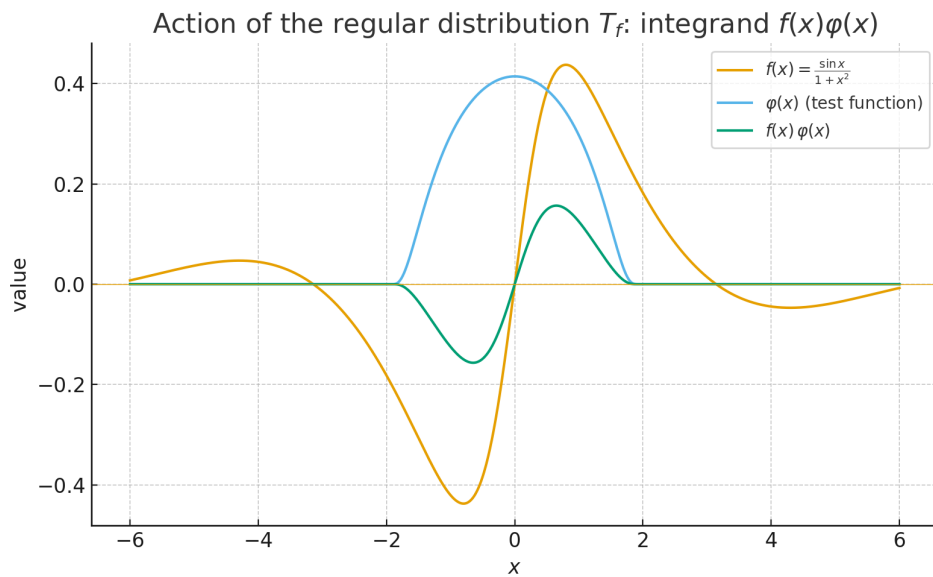


Figure 45: The function $f(x) = \sin(x)/(1 + x^2)$, a smooth compactly supported test function $\varphi(x)$, and their product $f(x)\varphi(x)$, which is the integrand in $\langle T_f, \varphi \rangle = \int f(x)\varphi(x) dx$.

Example B: Heaviside function and Dirac mass. Define the Heaviside function

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

It induces a regular distribution $T_H(\varphi) = \int_{\mathbb{R}} H(x)\varphi(x) dx$ of order 0. Using integration by parts one checks that

$$\langle H', \varphi \rangle := -\langle H, \varphi' \rangle = \varphi(0) = \langle \delta_0, \varphi \rangle,$$

so in $\mathcal{D}'(\mathbb{R})$ we have $H' = \delta_0$. Both T_H and δ_0 are tempered distributions.

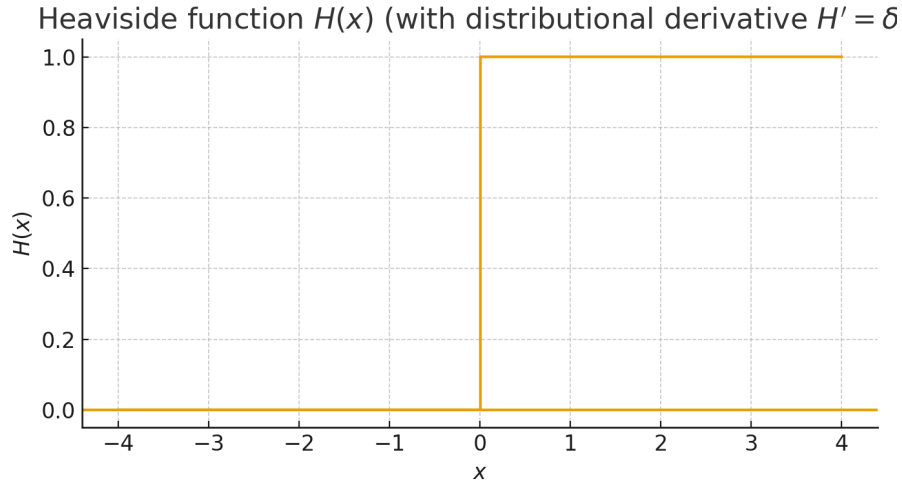


Figure 46: Heaviside function $H(x)$. Its distributional derivative is the Dirac mass δ_0 .

Example C: principal value $\text{P.V.}(1/x)$. The distribution

$$\langle \text{P.V.}(1/x), \varphi \rangle := \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx$$

is not given by a locally integrable function (since $1/x$ is not integrable near 0), hence it is not regular. However, the growth of $1/x$ at infinity is only of order $|x|^{-1}$, so the functional is continuous on the Schwartz space $\mathcal{S}(\mathbb{R})$, and thus

$$\text{P.V.}(1/x) \in \mathcal{S}'(\mathbb{R})$$

is a tempered distribution.

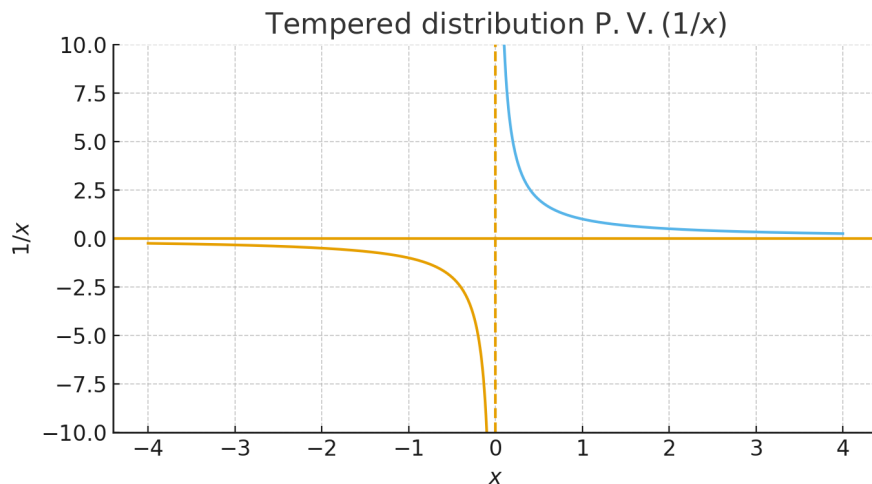


Figure 47: Graph of $1/x$ with a dashed vertical line at 0, indicating the singularity underlying the tempered distribution $\text{P.V.}(1/x)$.

Object	Regular?	Tempered?	Order
$f \in L^1_{\text{loc}}$, moderate growth	yes (T_f)	yes	0
H (Heaviside)	yes	yes	0
δ_0	no (not L^1_{loc})	yes	0
δ'_0	no	yes	1
P.V.($1/x$)	no	yes	1
e^{x^2} (as T_f)	yes	no (too fast growth)	0

Example D: Dirac mass δ_0 . The Dirac distribution acts by evaluation:

$$\langle \delta_0, \varphi \rangle = \varphi(0).$$

The following figure shows a test function φ together with the highlighted value $\varphi(0)$.

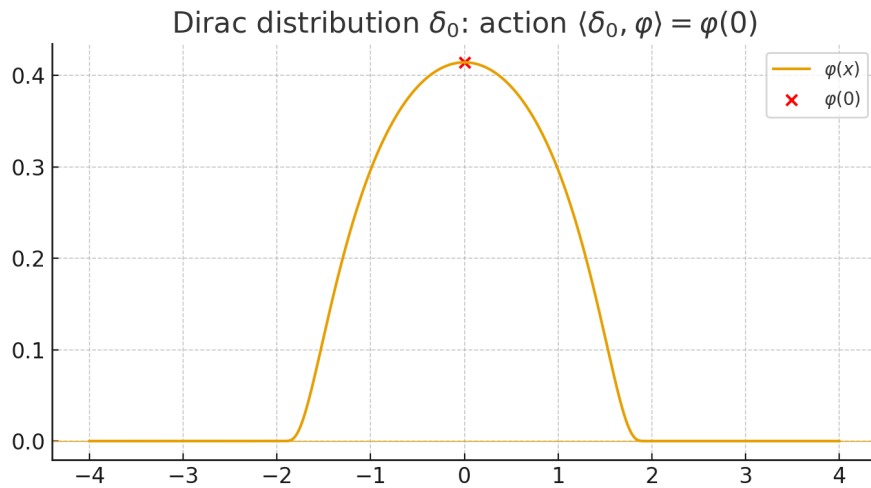


Figure 48: Test function φ and the point $\varphi(0)$ corresponding to the action of δ_0 .

Example E: derivative of Dirac δ'_0 . The action of the derivative of the Dirac mass is

$$\langle \delta'_0, \varphi \rangle = -\varphi'(0).$$

The next figure displays φ and φ' and highlights $\varphi'(0)$.

Example F: principal value P.V.($1/x$). The distribution P.V.($1/x$) is defined by

$$\langle \text{P.V.}(1/x), \varphi \rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

The graph below shows $\varphi(x)$ and the truncated integrand $\varphi(x)/x$ for $|x| > \varepsilon$.

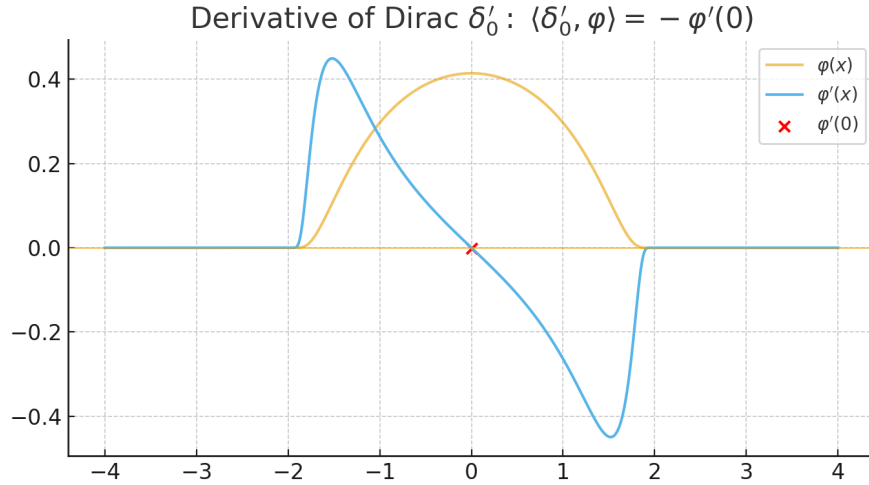


Figure 49: Test function φ , its derivative φ' , and the highlighted value $\varphi'(0)$ corresponding to the action of δ'_0 .

Exercise 2.8. Algebra and differentiation of regular distributions

(a) Show that if $f_1, f_2 \in C^0(\Omega)$ and $a \in C^\infty(\Omega)$, then

$$aT_{f_1} + T_{f_2} = T_{af_1+f_2}.$$

(b) Show that if $f \in C^k(\Omega)$ then

$$D^\alpha T_f = T_{D^\alpha f} \quad \text{for all multiindices } \alpha \text{ with } |\alpha| \leq k.$$

Deduce that $\iota \circ D^\alpha = D^\alpha \circ \iota$, where $\iota : C^k(\Omega) \rightarrow \mathcal{D}'(\Omega)$ is the natural embedding $f \mapsto T_f$.

(c) Deduce that if $f \in C^k(\Omega)$ then

$$\sum_{|\alpha| \leq k} a_\alpha D^\alpha T_f = T_{Lf},$$

where

$$Lf := \sum_{|\alpha| \leq k} a_\alpha D^\alpha f$$

and $a_\alpha \in C^\infty(\Omega)$.

Let $f \in C^0(\Omega)$ and

$$T_f(\varphi) := \int_{\Omega} f(x) \varphi(x) dx.$$

For $a \in C^\infty(\Omega)$ and $T \in \mathcal{D}'(\Omega)$ define $(aT)(\varphi) := T(a\varphi)$. For a multiindex α ,

$$\langle D^\alpha T, \varphi \rangle := (-1)^{|\alpha|} \langle T, D^\alpha \varphi \rangle.$$

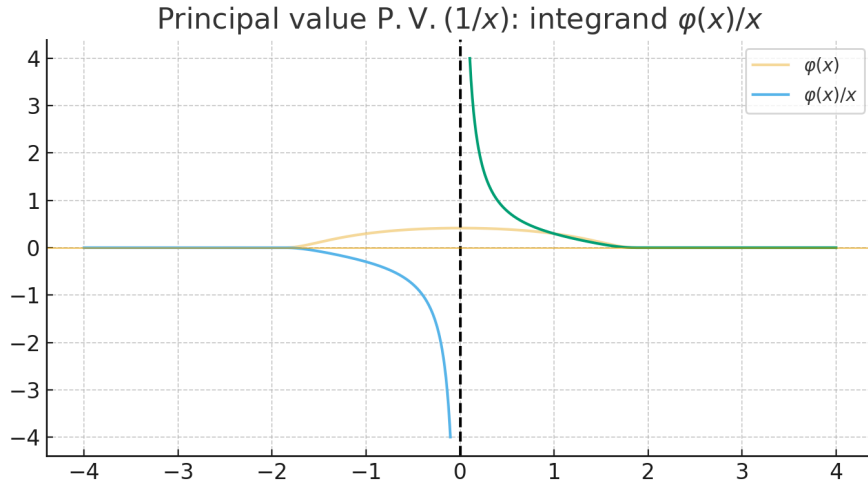


Figure 50: Test function $\varphi(x)$ and the integrand $\varphi(x)/x$ used in the principal value distribution.

(a) If $f_1, f_2 \in C^0(\Omega)$ and $a \in C^\infty(\Omega)$, show

$$aT_{f_1} + T_{f_2} = T_{af_1+f_2}.$$

solution

For $\varphi \in \mathcal{D}(\Omega)$,

$$\begin{aligned} \langle aT_{f_1} + T_{f_2}, \varphi \rangle &= \langle T_{f_1}, a\varphi \rangle + \langle T_{f_2}, \varphi \rangle \\ &= \int_{\Omega} f_1(x)a(x)\varphi(x) dx + \int_{\Omega} f_2(x)\varphi(x) dx \\ &= \int_{\Omega} (af_1 + f_2)(x)\varphi(x) dx = \langle T_{af_1+f_2}, \varphi \rangle. \end{aligned}$$

Thus $aT_{f_1} + T_{f_2} = T_{af_1+f_2}$.

(b) If $f \in C^k(\Omega)$, show that for $|\alpha| \leq k$,

$$D^\alpha T_f = T_{D^\alpha f}.$$

Deduce that $\iota \circ D^\alpha = D^\alpha \circ \iota$ for the embedding $\iota : C^k(\Omega) \rightarrow \mathcal{D}'(\Omega)$, $\iota(f) = T_f$.

solution

For $\varphi \in \mathcal{D}(\Omega)$,

$$\langle D^\alpha T_f, \varphi \rangle = (-1)^{|\alpha|} \langle T_f, D^\alpha \varphi \rangle = (-1)^{|\alpha|} \int_{\Omega} f D^\alpha \varphi.$$

Since $f \in C^k(\Omega)$ and φ has compact support, we may integrate by parts $|\alpha|$ times, with no boundary term:

$$(-1)^{|\alpha|} \int_{\Omega} f D^{\alpha} \varphi = \int_{\Omega} D^{\alpha} f \varphi = \langle T_{D^{\alpha} f}, \varphi \rangle.$$

Hence $D^{\alpha} T_f = T_{D^{\alpha} f}$.

For the embedding ι , we have

$$(\iota \circ D^{\alpha})(f) = T_{D^{\alpha} f} = D^{\alpha} T_f = (D^{\alpha} \circ \iota)(f),$$

so $\iota \circ D^{\alpha} = D^{\alpha} \circ \iota$ on $C^k(\Omega)$.

(c) Let $f \in C^k(\Omega)$ and

$$Lf := \sum_{|\alpha| \leq k} a_{\alpha} D^{\alpha} f, \quad a_{\alpha} \in C^{\infty}(\Omega).$$

Show that

$$\sum_{|\alpha| \leq k} a_{\alpha} D^{\alpha} T_f = T_{Lf}.$$

solution

By part (b), $D^{\alpha} T_f = T_{D^{\alpha} f}$. By part (a), $a_{\alpha} T_{D^{\alpha} f} = T_{a_{\alpha} D^{\alpha} f}$, hence

$$\sum_{|\alpha| \leq k} a_{\alpha} D^{\alpha} T_f = \sum_{|\alpha| \leq k} T_{a_{\alpha} D^{\alpha} f} = T_{\sum_{|\alpha| \leq k} a_{\alpha} D^{\alpha} f} = T_{Lf}.$$

Exercise 2.9. Convolution of regular and general distributions

(a) Show that for $f, g \in C_0^{\infty}(\mathbb{R}^n)$,

$$T_{f*g} = T_f * T_g.$$

(b) Show that convolution is linear in both of its arguments, i.e. if $u_i \in \mathcal{D}'(\mathbb{R}^n)$ and u_3, u_4 have compact support, then for any $a \in \mathbb{C}$,

$$(u_1 + au_2) * u_3 = u_1 * u_3 + a u_2 * u_3,$$

and

$$u_1 * (u_3 + au_4) = u_1 * u_3 + a u_1 * u_4.$$

Solution

(a) Let $f, g \in C_0^\infty(\mathbb{R}^n)$. We show that $T_{f*g} = T_f * T_g$.

For any $\varphi \in \mathcal{D}(\mathbb{R}^n)$,

$$\begin{aligned}
 \langle T_f * T_g, \varphi \rangle &= \left\langle T_f, x \mapsto \langle T_g, \varphi(x + \cdot) \rangle \right\rangle \\
 &= \int_{\mathbb{R}^n} f(x) \left(\int_{\mathbb{R}^n} g(y) \varphi(x + y) dy \right) dx \\
 &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(x) g(y) \varphi(x + y) dy dx \\
 &= \int_{\mathbb{R}^n} (f * g)(z) \varphi(z) dz \\
 &= \langle T_{f*g}, \varphi \rangle.
 \end{aligned}$$

Thus

$$T_f * T_g = T_{f*g}.$$

(b) Let $u_1, u_2, u_3, u_4 \in \mathcal{D}'(\mathbb{R}^n)$, with u_3, u_4 of compact support. For any $\varphi \in \mathcal{D}(\mathbb{R}^n)$,

$$\langle u_1 * u_3, \varphi \rangle = \left\langle u_1, x \mapsto \langle u_3, \varphi(x + \cdot) \rangle \right\rangle.$$

Then

$$\begin{aligned}
 \langle (u_1 + au_2) * u_3, \varphi \rangle &= \left\langle u_1 + au_2, x \mapsto \langle u_3, \varphi(x + \cdot) \rangle \right\rangle \\
 &= \langle u_1 * u_3, \varphi \rangle + a \langle u_2 * u_3, \varphi \rangle.
 \end{aligned}$$

Hence

$$(u_1 + au_2) * u_3 = u_1 * u_3 + a u_2 * u_3.$$

The same computation shows that

$$u_1 * (u_3 + au_4) = u_1 * u_3 + a u_1 * u_4.$$

□

Remarks on convolution of distributions and regularisation

(1) **Why is compact support needed?** For functions f, g the convolution

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy$$

is defined by an ordinary integral. For distributions $u, v \in \mathcal{D}'(\mathbb{R}^n)$ one instead defines

$$\langle u * v, \varphi \rangle := \left\langle u_x, \left\langle v_y, \varphi(x + y) \right\rangle \right\rangle, \quad \varphi \in \mathcal{D}(\mathbb{R}^n),$$

whenever this makes sense. The inner pairing

$$x \mapsto \langle v_y, \varphi(x + y) \rangle$$

must be a test function in x (smooth with compact support), so that the outer pairing with u is defined. This is guaranteed, for example, if v has compact support. Without such a hypothesis there is no reason for $x \mapsto \langle v_y, \varphi(x + y) \rangle$ to belong to $\mathcal{D}(\mathbb{R}^n)$, and the convolution $u * v$ may fail to be well-defined. This is why, in the general theory, one usually assumes that at least one of the distributions has compact support (or stronger decay conditions) when defining $u * v$.

(2) Convolution with test functions as regularisation. If $\phi \in C_0^\infty(\mathbb{R}^n)$ and $T \in \mathcal{D}'(\mathbb{R}^n)$, the convolution

$$(T * \phi)(x) := \langle T, \phi(x - \cdot) \rangle$$

is always well-defined and belongs to $C^\infty(\mathbb{R}^n)$. In fact, for every multiindex α ,

$$D^\alpha(T * \phi) = T * (D^\alpha \phi).$$

If $(\phi_\varepsilon)_{\varepsilon>0}$ is a mollifier (approximate identity), one moreover has

$$T * \phi_\varepsilon \longrightarrow T \quad \text{in } \mathcal{D}'(\mathbb{R}^n) \text{ as } \varepsilon \rightarrow 0.$$

Thus convolution with test functions provides a canonical way to *regularise* distributions:

$$T \in \mathcal{D}'(\mathbb{R}^n) \xrightarrow{* \phi_\varepsilon} T * \phi_\varepsilon \in C^\infty(\mathbb{R}^n), \quad T * \phi_\varepsilon \rightarrow T.$$

In the tempered setting, the same picture holds with Schwartz mollifiers $\eta_\varepsilon \in \mathcal{S}(\mathbb{R}^n)$ and tempered distributions $U \in \mathcal{S}'(\mathbb{R}^n)$.

Convolution with test functions. If $\phi \in C_0^\infty(\mathbb{R}^n)$ and $T \in \mathcal{D}'(\mathbb{R}^n)$, the convolution $T * \phi$ is defined by

$$(T * \phi)(x) := \langle T, \phi(x - \cdot) \rangle, \quad x \in \mathbb{R}^n.$$

Here $\phi(x - \cdot)$ denotes the test function of the dummy variable y given by $y \mapsto \phi(x - y)$. Thus we first fix x , then consider the test function $y \mapsto \phi(x - y)$, and finally let T act on it.

Examples.

1. If $T = \delta_0$, then

$$(\delta_0 * \phi)(x) = \langle \delta_0, \phi(x - \cdot) \rangle = \phi(x - 0) = \phi(x).$$

2. More generally, for the Dirac mass at $a \in \mathbb{R}^n$,

$$(\delta_a * \phi)(x) = \langle \delta_a, \phi(x - \cdot) \rangle = \phi(x - a).$$

3. If $T = T_f$ is the regular distribution induced by $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then

$$(T_f * \phi)(x) = \int_{\mathbb{R}^n} f(y) \phi(x - y) dy,$$

which is the classical convolution of f and ϕ .

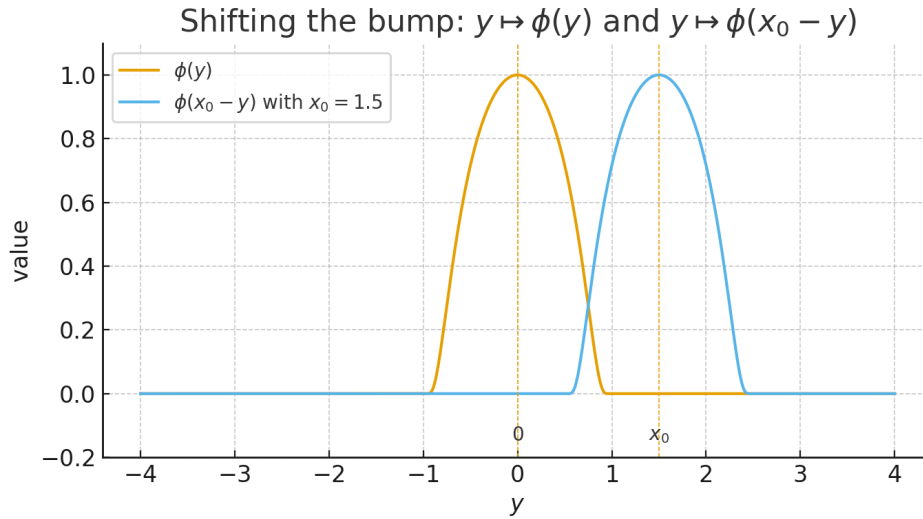


Figure 51: Geometric meaning of $\phi(x - \cdot)$. The blue curve is the bump $y \mapsto \phi(y)$, centred at 0. For a fixed x_0 , the orange curve is $y \mapsto \phi(x_0 - y)$: the same bump, reflected and shifted so that it is attached to the point x_0 . For each x , the test function $y \mapsto \phi(x - y)$ is what we feed into the distribution T in the definition $(T * \phi)(x) = \langle T, \phi(x - \cdot) \rangle$.

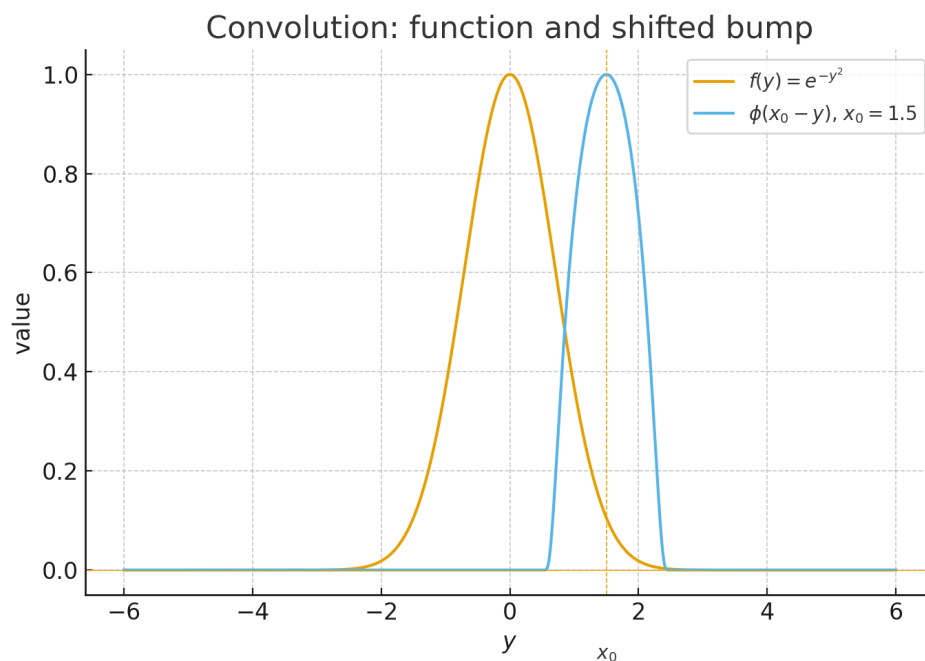


Figure 52: Convolution with a bump: the function $f(y) = e^{-y^2}$ (blue) and the shifted test function $y \mapsto \phi(x_0 - y)$ (orange) for a fixed x_0 . As x_0 varies, the bump slides along the y -axis.

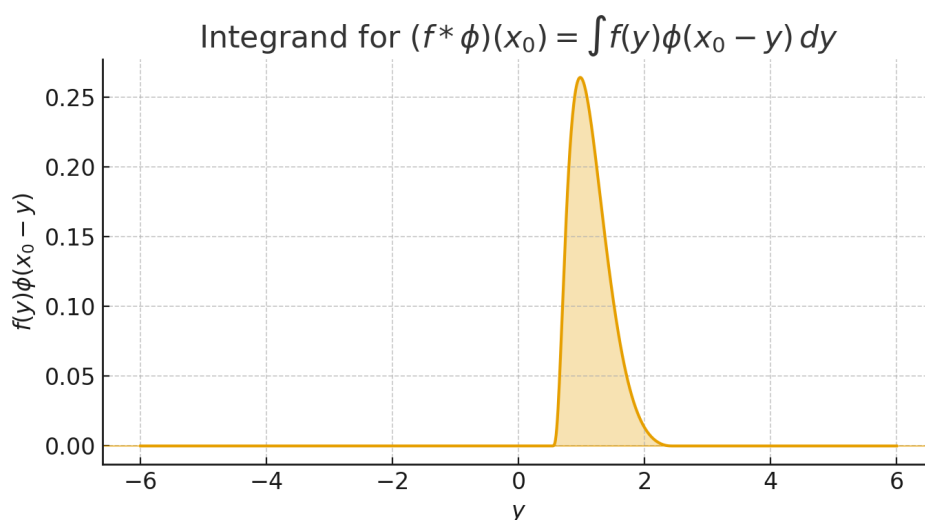


Figure 53: Integrand $f(y)\phi(x_0 - y)$ for $f(y) = e^{-y^2}$. The area under this curve equals $(f * \phi)(x_0)$. Convolution at x_0 is the weighted overlap of f with the shifted bump.

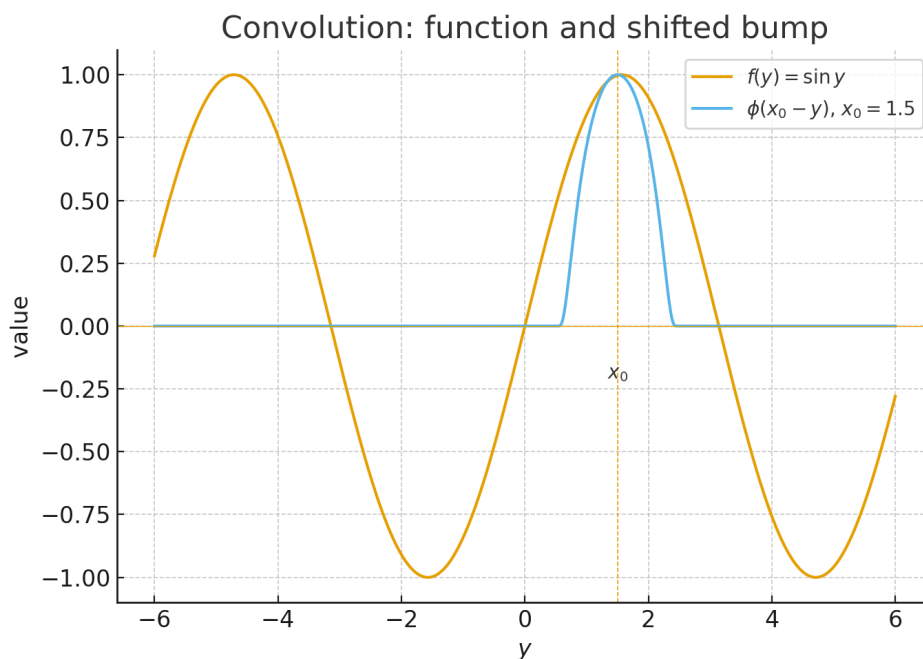


Figure 54: Convolution with a bump: the function $f(y) = \sin y$ (blue) and the shifted test function $y \mapsto \phi(x_0 - y)$ (orange) for a fixed x_0 . As x_0 moves, the bump samples local averages of $\sin y$.

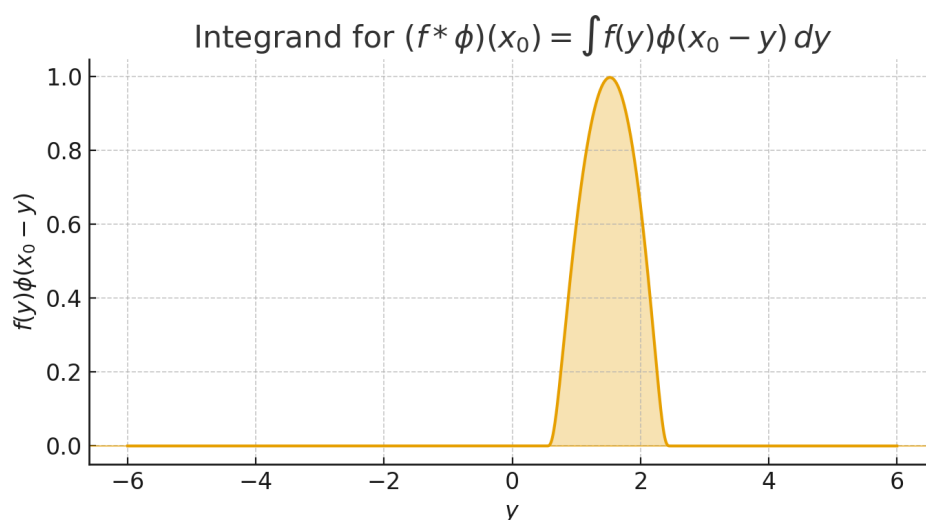


Figure 55: Integrand $f(y)\phi(x_0 - y)$ for $f(y) = \sin y$. The signed area under this curve equals $(f * \phi)(x_0)$. Convolution smooths oscillations by averaging $\sin y$ against the bump.

Exercise 2.10. Identity property of the Dirac distribution

(a) Show that if $\phi \in \mathcal{D}(\mathbb{R}^n)$ then

$$\delta_0 * \phi = \phi.$$

(b) Show that if $u \in \mathcal{D}'(\mathbb{R}^n)$ has compact support, then

$$\delta_0 * u = u.$$

Solution

(a) Let $\phi \in \mathcal{D}(\mathbb{R}^n)$. By definition of convolution between a distribution and a test function,

$$\langle \delta_0 * \phi, \psi \rangle = \langle \delta_0, x \mapsto \langle \phi, \psi(x + \cdot) \rangle \rangle, \quad \psi \in \mathcal{D}(\mathbb{R}^n).$$

Since δ_0 evaluates at 0,

$$\langle \delta_0 * \phi, \psi \rangle = \langle \phi, \psi(\cdot + 0) \rangle = \langle \phi, \psi \rangle.$$

Thus

$$\delta_0 * \phi = \phi.$$

(b) Let $u \in \mathcal{D}'(\mathbb{R}^n)$ have compact support. The convolution $\delta_0 * u$ is defined by

$$\langle \delta_0 * u, \psi \rangle = \langle \delta_0, x \mapsto \langle u, \psi(x + \cdot) \rangle \rangle, \quad \psi \in \mathcal{D}(\mathbb{R}^n).$$

Again using the defining property of δ_0 ,

$$\langle \delta_0 * u, \psi \rangle = \langle u, \psi(\cdot + 0) \rangle = \langle u, \psi \rangle.$$

Therefore,

$$\delta_0 * u = u.$$

□